

Transport by rough vector fields

Finn Rudolph

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Betreuer: Prof. Dr. Herbert Koch

Zweitgutachter: Prof. Dr. Juan J. L. Velázquez

MATHEMATISCHES INSTITUT

MATHEMATISCH-NATURWISSENSCHAFTLICHE FAKULTÄT DER
RHEINISCHEN FRIEDRICH-WILHELMS-UNIVERSITÄT BONN

Zusammenfassung

Ein klassisches Ergebnis zur Existenz und Eindeutigkeit von Lösungen gewöhnlicher Differentialgleichungen ist der Satz von Cauchy-Lipschitz. Wir betrachten die Differentialgleichung $\dot{\gamma} = b(t, \gamma(t))$, wobei $b : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ ein Vektorfeld ist und eine Anfangsbedingung $\gamma(0) \in \mathbb{R}^d$ gegeben ist. Für die Eindeutigkeit von Lösungen ist eine zentrale Voraussetzung des Satzes, dass b Lipschitz-stetig in der räumlichen Variable ist. In einigen Anwendungen ist diese Regularitätsannahme jedoch nicht erfüllt, beispielsweise in der Strömungsmechanik. DiPerna und Lions [DL89] haben den Satz von Cauchy-Lipschitz auf weniger reguläre Vektorfelder verallgemeinert. Der klassische punktweise Lösungsbegriff wird hierbei durch *Regular Lagrangian flows* ersetzt. Ziel dieser Arbeit ist es, die Theorie von DiPerna und Lions mit detaillierten Beweisen vorzustellen, sowie Anwendungen und Gegenbeispiele zu zeigen.

Die Methode von DiPerna und Lions basiert auf der Analyse einer zugehörigen Transportgleichung. In Abschnitt §2 entwickeln wir die Theorie renormalisierter Lösungen für die Transportgleichung. Daraus werden in Abschnitt §3 Existenz- und Eindeutigkeitsresultate für die gewöhnliche Differentialgleichung $\dot{\gamma} = b(t, \gamma(t))$ hergeleitet. Abschnitt §2.5 behandelt Anwendungen auf die zweidimensionale Eulergleichung. Zuletzt diskutieren wir drei Gegenbeispiele in Abschnitt §4.

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1 Introduction

Consider the ordinary differential equation

$$\begin{cases} \dot{\gamma}(t) = b(t, \gamma(t)) & \text{for every } t \in (0, T) \\ \gamma(0) = x^0 & \text{for some } x^0 \in \mathbb{R}^d \end{cases} \quad (1.1)$$

Here, $T > 0$ is a real number and $b : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a (possibly) time-dependent vector field. The classical existence and uniqueness result for (1.1) is the Cauchy-Lipschitz theorem. We denote by $\text{Lip}(\cdot)$ the Lipschitz constant of a Lipschitz-continuous function. The following is a global version of the theorem:

Theorem 1 (Cauchy-Lipschitz). *Let $b : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ be measurable. Assume that $b(t)$ is Lipschitz-continuous on $\mathbb{R}^d$ for every $t \in [0, T]$ with $\text{Lip}(b(t)) \in L^1([0, T])$. Then, for every $x^0 \in \mathbb{R}^d$, there exists a unique absolutely continuous solution $\gamma : [0, T] \rightarrow \mathbb{R}^d$ to (1.1).*

A natural question is whether the assumption of Lipschitz-continuity can be weakened. In [DL89], DiPerna and Lions extended the above result to vector fields that are not Lipschitz-continuous, but only belong to a Sobolev space. However, their results are of a different kind than Theorem 1: While Theorem 1 provides uniqueness for every initial condition x^0 individually, in the more general setting we only get uniqueness by adding some additional information over all initial conditions.

1.1 Flows with controlled compressibility

We will see that there exists a unique *regular Lagrangian flow* $X : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ under suitable assumptions on b . In the following, m^d denotes the d -dimensional Lebesgue measure. If $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is measurable, we denote by $f_{\#}m^d$ the pushforward measure of m^d along f , i.e. $f_{\#}m^d(A) = m^d(f^{-1}(A))$ for every m^d -measurable $A \subset \mathbb{R}^d$.

Definition 1 (Regular Lagrangian Flow). Let $b : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$. A measurable function $X : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a regular Lagrangian flow for b if

- F1) for m^d -almost every $x \in \mathbb{R}^d$, the map $t \mapsto X(t, x)$ is an absolutely continuous solution of (1.1) with $x^0 = x$,
- F2) there exists a $K \geq 0$ such that

$$X(t, \cdot)_{\#}m^d \leq Km^d \quad \text{for every } t \in [0, T].$$

The constant K is called the *compressibility constant* of X .

This is a common definition, see e.g. [CL08], p. 15 or [AT17], p. 731. Property F2) is the additional information over all initial conditions mentioned above. The following example shows that uniqueness can fail pointwise, but still there is a unique regular Lagrangian flow.

Remark 1 (Uniqueness of regular Lagrangian flows). We say that regular Lagrangian flows for a vector field b are unique, if any two regular Lagrangian flows X and Y satisfy $X(\cdot, x) = Y(\cdot, x)$ for almost every $x \in \mathbb{R}^d$.

Example. Take $d = 1$, $b(t, x) = \sqrt{|x|}$, and $x^0 \leq 0$. Then, for every $a \in [0, \infty]$, the following is a solution to (1.1).

$$\gamma_a(t) = \begin{cases} -(t/2 - \sqrt{-x^0})^2 & \text{if } t \leq 2\sqrt{-x^0} \\ 0 & \text{if } 2\sqrt{-x^0} < t \leq 2\sqrt{-x^0} + a \\ ((t-a)/2 - \sqrt{-x^0})^2 & \text{if } t > 2\sqrt{-x^0} + a \end{cases} \quad (1.2)$$

Informally speaking, since b is not Lipschitz-continuous, the solution is able to reach the point 0 and may stay there indefinitely. However, this is the only obstruction to uniqueness in this example. If we require that a solution γ satisfies

$$m^1(\{t \in [0, T] : \gamma(t) = 0\}) = 0,$$

then the only solution to (1.1) is γ_a with $a = 0$.

Now suppose that $X : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ is a regular Lagrangian flow for b . By F2), $m^1(X(t, \cdot)^{-1}(0)) = 0$ for every $t \in [0, T]$. Therefore

$$m^2(\{(t, x) \in [0, T] \times \mathbb{R} : X(t, x) = 0\}) = 0.$$

By Fubini's theorem, this implies that $m^1(\{t \in [0, T] : X(t, x) = 0\}) = 0$ for almost every $x \in \mathbb{R}$. Hence, for almost every $x \in \mathbb{R}$, $X(\cdot, x)$ must be equal to γ_a from (1.2) with $a = 0$ and $x^0 = x$. This shows that X is unique in the sense of Remark 1.

1.2 Flows in the Lipschitz case

For later reference and to motivate F2), we collect some properties of the flow map in the classical setting. We assume that $b(t)$ is Lipschitz-continuous on $\mathbb{R}^d$ and $\text{Lip}(b(t)) \in L^1([0, T])$. Then, by Theorem 1, there exists a map $X : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that for every $x \in \mathbb{R}^d$, $X(\cdot, x)$ is the unique absolutely continuous solution to (1.1) with $x^0 = x$.

The first observation is that for all $x, y \in \mathbb{R}^d$

$$\frac{d}{dt}|X(t, x) - X(t, y)| \leq |b(t, X(t, x)) - b(t, X(t, y))| \leq \text{Lip}(b(t))|X(t, x) - X(t, y)|. \quad (1.3)$$

Hence, by Grönwall's inequality (see [Eva10, p. 708]),

$$|X(t, x) - X(t, y)| \leq e^{\ell(t)}|x - y|, \quad \text{where } \ell(t) := \int_0^t \text{Lip } b(s) ds. \quad (1.4)$$

By reversing time, we can also show a similar reverse inequality. Let $t_0 \in [0, T]$ and $Y^{t_0} : [0, t_0] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ be the flow map of $-b(t_0 - t, x)$. Since

$$\frac{d}{dt}X(t_0 - t, x) = -b(t_0 - t, X(t_0 - t, x)) \quad \text{and} \quad X(t_0 - t, x)|_{t=0} = X(t_0, x),$$

both $X(t_0 - t, x)$ and $Y^{t_0}(t, X(t_0, x))$ are solutions to (1.1) with the vector field $-b(t_0 - t, x)$ and initial condition $X(t_0, x)$. By uniqueness of solutions to (1.1) in the Lipschitz case, $Y^{t_0}(t_0, X(t_0, x)) = x$ for every $x \in \mathbb{R}^d$. Hence $X(t_0) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is invertible. Moreover, by applying the computation in (1.3) to Y^{t_0} for every $t_0 \in [0, T]$,

$$e^{-\ell(t)}|x - y| \leq |X(t, x) - X(t, y)| \leq e^{\ell(t)}|x - y| \quad \text{for all } x, y \in \mathbb{R}^d. \quad (1.5)$$

The lower bound in (1.5) shows that F2) holds with compressibility constant

$$K = \exp(d\|\text{Lip } b(t)\|_{L^1([0, T])}).$$

The compressibility constant of X can be improved to depend only on the divergence of b . For this, we assume in addition that $b \in L^1_{\text{loc}}([0, T] \times \mathbb{R}^d)$. We claim that the measure $\mu(t) := X(t)_\# m^d$ satisfies the following equation in the distributional sense:

$$\frac{d\mu}{dt} + \text{div}(b\mu) = 0 \quad \text{on } (0, T) \times \mathbb{R}^d, \quad \mu|_{t=0} = m^d \quad \text{on } \mathbb{R}^d \quad (1.6)$$

To prove this, let $\phi \in C_c^\infty([0, T] \times \mathbb{R}^d)$. Then, by the chain rule

$$\frac{d}{dt}(\phi(t, X(t, x))) = \partial_t \phi(t, X(t, x)) + b(t, X(t, x)) \cdot \nabla \phi(t, X(t, x)) \quad \text{for a.e. } t \in (0, T).$$

Integrating over $[0, T] \times \mathbb{R}^d$ yields

$$-\int_{\mathbb{R}^d} \phi(0, x) dx = \int_0^T \int_{\mathbb{R}^d} \partial_t \phi(t, X(t, x)) dx dt + \int_0^T \int_{\mathbb{R}^d} b(t, X(t, x)) \cdot \nabla \phi(t, X(t, x)) dx dt$$

By (1.5), these integrals are well defined. By definition of the distributions in (1.6),

$$\frac{d\mu}{dt}[\phi] = - \int_0^T \int_{\mathbb{R}^d} \partial_t \phi(t, X(t, x)) dx dt$$

and

$$\operatorname{div}(b\mu)[\phi] = -\mu[b \cdot \nabla \phi] = - \int_0^T \int_{\mathbb{R}^d} b(t, X(t, x)) \cdot \nabla \phi(t, X(t, x)) dx dt.$$

Therefore, (1.6) holds.

Next, we look at the density of $\mu(t)$ with respect to m^d . By (1.5), $\mu(t)$ is absolutely continuous and $\mu(t) = h(t, \cdot) dm^d$ for some $h \in L^\infty([0, T] \times \mathbb{R}^d)$. Then also

$$\frac{dh}{dt} + \operatorname{div}(bh) = 0 \quad \text{on } (0, T) \times \mathbb{R}^d, \quad h|_{t=0} = 1 \quad \text{on } \mathbb{R}^d \quad (1.7)$$

in the weak sense, since h is the same distribution as μ . Now, we use the fact that weak solutions of the continuity equation (1.7) in $L^\infty([0, T] \times \mathbb{R}^d)$ are unique. This is not clear from the discussion so far, but it will be a consequence of the results in section §2.4. Therefore, h satisfies the following ODE along the characteristics of b :

$$\frac{d}{dt}(h(t, X(t, x))) = -h(t, X(t, x)) \operatorname{div} b(t, X(t, x))$$

The assumption $\operatorname{Lip}(b(t)) \in L^1([0, T])$ implies $\operatorname{div} b \in L^1(0, T; L^\infty(\mathbb{R}^d))$. Thus, by Grönwall's inequality,

$$e^{-D(t)} \leq h(t, x) \leq e^{D(t)}, \quad \text{where } D(t) := \int_0^t \|\operatorname{div} b(s)\|_{L^\infty(\mathbb{R}^d)} ds.$$

By definition of h , this implies

$$e^{-D(t)} m^d \leq X(t)_\# m^d \leq e^{D(t)} m^d \quad \text{for all } t \in [0, T]. \quad (1.8)$$

Thus, the divergence of b controls the rate of contraction and expansion of the flow X .

Remark 2 (Local flows). Corresponding results hold if b belongs only to $L^1(0, T; W_{\text{loc}}^{1, \infty}(\mathbb{R}^d))$. In this case, by the local version of the Cauchy-Lipschitz theorem, for every compact set $K \subset \mathbb{R}^d$, there exists a $0 < \tilde{T} \leq T$ and a map $X : [0, \tilde{T}] \times K \rightarrow \mathbb{R}^d$ such that for all $x \in K$, $X(\cdot, x)$ is the unique absolutely continuous solution to (1.1) with $x^0 = x$.

The analogue of (1.5) is that for all $x, y \in K$ and $t \in [0, \tilde{T}]$

$$e^{-\ell(t)} |x - y| \leq |X(t, x) - X(t, y)| \leq e^{\ell(t)} |x - y|, \quad \ell(t) := \int_0^t \operatorname{Lip}(b(s)|_{K'}) ds. \quad (1.9)$$

Here, $K' \subset \mathbb{R}^d$ is a compact set, for example $K' = X([0, \tilde{T}] \times K)$.

2 The transport equation

The ODE (1.1) is closely related to the following linear transport equation.

$$\begin{cases} \partial_t u - b \cdot \nabla u + cu = 0 & \text{in } (0, T) \times \mathbb{R}^d \\ u|_{t=0} = u^0 & \text{in } \mathbb{R}^d \end{cases} \quad (2.1)$$

Equations (1.1) and (2.1) are connected via the theory of characteristics, which will be explained in the proof of Proposition 1. We consider solutions $u \in L^\infty(0, T; L^p_{\text{loc}}(\mathbb{R}^d))$ for some $1 \leq p \leq \infty$ and understand (2.1) in the distributional sense, that is,

$$-\int_0^T \int_{\mathbb{R}^d} u \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} u^0 \phi(0, x) \, dx + \int_0^T \int_{\mathbb{R}^d} u(b \cdot \nabla \phi + (\text{div } b + c)\phi) \, dx \, dt = 0 \quad (2.2)$$

for every $\phi \in C_c^\infty([0, T] \times \mathbb{R}^d)$. For (2.2) to make sense, we assume $u^0 \in L^1_{\text{loc}}(\mathbb{R}^d)$, $b \in L^1(0, T; L^q_{\text{loc}}(\mathbb{R}^d; \mathbb{R}^d))$ and $c, \text{div } b \in L^1(0, T; L^q_{\text{loc}}(\mathbb{R}^d))$, where $1/p + 1/q = 1$.

We will use the following notation frequently: We say that u is a solution to (2.1) with $(\tilde{b}, \tilde{c}, \tilde{u}^0)$, if u satisfies (2.2), where b, c and u^0 are replaced by $\tilde{b}, \tilde{c}$ and $\tilde{u}^0$, respectively.

2.1 Existence

Before proving the existence of solutions to (2.1), let us introduce some notation. We denote by $C_b^1(\mathbb{R}^d; \mathbb{R}^m)$ the space of C^1 -functions for which the norm

$$\|f\|_{C_b^1(\mathbb{R}^d; \mathbb{R}^m)} := \|f\|_{L^\infty(\mathbb{R}^d; \mathbb{R}^m)} + \|Df\|_{L^\infty(\mathbb{R}^d; \mathbb{R}^{m \times d})}$$

is finite. The norm on $\mathbb{R}^{m \times d}$ is the operator norm of linear maps $\mathbb{R}^d \rightarrow \mathbb{R}^m$. We write $C_c^1(\mathbb{R}^d; \mathbb{R}^m)$ for the subspace of compactly supported functions, equipped with the same norm.

For $x \in \mathbb{R}^d$ and $R > 0$, we denote by $B_R(x) := \{y \in \mathbb{R}^d : |x - y| < R\}$ the ball of radius R centered at x . We sometimes abbreviate $B_R(0)$ by B_R . We also sometimes abbreviate $L^\gamma(0, T; L^p(\mathbb{R}^d))$ by $L_t^\gamma L_x^p$ and $C([0, T]; L^p(\mathbb{R}^d))$ by $C_t L_x^p$ ($1 \leq \gamma, p \leq \infty$).

In the proof of existence of weak solutions to the transport equation (2.1), we will approximate b by a sequence of vector fields $b_n \in L^1(0, T; C_b^1(\mathbb{R}^d; \mathbb{R}^d))$. The next lemma shows that this is possible while controlling the divergence of b_n uniformly.

Lemma 1. *Let $1 \leq q < \infty$ and $b \in L^1(0, T; L^q_{\text{loc}}(\mathbb{R}^d; \mathbb{R}^d))$. If $\text{div } b \in L^1(0, T; L^\infty(\mathbb{R}^d))$, then there exists a sequence of vector fields $b_n \in L^1(0, T; C_b^1(\mathbb{R}^d; \mathbb{R}^d))$ such that*

$$b_n \rightarrow b, \quad \text{div } b_n \rightarrow \text{div } b \quad \text{in } L^1(0, T; L^q_{\text{loc}}(\mathbb{R}^d)) \text{ as } n \rightarrow \infty, \quad (2.3)$$

and

$$\|\text{div } b_n(t)\|_{L^\infty_x(\mathbb{R}^d)} \leq C \|\text{div } b(t)\|_{L^\infty_x(\mathbb{R}^d)} \quad \text{for a.e. } t \in [0, T], \quad (2.4)$$

for some constant C independent of b .

Proof. Let ρ_ε be regularizing kernels defined by

$$\rho_\varepsilon(x) := \varepsilon^{-d} \rho(x/\varepsilon) \quad \text{for } \varepsilon > 0, \quad \text{for some } \rho \in C_c^\infty(\mathbb{R}^d), \quad \rho \geq 0, \quad \int_{\mathbb{R}^d} \rho \, dx = 1. \quad (2.5)$$

Also, let η_R be smooth cutoff functions defined by

$$\begin{aligned} \eta_R(x) &:= \eta(|x|/R) \quad \text{for } R > 0, \\ \text{for some } \eta &\in C_c^\infty([0, \infty)), \quad \eta' \leq 0, \quad \eta|_{[0, 5/4]} = 1, \quad \text{supp } \eta \subset [0, 2]. \end{aligned} \quad (2.6)$$

We set $\tilde{b}_n := (b * \rho_{1/n})\eta_n$. The symbol “ $*$ ” means convolution only in the spatial variable. This will always be the convention. Since $q < \infty$, by standard properties of convolutions

$$\tilde{b}_n \rightarrow b, \quad \operatorname{div} \tilde{b}_n \rightarrow \operatorname{div} b \quad \text{in } L^1(0, T; L^q_{\operatorname{loc}}(\mathbb{R}^d)) \text{ as } n \rightarrow \infty. \quad (2.7)$$

However,

$$\operatorname{div} \tilde{b}_n = (\operatorname{div} b * \rho_{1/n})\eta_n + (b * \rho_{1/n}) \cdot \nabla \eta_n. \quad (2.8)$$

Due to the second term, (2.4) need not hold for $\tilde{b}_n$. In fact $\operatorname{div} \tilde{b}_n(t)$ need not even be bounded in $L^\infty(\mathbb{R}^d)$ for any t . Thus, we have to modify $\tilde{b}_n$.

Let

$$I_n(t) := \int_{\mathbb{R}^d} (b * \rho_{1/n}) \cdot \nabla \eta_n \, dx.$$

Also, let $\zeta_n \in C_c^\infty(B_{2n} \setminus \overline{B_n})$ such that $\int \zeta_n \, dx = 1$ and $\|\zeta_n\|_{L^\infty} \leq 2/m^d(B_{2n} \setminus B_n)$. We set

$$f_n(t, x) := (b * \rho_{1/n}) \cdot \nabla \eta_n - I_n(t)\zeta_n(x).$$

Then $f_n \in L^1(0, T; L^\infty(\mathbb{R}^d))$ and $f_n(t, \cdot) \in C_c^\infty(B_{2n} \setminus \overline{B_n})$ for a.e. $t \in [0, T]$. Furthermore

$$\int_{B_{2n} \setminus B_n} f_n(t, x) \, dx = 0 \quad \text{for a.e. } t \in [0, T].$$

Let $L_0^2(B_{2n} \setminus B_n)$ be the subspace of functions $f \in L^2(B_{2n} \setminus B_n)$ such that $\int f \, dx = 0$. If $d \geq 2$, by Theorem III.3.1 of [Gal11], there exists a bounded linear operator

$$\mathcal{B} : L_0^2(B_{2n} \setminus B_n) \rightarrow W_0^{1,2}(B_{2n} \setminus B_n; \mathbb{R}^d),$$

such that $\operatorname{div} \mathcal{B}(f) = f$ for every $f \in L_0^2(B_{2n} \setminus B_n)$. Moreover, by Theorem III.3.3 of [Gal11], $\mathcal{B}(f) \in C_c^\infty(B_{2n} \setminus \overline{B_n}; \mathbb{R}^d)$ if $f \in C_c^\infty(B_{2n} \setminus \overline{B_n})$.

We define $v_n \in L^1(0, T; W_0^{1,2}(B_{2n} \setminus B_n; \mathbb{R}^d))$ by $v_n(t) = \mathcal{B}(f_n(t))$. Since $\mathcal{B}$ is bounded and linear, v_n is measurable on $[0, T] \times \mathbb{R}^d$. Also

$$v_n(t, \cdot) \in C_c^\infty(B_{2n} \setminus \overline{B_n}; \mathbb{R}^d), \quad \operatorname{div} v_n(t) = f_n(t) \quad \text{for a.e. } t \in [0, T].$$

Let $b_n := \tilde{b}_n - v_n$. Now

$$\operatorname{div} b_n = (\operatorname{div} b * \rho_{1/n})\eta_n + I_n(t)\zeta_n(x).$$

By Hölder's inequality,

$$|I_n(t)| = \left| \int_{\mathbb{R}^d} (\operatorname{div} b(t) * \rho_{1/n})\eta_n \, dx \right| \leq m^d(B_{2n}) \|\operatorname{div} b(t)\|_{L^\infty(\mathbb{R}^d)}.$$

Thus

$$\|\operatorname{div} b_n(t)\|_{L_x^\infty} \leq \|\operatorname{div} b\|_{L_x^\infty} + \frac{2m^d(B_{2n})}{m^d(B_{2n} \setminus B_n)} \|\operatorname{div} b\|_{L_x^\infty} \leq 5\|\operatorname{div} b\|_{L_x^\infty},$$

so (2.4) holds. Since $\operatorname{supp} v_n \subset [0, T] \times B_{2n} \setminus \overline{B_n}$, (2.3) follows from (2.7).

If $d = 1$, then $\partial_x b = \operatorname{div} b \in L^\infty(\mathbb{R})$ for almost every $t \in [0, T]$. Hence, $b(t, \cdot)$ is Lipschitz continuous (after modifying b on a m^2 -negligible set). Because $b \in L^1(0, T; L^1([-1, 1]))$, the Lipschitz continuity in space implies $b(t, 0) \in L^1((0, T))$. Thus, we may assume without loss

of generality that $b(t, 0) = 0$ for a.e. $t \in [0, T]$.¹ In this case, $|b(t, x)| \leq |x| \|\operatorname{div} b(t)\|_{L_x^\infty}$. Since $\operatorname{supp} \nabla \eta_n \subset B_{2n} \setminus B_n$ and $\|\nabla \eta_n\|_{L^\infty} \leq \|\nabla \eta\|_{L^\infty}/n$,

$$\begin{aligned} |(b(t) * \rho_{1/n})(x) \cdot \nabla \eta_n(x)| &\leq |x| \|\operatorname{div} b(t)\|_{L_x^\infty(\mathbb{R})} |\nabla \eta_n(x)| \\ &\leq 2 \|\nabla \eta\|_{L^\infty(\mathbb{R})} \|\operatorname{div} b(t)\|_{L_x^\infty(\mathbb{R})}. \end{aligned}$$

Thus, by (2.8)

$$\|\operatorname{div} \tilde{b}_n(t)\|_{L_x^\infty} \leq \left(1 + 2 \|\nabla \eta\|_{L^\infty(\mathbb{R})}\right) \|\operatorname{div} b(t)\|_{L_x^\infty}$$

for a.e. $t \in [0, T]$. We set $b_n = \tilde{b}_n$. □

Proposition 1. *Let $1 < p \leq \infty$, $1/p + 1/q = 1$. Let $u^0 \in L^p(\mathbb{R}^d)$ and*

$$b \in L^1(0, T; L_{\operatorname{loc}}^q(\mathbb{R}^d; \mathbb{R}^d)), \quad c, \operatorname{div} b \in L^1(0, T; L^\infty(\mathbb{R}^d)).$$

Then there exists a weak solution $u \in L^\infty(0, T; L^p(\mathbb{R}^d))$ to (2.1) for the initial data u^0 .

Proof. This proof follows that of Proposition II.1 in [DL89].

Step 1: Existence in the smooth case. For now, assume in addition to the above assumptions that $b \in L^1(0, T; C_b^1(\mathbb{R}^d; \mathbb{R}^d))$, $c \in L^1(0, T; C_b^1(\mathbb{R}^d))$ and $u^0 \in C_c^1(\mathbb{R}^d)$. By Theorem 1, there exists a unique $X : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ satisfying $\partial_t X(t, x) = -b(t, X(t, x))$ and $X(0, \cdot) = \operatorname{id}_{\mathbb{R}^d}$. Moreover $X(t, \cdot) \in C^1(\mathbb{R}^d)$ for every $t \in [0, T]$ by Proposition A.1. We define

$$v(t, x) := u^0(x) \exp\left(-\int_0^t c(s, X(s, x)) ds\right). \quad (2.9)$$

For every $x \in \mathbb{R}^d$, $v(\cdot, x)$ is an absolutely continuous solution of

$$\partial_t v(t, x) = -c(t, X(t, x)) v(t, x), \quad v(0, x) = u^0(x). \quad (2.10)$$

Because c need not be continuous in t , in general $v(\cdot, x) \notin C^1([0, T])$.

Next, we note that by Proposition A.1, $DX(t, x)$ is absolutely continuous in t and satisfies

$$\partial_t DX(t, x) = -Db(t, X(t, x))DX(t, x) \quad \text{for a.e. } t \in (0, T). \quad (2.11)$$

Since $Db \in L^1(0, T; C_b(\mathbb{R}^d))$, (2.9) and (2.11) imply $v \in L^\infty(0, T; C_c^1(\mathbb{R}^d))$. Also, note that $\operatorname{supp} v(t, \cdot) \subset \operatorname{supp} u^0$ for every $t \in [0, T]$.

By (2.11) and the formula for the derivative of the determinant,

$$\partial_t (\det DX(t, x)) = -\operatorname{div} b(t, X(t, x)) \det DX(t, x) \quad \text{for a.e. } t \in (0, T).$$

Since $\operatorname{div} b \in L^1(0, T; C_b(\mathbb{R}^d))$, it follows that $X(t, \cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a diffeomorphism for every $t \in [0, T]$. We define $u(t, x) := v(t, X(t, \cdot)^{-1}(x))$. Since $X(t, \cdot)$ and $X(t, \cdot)^{-1}$ are Lipschitz continuous uniformly in time, $u \in L^\infty(0, T; C_c^1(\mathbb{R}^d))$ and $\operatorname{supp} u$ is compact. Let $\phi \in C_c^\infty([0, T] \times \mathbb{R}^d)$ and $t \in [0, T]$. Then

$$\int_{\mathbb{R}^d} u(t, x) \phi(t, x) dx = \int_{\mathbb{R}^d} v(t, x) \phi(t, X(t, x)) \det DX(t, x) dx \quad (2.12)$$

¹This reduction is the only reason why the lemma states $b_n \in L^1(0, T; C_b^1)$ instead of $b_n \in L^1(0, T; C_c^1)$.

by changing variables $x = X(t, x')$. Since each of the three factors on the right-hand side in (2.12) is absolutely continuous with respect to t , for almost every $t \in [0, T]$

$$\begin{aligned} & \frac{d}{dt} \int_{\mathbb{R}^d} v(t, x) \phi(t, X(t, x)) \det DX(t, x) dx \\ &= \int_{\mathbb{R}^d} \left(-c(t, X(t, x)) v(t, x) \phi(t, X(t, x)) \det DX(t, x) \right. \\ & \quad + v(t, x) (\partial_t \phi(t, X(t, x)) - \nabla \phi(t, X(t, x)) \cdot b(t, X(t, x))) \det DX(t, x) \\ & \quad \left. - v(t, x) \phi(t, X(t, x)) \operatorname{div} b(t, X(t, x)) \det DX(t, x) \right) dx. \end{aligned}$$

By changing variables again, we get

$$\begin{aligned} & \frac{d}{dt} \int_{\mathbb{R}^d} u(t, x) \phi(t, x) dx \\ &= \int_{\mathbb{R}^d} -\phi(t, x) u(t, x) (c(t, x) + \operatorname{div} b(t, x)) - u(t, x) b(t, x) \cdot \nabla \phi(t, x) + u(t, x) \partial_t \phi(t, x) dx. \end{aligned}$$

Thus, u is a weak solution to the transport equation.

We now derive L^p -estimates of u in terms of u^0 . In the case $p = \infty$, by (2.9)

$$\|u(t, \cdot)\|_{L^\infty(\mathbb{R}^d)} \leq \exp\left(\|c\|_{L^1(0, T; L^\infty(\mathbb{R}^d))}\right) \|u^0\|_{L^\infty(\mathbb{R}^d)}. \quad (2.13)$$

In the case $1 < p < \infty$, we note that u is weakly differentiable in time, since u satisfies the transport equation and $u \in L^\infty(0, T; C_c^1(\mathbb{R}^d))$. By the chain rule,

$$\partial_t |u|^p - b \cdot \nabla |u|^p + pc|u|^p = 0. \quad (2.14)$$

Integrating this equation over $\mathbb{R}^d$ and using integration by parts yields

$$\partial_t \int_{\mathbb{R}^d} |u|^p dx + \int_{\mathbb{R}^d} |u|^p \operatorname{div} b + pc|u|^p dx = 0 \quad \text{for a.e. } t \in (0, T).$$

Moving the time derivative in front of the integral is possible by the dominated convergence theorem, since $-b \cdot \nabla |u|^p + pc|u|^p \in L_t^1 L_x^\infty$ and compactly supported. Thus

$$\partial_t \int_{\mathbb{R}^d} |u|^p dx \leq \|\operatorname{div} b + pc\|_{L^\infty(\mathbb{R}^d)} \int_{\mathbb{R}^d} |u|^p dx \quad \text{for a.e. } t \in (0, T).$$

By Grönwall's inequality, we conclude

$$\|u(t, \cdot)\|_{L^p(\mathbb{R}^d)} \leq \exp\left(\|(\operatorname{div} b)/p + c\|_{L^1(0, T; L^\infty(\mathbb{R}^d))}\right) \|u^0\|_{L^p(\mathbb{R}^d)}. \quad (2.15)$$

Step 2: Existence in the general case. Let $b_n \in L^1(0, T; C_b^1(\mathbb{R}^d; \mathbb{R}^d))$ be a sequence from Lemma 1. For every $n \in \mathbb{N}$, we set

$$c_n := (c * \rho_{1/n}) \eta_n, \quad u_n^0 := (u^0 * \rho_{1/n}) \eta_n.$$

By Step 1, there exists a weak solution u_n of

$$\partial_t u_n - b_n \cdot \nabla u_n + c_n u_n = 0, \quad u_n|_{t=0} = u_n^0.$$

Since $\|\operatorname{div} b_n\|_{L_t^1 L_x^\infty} \leq C \|\operatorname{div} b\|_{L_t^1 L_x^\infty}$, by Lemma 1, (2.13) and (2.15) show that u_n is bounded in $L^\infty(0, T; L^p(\mathbb{R}^d))$.

By the Banach-Alaoglu theorem, a subsequence of u_n converges weak-* in $L^\infty(0, T; L^p(\mathbb{R}^d))$ to some u . We just consider this subsequence from now on and also denote it by u_n . Let $\phi \in C_c^\infty([0, T] \times \mathbb{R}^d)$. Then for every $n \in \mathbb{N}$

$$\begin{aligned} & - \int_0^T \int_{\mathbb{R}^d} u_n \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} u_n^0 \phi(0, x) \, dx \\ & + \int_0^T \int_{\mathbb{R}^d} u_n (b_n \cdot \nabla \phi + (\operatorname{div} b_n + c_n) \phi) \, dx \, dt = 0. \end{aligned} \quad (2.16)$$

Since $q < \infty$, by standard properties of convolutions, $c_n \rightarrow c$ in $L^1(0, T; L_{\text{loc}}^q(\mathbb{R}^d))$. By Lemma 1, also $b_n \rightarrow b$ and $\operatorname{div} b_n \rightarrow \operatorname{div} b$ in $L^1(0, T; L_{\text{loc}}^q(\mathbb{R}^d))$. Thus, by letting $n \rightarrow \infty$ in (2.16), we conclude that u satisfies (2.2). $\square$

Remark 3. The existence result also holds for the transport equation with a right-hand side $f \in L^1(0, T; L^p(\mathbb{R}^d))$:

$$\partial_t u - b \cdot \nabla u + cu = f, \quad u|_{t=0} = u^0.$$

The proof is very similar. For Step 1, we assume $f \in L^1(0, T; C_b^1(\mathbb{R}^d))$. Instead of (2.10), one solves a linear, inhomogeneous ODE along the characteristics, and (2.13), (2.15) are replaced by similar estimates with a contribution from f . In Step 2, we set $f_n = (f * \rho_{1/n})\eta_n$.

2.2 Regularization

The following lemma will be an important tool in subsequent proofs, in particular for the uniqueness results in §2.4. It shows that, if b belongs to a suitable Sobolev space, the regularization $u * \rho_\varepsilon$ of a solution u is still an approximate solution. The lemma is from [DL89], Theorem II.1.

Lemma 2. *Let $1 \leq p, \alpha \leq \infty$ and $1 \leq \lambda < \infty$ such that $1/p + 1/\alpha \leq 1/\lambda$. Assume that*

$$b \in L^1(0, T; W_{\text{loc}}^{1, \alpha}(\mathbb{R}^d)), \quad c \in L^1(0, T; L_{\text{loc}}^\alpha(\mathbb{R}^d))$$

*and that $u \in L^\infty(0, T; L_{\text{loc}}^p(\mathbb{R}^d))$ solves (2.1) for some initial data $u^0 \in L^p(\mathbb{R}^d)$. Then $u_\varepsilon := u * \rho_\varepsilon$ satisfies*

$$\partial_t u_\varepsilon - b \cdot \nabla u_\varepsilon + cu_\varepsilon = r_\varepsilon, \quad u_\varepsilon|_{t=0} = u^0 * \rho_\varepsilon, \quad (2.17)$$

where $r_\varepsilon \rightarrow 0$ in $L^1(0, T; L_{\text{loc}}^\lambda(\mathbb{R}^d))$.

The proof of Lemma 2 relies on estimating the commutator between the directional derivative along b and convolution with ρ_ε . Such an estimate is established in Lemma 3. In the proof of the lemma and in later proofs, we use $C_{x, y, \dots}$ to denote a constant depending only on $x, y, \dots$ (where x, y can be numbers, functions etc.)

Lemma 3 (Commutator estimate). *Let p, α, λ be as in Lemma 2. If $b \in W_{\text{loc}}^{1, \alpha}(\mathbb{R}^d; \mathbb{R}^d)$ and $w \in L_{\text{loc}}^p(\mathbb{R}^d)$, then*

$$(b \cdot \nabla w) * \rho_\varepsilon - b \cdot \nabla(w * \rho_\varepsilon) \rightarrow 0 \quad \text{in } L_{\text{loc}}^\lambda(\mathbb{R}^d)$$

as $\varepsilon \rightarrow 0$.

Proof. This proof is from [DL89], p. 516, Lemma II.1. We first write out the convolution explicitly.

$$\begin{aligned} (b \cdot \nabla w) * \rho_\varepsilon - b \cdot \nabla(w * \rho_\varepsilon) &= - \int_{\mathbb{R}^d} w(y) \operatorname{div}_y(b(y) \rho_\varepsilon(x - y)) + b(x) \cdot \nabla \rho_\varepsilon(x - y) w(y) \, dy \\ &= \int_{\mathbb{R}^d} w(y) (b(y) - b(x)) \cdot \nabla \rho_\varepsilon(x - y) \, dy - (w \operatorname{div} b) * \rho_\varepsilon \end{aligned} \quad (2.18)$$

Since $\lambda < \infty$, in view of the assumed integrability on w and ∇b , we have $(w \operatorname{div} b) * \rho_\varepsilon \rightarrow w \operatorname{div} b$ in $L_{\text{loc}}^\lambda(\mathbb{R}^d)$. Thus, our goal is to show that the first term also converges to $w \operatorname{div} b$ in $L_{\text{loc}}^\lambda(\mathbb{R}^d)$.

Step 1: The smooth case. For now, assume that in addition $b \in C^1(\mathbb{R}^d; \mathbb{R}^d)$ and $w \in C^1(\mathbb{R}^d)$. Then an integration by parts yields

$$\begin{aligned} & \int_{\mathbb{R}^d} w(y)(b(y) - b(x)) \cdot \nabla \rho_\varepsilon(x - y) dy \\ &= \int_{\mathbb{R}^d} \nabla w(y) \cdot (b(y) - b(x)) \rho_\varepsilon(x - y) dy + \int_{\mathbb{R}^d} w(y) \operatorname{div} b(y) \rho_\varepsilon(x - y) dy \\ &= (\nabla w \cdot b) * \rho_\varepsilon - (\nabla w * \rho_\varepsilon) \cdot b + (w \operatorname{div} b) * \rho_\varepsilon. \end{aligned}$$

The first and third term converge to $\nabla w \cdot b$ and $w \operatorname{div} b$ in $L_{\text{loc}}^\lambda(\mathbb{R}^d)$, respectively. To see that the second term converges to $\nabla w \cdot b$, note that b and w belong to C^1 , hence $\nabla w \in L_{\text{loc}}^\lambda(\mathbb{R}^d)$ and $b \in L_{\text{loc}}^\infty(\mathbb{R}^d)$.

Step 2: Estimating the first term in (2.18). To reduce to the smooth case, we will need an estimate of

$$A := \left\| \int_{\mathbb{R}^d} w(y)(b(y) - b(x)) \cdot \nabla \rho_\varepsilon(x - y) dy \right\|_{L_x^\lambda(B_R)}.$$

By changing variables $z = (y - x)/\varepsilon$ in integral over y ,

$$A = \varepsilon^d \left\| \int_{\mathbb{R}^d} w(x + \varepsilon z)(b(x + \varepsilon z) - b(x)) \cdot \nabla \rho_\varepsilon(\varepsilon z) dz \right\|_{L_x^\lambda(B_R)}.$$

By the definition of ρ_ε in (2.5), $\operatorname{supp} \nabla \rho_\varepsilon \subset B_{\varepsilon C_\rho}$ for some $C_\rho > 0$. Thus the integrand is nonzero only for $z \in B_{C_\rho}$. Moreover $|\nabla \rho_\varepsilon| \leq \tilde{C}_\rho \varepsilon^{-d-1}$, hence

$$A \leq \frac{\tilde{C}_\rho}{\varepsilon} \left\| \int_{B_{C_\rho}} |w(x + \varepsilon z)| |b(x + \varepsilon z) - b(x)| dz \right\|_{L_x^\lambda(B_R)}.$$

By Minkowski's inequality and Hölder's inequality, and since $1/\lambda \geq 1/\alpha + 1/p$,

$$\begin{aligned} A &\leq \frac{\tilde{C}_\rho}{\varepsilon} \int_{B_{C_\rho}} \| |w(x + \varepsilon z)| |b(x + \varepsilon z) - b(x)| \|_{L_x^\lambda(B_R)} dz \\ &\leq \frac{C_{\rho, \lambda, R, d}}{\varepsilon} \int_{B_{C_\rho}} \|w(x + \varepsilon z)\|_{L_x^p(B_R)} \|b(x + \varepsilon z) - b(x)\|_{L_x^\alpha(B_R)} dz. \end{aligned}$$

For small enough ε , we have $\varepsilon C_\rho < 1$, hence

$$A \leq \frac{C_{\rho, \lambda, R, d}}{\varepsilon} \|w\|_{L^p(B_{R+1})} \int_{B_{C_\rho}} \|b(x + \varepsilon z) - b(x)\|_{L_x^\alpha(B_R)} dz. \quad (2.19)$$

Next, we use the fact that for every $h \in \mathbb{R}^d$

$$\|b(\cdot + h) - b\|_{L^\alpha(B_R)} \leq |h| \|\nabla b\|_{L^\alpha(B_{R+2|h|})}. \quad (2.20)$$

Combining (2.19) with (2.20) yields

$$A \leq C \|w\|_{L^p(B_{R+1})} \|\nabla b\|_{L^\alpha(B_{R+1+C_\rho})}. \quad (2.21)$$

for some C not depending on w and b .

Step 3: The general case. Let $b_\delta := b * \rho_\delta$, $w_\gamma := w * \rho_\gamma$ for some $\delta, \gamma > 0$ and write

$$\int_{\mathbb{R}^d} w(y)(b(y) - b(x)) \cdot \nabla \rho_\varepsilon(x - y) dy = D + E + F, \quad \text{where} \quad (2.22)$$

$$\begin{aligned} D &:= \int_{\mathbb{R}^d} w_\gamma(y)(b_\delta(y) - b_\delta(x)) \cdot \nabla \rho_\varepsilon(x - y) dy \\ E &:= \int_{\mathbb{R}^d} w(y)(b(y) - b_\delta(y) - b(x) + b_\delta(x)) \cdot \nabla \rho_\varepsilon(x - y) dy \\ F &:= \int_{\mathbb{R}^d} (w(y) - w_\gamma(y))(b_\delta(y) - b_\delta(x)) \cdot \nabla \rho_\varepsilon(x - y) dy. \end{aligned}$$

Let $R > 0$. We first consider the case $p < \infty$ and $\alpha < \infty$. Then $E \rightarrow 0$ as $\delta \rightarrow 0$ and $F \rightarrow 0$ as $\gamma \rightarrow 0$ in $L^\lambda(B_R)$ by (2.21). Moreover, both converge uniformly in ε and F converges uniformly in δ . By writing

$$\begin{aligned} &\left\| \int_{\mathbb{R}^d} w(y)(b(y) - b(x)) \cdot \nabla \rho_\varepsilon(x - y) dy - w \operatorname{div} b \right\|_{L^\lambda(B_R)} \\ &\leq \|w_\gamma \operatorname{div} b_\delta - w \operatorname{div} b\|_{L^\lambda(B_R)} + \|D - w_\gamma \operatorname{div} b_\delta\|_{L^\lambda(B_R)} + \|E\|_{L^\lambda(B_R)} + \|F\|_{L^\lambda(B_R)}, \end{aligned} \quad (2.23)$$

we may finish the argument in the case $p < \infty$, $\alpha < \infty$ as follows. Choose δ and γ such that $\|w_\gamma \operatorname{div} b_\delta - w \operatorname{div} b\|_{L^\lambda(B_R)}$ is small and $\|E\|_{L^\lambda(B_R)}$, $\|F\|_{L^\lambda(B_R)}$ are small. By Step 1, we may then choose ε such that $\|D - w_\gamma \operatorname{div} b_\delta\|_{L^\lambda(B_R)}$ is small.

Next, we consider the case $p = \infty$. By (2.21), applied with the λ -norm for $w - w_\gamma$ and the ∞ -norm for b_δ ,

$$\|F\|_{L^\lambda(B_R)} \leq C \|w - w_\gamma\|_{L^\lambda(B_R)} \|\nabla b_\delta\|_{L^\infty(B_R)}. \quad (2.24)$$

Thus still $F \rightarrow 0$ as $\gamma \rightarrow 0$ in $L^\lambda(B_R)$, but no longer uniformly in δ . Hence, γ now has to be chosen after δ . This is not a problem for E , since E is independent of γ . To see that it is also not a problem for the first term on the right hand side of (2.23), note that

$$\|w_\gamma \operatorname{div} b_\delta - w \operatorname{div} b\|_{L^\lambda(B_R)} \leq \|w_\gamma(\operatorname{div} b_\delta - \operatorname{div} b)\|_{L^\lambda(B_R)} + \|(w_\gamma - w) \operatorname{div} b\|_{L^\lambda(B_R)}.$$

Since $\operatorname{div} b_\delta \rightarrow \operatorname{div} b$ in $L^\lambda(B_R)$ and w_γ is bounded in $L^\infty(B_R)$, the first term converges to 0 as $\delta \rightarrow 0$, uniformly in γ . The second term converges to 0 since w_γ is bounded in $L^\infty(B_R)$ and $w_\gamma \rightarrow w$ almost everywhere.

The last step of the proof in the case $\alpha = \infty$ works symmetrically to $p = \infty$. Now, δ has to be chosen after γ . Note that for (2.24), the smoothness of b_δ was important. Thus the definitions of E and F must be adapted: w is replaced by w_γ in E , and b_δ is replaced by b in F . Then, one obtains an estimate for E similar to (2.24). $\square$

Remark 4. Lemma 3 yields the following analogous result for time-dependent vector fields. If $b \in L^1(0, T; W_{\operatorname{loc}}^{1, \alpha}(\mathbb{R}^d; \mathbb{R}^d))$ and $w \in L^\infty(0, T; L_{\operatorname{loc}}^p(\mathbb{R}^d))$, then

$$(b \cdot \nabla w) * \rho_\varepsilon - b \cdot \nabla(w * \rho_\varepsilon) \rightarrow 0 \quad \text{in } L^1(0, T; L_{\operatorname{loc}}^\lambda(\mathbb{R}^d)) \quad (2.25)$$

as $\varepsilon \rightarrow 0$. As always, the convolution applies only to the spatial variables.

To prove (2.25), let $R > 0$. By Lemma 3,

$$(b(t) \cdot \nabla w(t)) * \rho_\varepsilon - b(t) \cdot \nabla(w(t) * \rho_\varepsilon) \rightarrow 0 \quad \text{in } L^\lambda(B_R)$$

for almost every $t \in [0, T]$. Moreover, by (2.18) and (2.21),

$$\begin{aligned} &\|(b(t) \cdot \nabla w(t)) * \rho_\varepsilon - b(t) \cdot \nabla(w(t) * \rho_\varepsilon)\|_{L^\lambda(B_R)} \\ &\leq C \|w(t)\|_{L^p(B_{R+1})} \|\nabla b(t)\|_{L^\alpha(B_{R+1+C_\rho})} + \|w(t) \operatorname{div} b(t)\|_{L^\lambda(B_{R+1})} \end{aligned}$$

for almost every $t \in [0, T]$. Thus we conclude (2.25) by the dominated convergence theorem, since the right-hand side is independent of ε and in $L^1([0, T])$.

Proof of Lemma 2. Let $\tilde{\rho}_\varepsilon(x) = \rho_\varepsilon(-x)$ and $\phi \in C_c^\infty([0, T] \times \mathbb{R}^d)$. Note that $\phi * \tilde{\rho}_\varepsilon$ is also in $C_c^\infty([0, T] \times \mathbb{R}^d)$. Using that u satisfies (2.2), we compute

$$\begin{aligned} - \int_0^T \int_{\mathbb{R}^d} u_\varepsilon \partial_t \phi \, dx \, dt &= - \int_0^T \int_{\mathbb{R}^d} u \partial_t \phi * \tilde{\rho}_\varepsilon \, dx \, dt \\ &= \int_{\mathbb{R}^d} u^0 \phi(0, \cdot) * \tilde{\rho}_\varepsilon \, dx - \int_0^T \int_{\mathbb{R}^d} u \operatorname{div}(b(\phi * \tilde{\rho}_\varepsilon)) + u c(\phi * \tilde{\rho}_\varepsilon) \, dx \, dt. \end{aligned} \quad (2.26)$$

By definition, the integral of $-u \operatorname{div}(b(\phi * \tilde{\rho}_\varepsilon))$ is the distribution $(b \cdot \nabla u) * \rho_\varepsilon$ evaluated at ϕ . Thus, by Lemma 3 and Remark 4

$$- \int_0^T \int_{\mathbb{R}^d} u \operatorname{div}(b(\phi * \tilde{\rho}_\varepsilon)) \, dx \, dt = \int_0^T \int_{\mathbb{R}^d} (b \cdot \nabla u_\varepsilon + s_\varepsilon) \phi \, dx \, dt$$

where $s_\varepsilon \rightarrow 0$ in $L^1(0, T; L_{\text{loc}}^\lambda(\mathbb{R}^d))$. Substituting this into (2.26), we obtain

$$\begin{aligned} - \int_0^T \int_{\mathbb{R}^d} u_\varepsilon \partial_t \phi \, dx \, dt &= \int_{\mathbb{R}^d} u^0 * \rho_\varepsilon \phi(0, x) \, dx \\ &\quad + \int_0^T \int_{\mathbb{R}^d} (b \cdot \nabla u_\varepsilon - c u_\varepsilon) \phi \, dx \, dt + \int_0^T \int_{\mathbb{R}^d} (s_\varepsilon + c u_\varepsilon - (c u) * \rho_\varepsilon) \phi \, dx \, dt. \end{aligned}$$

Because $(c u) * \rho_\varepsilon - c u_\varepsilon \rightarrow 0$ in $L^1(0, T; L_{\text{loc}}^\lambda(\mathbb{R}^d))$, we may set $r_\varepsilon := s_\varepsilon + c u_\varepsilon - (c u) * \rho_\varepsilon$, which completes the proof. $\square$

Remark 5. A corresponding result also holds for the transport equation with a right-hand side. We keep the notation and assumptions of Lemma 2. If $f \in L^1(0, T; L_{\text{loc}}^\lambda(\mathbb{R}^d))$ and u satisfies (2.1) with f on the right-hand side, then also

$$\partial_t u_\varepsilon - b \cdot \nabla u_\varepsilon + c u_\varepsilon = f + r_\varepsilon, \quad u_\varepsilon|_{t=0} = u^0 * \rho_\varepsilon$$

for some $r_\varepsilon \rightarrow 0$ in $L^1(0, T; L_{\text{loc}}^\lambda(\mathbb{R}^d))$. This is a direct consequence of the above proof, because $\lambda < \infty$, hence $f * \rho_\varepsilon \rightarrow f$ in $L^1(0, T; L_{\text{loc}}^\lambda(\mathbb{R}^d))$.

The regularization result, Lemma 2, allows us to show that the chain rule holds for solutions of the transport equation, in the following sense.

Lemma 4. *Let $1 \leq p \leq \infty$, $1/p + 1/q = 1$. Assume*

$$b \in L^1(0, T; W_{\text{loc}}^{1,q}(\mathbb{R}^d; \mathbb{R}^d)), \quad c \in L^1(0, T; L_{\text{loc}}^q(\mathbb{R}^d)) \quad (2.27)$$

and let $u \in L^\infty(0, T; L_{\text{loc}}^p(\mathbb{R}^d))$ be a solution of (2.1) for the initial data $u^0 \in L_{\text{loc}}^1(\mathbb{R}^d)$. If $\beta \in C^1(\mathbb{R})$ and β' is bounded, then $\beta(u)$ satisfies

$$\partial_t \beta(u) - b \cdot \nabla \beta(u) + c u \beta'(u) = 0, \quad \beta(u)|_{t=0} = \beta(u^0) \quad (2.28)$$

in the weak sense.

That (2.28) holds in the weak sense means that for every $\phi \in C_c^\infty([0, T] \times \mathbb{R}^d)$

$$\begin{aligned} - \int_0^T \int_{\mathbb{R}^d} \beta(u) \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} \beta(u^0) \phi(0, x) \, dx \\ + \int_0^T \int_{\mathbb{R}^d} \beta(u) (\phi \operatorname{div} b + b \cdot \nabla \phi) + c u \beta'(u) \phi \, dx \, dt = 0. \end{aligned} \quad (2.29)$$

Proof. By Lemma 2, $u_\varepsilon = u * \rho_\varepsilon$ satisfies

$$\partial_t u_\varepsilon - b \cdot \nabla u_\varepsilon + c u_\varepsilon = r_\varepsilon, \quad u_\varepsilon|_{t=0} = u_\varepsilon^0,$$

where $u_\varepsilon^0 = u^0 * \rho_\varepsilon$ and $r_\varepsilon \rightarrow 0$ in $L^1_{\text{loc}}([0, T] \times \mathbb{R}^d)$. Since $u_\varepsilon \in W^{1,1}_{\text{loc}}((0, T) \times \mathbb{R}^d)$ and β' is bounded, by the chain rule for Sobolev functions

$$\partial_t \beta(u_\varepsilon) - b \cdot \nabla \beta(u_\varepsilon) + c u_\varepsilon \beta'(u_\varepsilon) = r_\varepsilon \beta'(u_\varepsilon), \quad \beta(u_\varepsilon)|_{t=0} = \beta(u_\varepsilon^0).$$

Testing against $\phi \in C_c^\infty([0, T] \times \mathbb{R}^d)$ yields

$$\begin{aligned} & - \int_0^T \int_{\mathbb{R}^d} \beta(u_\varepsilon) \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} \beta(u_\varepsilon^0) \phi(0, x) \, dx \\ & + \int_0^T \int_{\mathbb{R}^d} \beta(u_\varepsilon) (\phi \operatorname{div} b + b \cdot \nabla \phi) + c u_\varepsilon \beta'(u_\varepsilon) \phi \, dx \, dt = \int_0^T \int_{\mathbb{R}^d} r_\varepsilon \beta'(u_\varepsilon) \phi \, dx \, dt. \end{aligned} \quad (2.30)$$

We let $\varepsilon \rightarrow 0$ in this equation with the aim of obtaining (2.29). Since β' is bounded and $r_\varepsilon \rightarrow 0$ in $L^1_{\text{loc}}([0, T] \times \mathbb{R}^d)$, the right-hand side converges to 0 as $\varepsilon \rightarrow 0$. For the $c u_\varepsilon \beta'(u_\varepsilon)$ term, we want to show that

$$\int_0^T \int_{\mathbb{R}^d} c u_\varepsilon \beta'(u_\varepsilon) \phi \, dx \, dt \rightarrow \int_0^T \int_{\mathbb{R}^d} c u \beta'(u) \phi \, dx \, dt \quad \text{as } \varepsilon \rightarrow 0. \quad (2.31)$$

In the case $p < \infty$, we write

$$c u_\varepsilon \beta'(u_\varepsilon) - c u \beta'(u) = c(u_\varepsilon - u) \beta'(u_\varepsilon) + c u (\beta'(u_\varepsilon) - \beta'(u)).$$

Since $u_\varepsilon(t) \rightarrow u(t)$ in $L^p_{\text{loc}}(\mathbb{R}^d)$ for a.e. $t \in (0, T)$ and β' is bounded, $\int_0^T \int_{\mathbb{R}^d} c(u_\varepsilon - u) \beta'(u_\varepsilon) \, dx \, dt$ converges to 0 as $\varepsilon \rightarrow 0$. Since $u_\varepsilon \rightarrow u$ pointwise almost everywhere, also $\int_0^T \int_{\mathbb{R}^d} c u (\beta'(u_\varepsilon) - \beta'(u)) \, dx \, dt$ converges to 0 as $\varepsilon \rightarrow 0$. In the case $p = \infty$, we conclude (2.31) directly from dominated convergence.

For the remaining terms in (2.30), if $p < \infty$, we use that $\beta(u_\varepsilon) \rightarrow \beta(u)$ in $L^p_{\text{loc}}(\mathbb{R}^d)$ since β' is bounded. If $p = \infty$, we apply the dominated convergence theorem, using that $u_\varepsilon \rightarrow u$ almost everywhere on $[0, T] \times \mathbb{R}^d$. $\square$

The same proof shows a similar result for the transport equation with a right-hand side $f \in L^1(0, T; L^1_{\text{loc}}(\mathbb{R}^d))$. In this case, there is a term $f \beta'(u)$ on the right-hand side in (2.28).

2.3 Renormalized solutions

We introduce here a new notion of a solution for the transport equation (2.1), motivated by formula (2.28). First, we set up some notation. If $A \subset \mathbb{R}^d$ is measurable, we denote by $L(A)$ the set of measurable functions $A \rightarrow \mathbb{R}$. Let $f_n \in L(A)$ for $n \in \mathbb{N}$ and $f \in L(A)$. We say that $f_n \rightarrow f$ in $L(A)$ if $f_n|_K$ converges to $f|_K$ in measure for every compact $K \subset A$. Using this notion of convergence, we also define the space $C([0, T]; L(A))$.

Definition 2 (Admissible function). We call $\beta \in C^1(\mathbb{R})$ an admissible function, if β and $\beta' \cdot (1 + |x|)$ are bounded on $\mathbb{R}$.

Definition 3 (Renormalized solution). Let $b \in L^1_{\text{loc}}([0, T] \times \mathbb{R}^d; \mathbb{R}^d)$ and $c, \operatorname{div} b \in L^1_{\text{loc}}([0, T] \times \mathbb{R}^d)$. Let $u^0 \in L(\mathbb{R}^d)$ and $u \in L([0, T] \times \mathbb{R}^d)$. If, for every admissible function β ,

$$\partial_t \beta(u) - b \cdot \nabla \beta(u) + c u \beta'(u) = 0, \quad \beta(u)|_{t=0} = \beta(u^0) \quad (2.32)$$

holds in the weak sense, then we call u a renormalized solution of (2.1) corresponding to the initial condition u^0 .

Definition 3 provides a notion of a solution for very general initial data $u^0 \in L(\mathbb{R}^d)$, for which the weak definition (2.2) is not well defined. This will become important in the application to ordinary differential equations. Moreover, renormalized solutions have several features of smooth solutions, such as the conservation of L^p -norms in the case $\operatorname{div} b = 0$ and $c = 0$ (see Proposition 3).

The following result establishes the relation between weak and renormalized solutions. It corresponds to part 1) of Theorem II.3 in [DL89].

Proposition 2 (Consistency). *Let $1 \leq p \leq \infty$, $1/p + 1/q = 1$. Let $u \in L^\infty(0, T; L^p_{\operatorname{loc}}(\mathbb{R}^d))$ and $u^0 \in L^1_{\operatorname{loc}}(\mathbb{R}^d)$. Assume that $c, \operatorname{div} b \in L^1(0, T; L^q_{\operatorname{loc}}(\mathbb{R}^d))$.*

- 1) *If $b \in L^1(0, T; L^q_{\operatorname{loc}}(\mathbb{R}^d; \mathbb{R}^d))$ and u is a renormalized solution of (2.1), then u is a weak solution.*
- 2) *If $b \in L^1(0, T; W^{1,q}_{\operatorname{loc}}(\mathbb{R}^d; \mathbb{R}^d))$ and u is a weak solution of (2.1), then u is a renormalized solution.*

Proof. Part 2) is a consequence of Lemma 4. For part 1), we choose a sequence of admissible functions β_n such that $\beta_n \rightarrow \operatorname{id}_{\mathbb{R}}$, $\beta'_n \rightarrow 1$ pointwise, $|\beta_n(\sigma)| \leq |\sigma|$ and $|\beta'_n(\sigma)| \leq 1$ for every $\sigma \in \mathbb{R}$. For example, one may take $\beta_n(\sigma) = n \tanh(\sigma/n)$. By (2.32), for every $\phi \in C_c^\infty([0, T] \times \mathbb{R}^d)$

$$\begin{aligned} & - \int_0^T \int_{\mathbb{R}^d} \beta_n(u) \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} \beta_n(u^0) \phi(0, x) \, dx \\ & + \int_0^T \int_{\mathbb{R}^d} \beta_n(u) (\phi \operatorname{div} b + b \cdot \nabla \phi + cu \beta'_n(u) \phi) \, dx \, dt = 0. \end{aligned}$$

We let $n \rightarrow \infty$ and use the dominated convergence theorem to conclude (2.2). For the third integral, $|u|(|\phi \operatorname{div} b| + |b| |\nabla \phi| + |\phi c|)$ is a majorant, since $|\beta_n(u)| \leq |u|$ and $|\beta'_n(u)| \leq 1$. The majorant is integrable, because $b \in L^1(0, T; L^q_{\operatorname{loc}}(\mathbb{R}^d; \mathbb{R}^d))$ and $c, \operatorname{div} b \in L^1(0, T; L^q_{\operatorname{loc}}(\mathbb{R}^d))$. $\square$

Remark 6. Similarly, one can define renormalized solutions for the transport equation with a right-hand side $f \in L^1_{\operatorname{loc}}([0, T] \times \mathbb{R}^d)$. In this case, we require that

$$\partial_t \beta(u) - b \cdot \nabla \beta(u) + cu \beta'(u) = f \beta'(u), \quad \beta(u)|_{t=0} = \beta(u^0)$$

for every admissible function β . Proposition 2 still holds for $f \in L^1_{\operatorname{loc}}([0, T] \times \mathbb{R}^d)$ by the same proof.

2.4 Uniqueness

In this section, we prove the uniqueness of renormalized and weak solutions in $L^\infty(0, T; L^p(\mathbb{R}^d))$ for $1 \leq p \leq \infty$. For all uniqueness results, we will assume that b satisfies the following growth condition:

$$\frac{|b|}{1 + |x|} \in L^1(0, T; L^1(\mathbb{R}^d)) + L^1(0, T; L^\infty(\mathbb{R}^d)) \quad (\star)$$

2.4.1 *The case $p < \infty$.* The following Proposition collects several properties of renormalized solutions. Uniqueness for $p < \infty$ will follow immediately from it.

Proposition 3. *Let $b \in L^1_{\operatorname{loc}}([0, T] \times \mathbb{R}^d; \mathbb{R}^d)$ and $c, \operatorname{div} b \in L^1_{\operatorname{loc}}([0, T] \times \mathbb{R}^d)$. Let $1 \leq p \leq \infty$, let $f \in L^1(0, T; L^p(\mathbb{R}^d))$ and let $u \in L^\infty(0, T; L^p(\mathbb{R}^d))$ be a renormalized solution of*

$$\partial_t u - b \cdot \nabla u + cu = f, \quad u|_{t=0} = u^0,$$

where $u^0 \in L^p(\mathbb{R}^d)$.

- 1) If $p = \infty$, then $u \in C([0, T]; L_{\text{loc}}^r(\mathbb{R}^d))$ for every $1 \leq r < \infty$.
- 2) Assume $p < \infty$, $(\star)$, and that $c, \operatorname{div} b \in L^1(0, T; L^\infty(\mathbb{R}^d))$. Then $u \in C([0, T]; L^p(\mathbb{R}^d))$ and for a.e. $t \in (0, T)$

$$\frac{d}{dt} \int_{\mathbb{R}^d} |u|^p dx = - \int_{\mathbb{R}^d} (\operatorname{div} b + pc) |u|^p dx + p \int_{\mathbb{R}^d} f |u|^{p-2} u dx. \quad (2.33)$$

Furthermore, for every admissible β such that $0 \notin \operatorname{supp} \beta$,

$$\frac{d}{dt} \int_{\mathbb{R}^d} \beta(u) dx = - \int_{\mathbb{R}^d} (\beta(u) \operatorname{div} b + cu\beta'(u)) dx + \int_{\mathbb{R}^d} f\beta'(u) dx \quad (2.34)$$

for a.e. $t \in (0, T)$.

The proof of Proposition 3 uses the following Lemma.

Lemma 5. Let (X, μ) be a measure space and $1 \leq p < \infty$. Let $f_n, f \in L^p(\mu)$ for $n \in \mathbb{N}$. If $f_n \rightarrow f$ μ -a.e. and $\|f_n\|_{L^p(\mu)} \rightarrow \|f\|_{L^p(\mu)}$ as $n \rightarrow \infty$, then $f_n \rightarrow f$ in $L^p(\mu)$.

Proof. Since the function $x \mapsto |x|^p$ is convex on $\mathbb{R}$, the following pointwise inequality holds.

$$|f_n - f|^p \leq 2^{p-1}(|f_n|^p + |f|^p)$$

Thus, by Fatou's Lemma and the almost everywhere convergence of f_n to f ,

$$\liminf_{n \rightarrow \infty} \int_X 2^{p-1}(|f_n|^p + |f|^p) - |f_n - f|^p d\mu \geq 2^p \int_X |f|^p d\mu.$$

We now use the following fact: If a_n, b_n are sequences of real numbers, and $a_n \rightarrow a$ as $n \rightarrow \infty$, then $\liminf_{n \rightarrow \infty} (a_n + b_n) = a + \liminf_{n \rightarrow \infty} b_n$. By applying this to the sequence $\|f_n\|_{L^p(\mu)}^p$,

$$\liminf_{n \rightarrow \infty} \int_X 2^{p-1}(|f_n|^p + |f|^p) - |f_n - f|^p d\mu = 2^p \int_X |f|^p d\mu - \limsup_{n \rightarrow \infty} \int_X |f_n - f|^p d\mu. \quad \square$$

Proof of Proposition 3. Let β be an admissible function and $\xi \in C_c^\infty(\mathbb{R}^d)$, $\psi \in C_c^\infty([0, T])$. Since u is a renormalized solution,

$$\begin{aligned} & - \int_0^T \int_{\mathbb{R}^d} \beta(u) \xi \partial_t \psi dx dt - \psi(0) \int_{\mathbb{R}^d} \beta(u^0) \xi dx \\ & + \int_0^T \int_{\mathbb{R}^d} \psi \beta(u) (\xi \operatorname{div} b + b \cdot \nabla \xi) + \psi \xi c u \beta'(u) dx dt = \int_0^T \int_{\mathbb{R}^d} \psi \xi f \beta'(u) dx dt. \end{aligned} \quad (2.35)$$

Since $b, c, f \in L^1(0, T; L_{\text{loc}}^1)$ and $\xi \in C_c^\infty(\mathbb{R}^d)$, (2.35) implies

$$\int_{\mathbb{R}^d} \beta(u) \xi dx \in W^{1,1}((0, T)) \hookrightarrow C([0, T]). \quad (2.36)$$

Since $\{\beta(u(t)) : t \in [0, T]\}$ is bounded in $L^\infty(\mathbb{R}^d)$, the Banach-Alaoglu theorem and (2.36) imply $\beta(u(s)) \rightharpoonup^* \beta(u(t))$ in $L^\infty(\mathbb{R}^d)$ if $s \rightarrow t$ in $[0, T]$. Since β^2 is admissible, also $\beta(u(s))^2 \rightharpoonup^* \beta(u(t))^2$ in $L^\infty(\mathbb{R}^d)$ if $s \rightarrow t$ in $[0, T]$. Let $K \subset \mathbb{R}^d$ be compact. Since $m^d(K) < \infty$, the characteristic function χ_K lies in $L^1(\mathbb{R}^d)$, so $\|\beta(u(t))\|_{L^2(K)} \in C([0, T])$. Thus $\beta(u) \in C([0, T]; L^2(K))$, since $L^2(K)$ is uniformly convex. By iterating this argument for $L_{\text{loc}}^{2^k}$, $k \in \mathbb{N}$, we conclude

$$\beta(u) \in C([0, T]; L_{\text{loc}}^r(\mathbb{R}^d)) \quad \text{for every } 1 \leq r < \infty. \quad (2.37)$$

Part 1). If $u \in L^\infty([0, T] \times \mathbb{R}^d)$, we can choose β admissible such that $\beta(u) = u$ almost everywhere. So part 1) follows from (2.37).

Part 2). We first show (2.34), so let β be admissible and assume that $0 \notin \text{supp } \beta$. We choose $\xi = \eta_R$ in (2.35), where η_R is defined in (2.6). We then let $R \rightarrow \infty$ with the goal of obtaining

$$\begin{aligned} & - \int_0^T \int_{\mathbb{R}^d} \beta(u) \partial_t \psi \, dx \, dt - \psi(0) \int_{\mathbb{R}^d} \beta(u^0) \, dx \\ & + \int_0^T \int_{\mathbb{R}^d} \psi (\beta(u) \operatorname{div} b + cu \beta'(u)) \, dx \, dt = \int_0^T \int_{\mathbb{R}^d} \psi f \beta'(u) \, dx \, dt. \end{aligned} \quad (2.38)$$

Since $p < \infty$ and $0 \notin \text{supp } \beta$,

$$\beta(u), \, u\beta'(u), \, \beta'(u) \in L^\infty(0, T; L^1(\mathbb{R}^d)) \cap L^\infty(0, T; L^\infty(\mathbb{R}^d)). \quad (2.39)$$

Therefore, all terms in (2.35) except the one with $\nabla \eta_R$ converge as desired by dominated convergence. For the $\nabla \eta_R$ -term, we note that $|\nabla \eta_R| \leq C_\eta/R$, that $\text{supp } \eta_R \subset B_{2R} \setminus B_R$ and that $1/R \leq 2/(1 + |x|)$ for $x \in B_{2R} \setminus B_R$. Hence

$$\left| \int_0^T \int_{\mathbb{R}^d} \psi \beta(u) b \cdot \nabla \eta_R \, dx \, dt \right| \leq 2C_\eta \int_0^T \int_{B_{2R} \setminus B_R} |\psi| |\beta(u)| \frac{|b|}{1 + |x|} \, dx \, dt.$$

By the growth condition $(\star)$ on b and by (2.39), the integrand on the right-hand side is in $L^1([0, T] \times \mathbb{R}^d)$. Hence the integral converges to 0 as $R \rightarrow \infty$. Thus (2.38) holds, and since $\psi \in C_c^\infty([0, T])$ is arbitrary, also (2.34).

To show (2.33), we choose a sequence of admissible functions β_n such that $0 \notin \text{supp } \beta_n$, $\beta_n(\sigma) \geq 0$, $\sigma \beta'_n(\sigma) \geq 0$ for all $\sigma \in \mathbb{R}$ and

$$\beta_n(\sigma), \, \sigma \beta'_n(\sigma) \leq C|\sigma|^p, \quad \beta_n(\sigma) \xrightarrow{n} |\sigma|^p, \quad \beta'_n(\sigma) \xrightarrow{n} p\sigma|\sigma|^{p-2} \quad \text{for all } \sigma \in \mathbb{R} \setminus \{0\}.$$

Here, C must be independent of n and σ . Note that because $0 \notin \text{supp } \beta_n$, we automatically have $\beta_n(0) \rightarrow 0$ and $\beta'_n(0) \rightarrow 0$ as $n \rightarrow \infty$. An example for a sequence with these properties is

$$\beta_n(\sigma) = |n \tanh(\sigma/n)|^p (1 - \eta_{1/n}(\sigma)),$$

where $\eta_{1/n}$ is a smooth cutoff function as in (2.6) with $d = 1$. We now choose $\beta = \beta_n$ in (2.38) and let $n \rightarrow \infty$, aiming to show

$$\begin{aligned} & - \int_0^T \int_{\mathbb{R}^d} |u|^p \partial_t \psi \, dx \, dt - \psi(0) \int_{\mathbb{R}^d} |u^0|^p \, dx \\ & + \int_0^T \int_{\mathbb{R}^d} \psi (\operatorname{div} b + pc) |u|^p \, dx \, dt = \int_0^T \int_{\mathbb{R}^d} \psi f |u|^{p-2} u \, dx \, dt. \end{aligned} \quad (2.40)$$

Since $u \in L_t^\infty L_x^p$ and we assume $c, \operatorname{div} b \in L_t^1 L_x^\infty$, all integrals on the left-hand side converge by dominated convergence. For the right-hand side, notice that $\beta'_n(u) \leq C|u|^{p-1}$. Since $|u|^{p-1} \in L_t^\infty L_x^{p/(p-1)}$, by Hölder's inequality $f|u|^{p-1} \in L_t^1 L_x^1$. Thus dominated convergence also applies. This concludes the proof of (2.40) and hence of (2.33).

To show $u \in C([0, T]; L^p(\mathbb{R}^d))$, let $t_n \rightarrow t$ be a convergent sequence in $[0, T]$. By (2.33), $\|u(t)\|_{L^p(\mathbb{R}^d)} \in C([0, T])$. So by Lemma 5, we only need to show that $u(t_n)$ converges to $u(t)$ almost everywhere on $\mathbb{R}^d$. Furthermore, it suffices to show this for a subsequence, since $t_n \rightarrow t$ is an arbitrary convergent sequence. By a diagonal argument, it suffices to show that for every $R > 0$, there exists a subsequence t_{k_n} of t_n such that

$$u(t_{k_n}) \rightarrow u(t) \quad \text{a.e. on } B_R. \quad (2.41)$$

Let β be an admissible function such that $\beta'(\sigma) > 0$ for all $\sigma \in \mathbb{R}$. By (2.37),

$$\beta(u(t_n)) \rightarrow \beta(u(t)) \quad \text{in } L^r(B_R) \text{ as } n \rightarrow \infty.$$

Thus, $\beta(u(t_{k_n})) \rightarrow \beta(u(t))$ a.e. on B_R for some subsequence t_{k_n} of t_n . Since $\beta' > 0$, this implies (2.41). $\square$

Corollary 1 (Uniqueness for $p < \infty$). *Let $1 \leq p < \infty$, $1/p + 1/q = 1$. Assume $(\star)$ and*

$$b \in L^1(0, T; W_{\text{loc}}^{1,q}(\mathbb{R}^d; \mathbb{R}^d)), \quad c, \operatorname{div} b \in L^1(0, T; L^\infty(\mathbb{R}^d)).$$

If $u \in L^\infty(0, T; L^p(\mathbb{R}^d))$ is a solution of (2.1) for $u^0 = 0$, then $u = 0$.

Proof. By Proposition 2, u is a renormalized solution. Thus (2.33) and Grönwall's inequality imply $u = 0$. $\square$

Remark 7. Proposition 3 also shows the following property of renormalized solutions, which is familiar from smooth solutions: If $c = 0$ and $\operatorname{div} b = 0$, the measures of level sets of u are constant. To show this, let β be admissible such that $0 \notin \operatorname{supp} \beta$. By (2.34), for almost every $t \in (0, T)$

$$\int_{\mathbb{R}^d} \beta(u(t)) dx = \int_{\mathbb{R}^d} \beta(u^0) dx.$$

Let $0 < a < b$. Then, letting $\beta \uparrow \chi_{(a,b)}$ in the above equation and the monotone convergence theorem imply that $m^d(\{a < u(t) < b\}) = m^d(\{a < u^0 < b\})$ for almost every $t \in (0, T)$.

2.4.2 The case $p = \infty$. The uniqueness of weak solutions for $p = \infty$ will be a consequence of the following theorem. It is the same as Theorem II.6 of [DL89], except that we allow a right-hand side for both renormalized solutions.

Theorem 2 (Integration by parts along the flow). *Assume $(\star)$ and*

$$b \in L^1(0, T; W_{\text{loc}}^{1,1}(\mathbb{R}^d; \mathbb{R}^d)), \quad c, \operatorname{div} b \in L^1(0, T; L^\infty(\mathbb{R}^d)).$$

Let $1 \leq p \leq \infty$, $1/p + 1/q = 1$ and let $f \in L^1(0, T; L^p(\mathbb{R}^d))$, $g \in L^1(0, T; L^q(\mathbb{R}^d))$. If $u \in L^\infty(0, T; L^p(\mathbb{R}^d))$ and $v \in L^\infty(0, T; L^q(\mathbb{R}^d))$ are renormalized solutions of

$$\partial_t u - b \cdot \nabla u + cu = f, \quad \partial_t v - b \cdot \nabla v - (\operatorname{div} b + c)v = g,$$

then

$$\int_0^T \int_{\mathbb{R}^d} ug dx dt = - \int_0^T \int_{\mathbb{R}^d} fv dx dt + \int_{\mathbb{R}^d} u(T)v(T) dx - \int_{\mathbb{R}^d} u(0)v(0) dx. \quad (2.42)$$

The boundary terms on right-hand side of (2.42) make sense due to the time-continuity of u and v , which was shown in Proposition 3.

Proof. This proof is from [DL89], Theorem II.6. Let β be an admissible function and set $\tilde{u} := \beta(u)$, $\tilde{v} := \beta(v)$. By definition of a renormalized solution, $\tilde{u}$ and $\tilde{v}$ satisfy

$$\begin{aligned} \partial_t \tilde{u} - b \cdot \nabla \tilde{u} + cu\beta'(u) &= f\beta'(u) \\ \partial_t \tilde{v} - b \cdot \nabla \tilde{v} - (c + \operatorname{div} b)v\beta'(v) &= g\beta'(v). \end{aligned}$$

Let $\tilde{u}_\varepsilon := \tilde{u} * \rho_\varepsilon$ and $\tilde{v}_\varepsilon := \tilde{v} * \rho_\varepsilon$. By Lemma 2 and Remark 5

$$\begin{aligned} \partial_t \tilde{u}_\varepsilon - b \cdot \nabla \tilde{u}_\varepsilon + cu\beta'(u) &= f\beta'(u) + r_\varepsilon, \\ \partial_t \tilde{v}_\varepsilon - b \cdot \nabla \tilde{v}_\varepsilon - (c + \operatorname{div} b)v\beta'(v) &= g\beta'(v) + s_\varepsilon, \end{aligned} \quad (2.43)$$

where $r_\varepsilon, s_\varepsilon \rightarrow 0$ in $L^1(0, T; L^1_{\text{loc}}(\mathbb{R}^d))$ as $\varepsilon \rightarrow 0$. For $R > 0$, let η_R be a smooth cutoff function defined in (2.6). Due to the regularization, the equations in (2.43) are no longer just equations of distributions, but also of $L^1_{\text{loc}}([0, T] \times \mathbb{R}^d)$ -functions. Hence, we can multiply the equation for $\tilde{u}_\varepsilon$ by $\tilde{v}_\varepsilon \eta_R$ and integrate over $[0, T] \times \mathbb{R}^d$:

$$\begin{aligned} \int_0^T \int_{\mathbb{R}^d} \eta_R \tilde{v}_\varepsilon \partial_t \tilde{u}_\varepsilon \, dx \, dt + \int_0^T \int_{\mathbb{R}^d} \eta_R \tilde{v}_\varepsilon (-b \cdot \nabla \tilde{u}_\varepsilon + cu\beta'(u)) \, dx \, dt \\ = \int_0^T \int_{\mathbb{R}^d} \eta_R \tilde{v}_\varepsilon (f\beta'(u) + r_\varepsilon) \, dx \, dt \end{aligned} \quad (2.44)$$

If $\eta_R \tilde{v}_\varepsilon$ would belong to $C^1([0, T] \times \mathbb{R}^d)$, we could conclude from the trace theorem that

$$\begin{aligned} \int_0^T \int_{\mathbb{R}^d} \eta_R \tilde{v}_\varepsilon \partial_t \tilde{u}_\varepsilon \, dx \, dt = - \int_0^T \int_{\mathbb{R}^d} \eta_R \tilde{u}_\varepsilon \partial_t \tilde{v}_\varepsilon \, dx \, dt \\ + \int_{\mathbb{R}^d} \eta_R \tilde{v}_\varepsilon(T) \tilde{u}_\varepsilon(T) \, dx - \int_{\mathbb{R}^d} \eta_R \tilde{v}_\varepsilon(0) \tilde{u}_\varepsilon(0) \, dx. \end{aligned} \quad (2.45)$$

However, because $\eta_R \tilde{v}_\varepsilon \notin C^1([0, T] \times \mathbb{R}^d)$, some additional arguments are needed. Let $Z = (0, T) \times B_{2R}$ and $F \in C_c^\infty([0, T] \times B_{2R})$. Because $\tilde{u}_\varepsilon \in W^{1,1}(Z)$, by the trace theorem

$$\int_Z F \partial_t \tilde{u}_\varepsilon \, dx \, dt = - \int_Z \tilde{u}_\varepsilon \partial_t F \, dx \, dt + \int_{B_{2R}} F(T, x) \tilde{u}_\varepsilon(T, x) \, dx - \int_{B_{2R}} F(0, x) \tilde{u}_\varepsilon(0, x) \, dx. \quad (2.46)$$

Next, we choose a sequence $F_n \in C_c^\infty(\mathbb{R} \times B_{2R})$ such that

$$F_n \rightarrow \eta_R \tilde{v}_\varepsilon \quad \text{in } W^{1,1}(Z), \quad F_n \rightharpoonup^* \eta_R \tilde{v}_\varepsilon \quad \text{in } L^\infty(Z). \quad (2.47)$$

This works as follows. Because Z has Lipschitz boundary, by the Whitney extension theorem, there exists a $h \in W^{1,1}(\mathbb{R}^{d+1})$ such that $h|_Z = \eta_R \tilde{v}_\varepsilon$. Since $\text{supp } \eta_R \subset B_{2R}$, we may assume $\text{supp } h \subset \mathbb{R} \times B_{2R}$ after multiplying h by a smooth cutoff function. Since $\eta_R \tilde{v}_\varepsilon \in L^\infty(Z)$, we may also assume $h \in L^\infty(\mathbb{R}^{d+1})$ by composing with a function in $C_b^1(\mathbb{R})$ which is the identity on a suitably large interval. Then, there exists a sequence $F_n \in C_c^\infty(\mathbb{R} \times B_{2R})$, bounded in L^∞ , such that $F_n \rightarrow h$ in $W^{1,1}(\mathbb{R}^{d+1})$ as $n \rightarrow \infty$. After extraction of a subsequence, also $F_n \rightharpoonup^* h$ in $L^\infty(\mathbb{R}^{d+1})$.

We now set $F = F_n$ in (2.46) and let $n \rightarrow \infty$. The left-hand side of (2.46) converges to the left-hand side of (2.45) by the weak-* convergence of F_n and since $\partial_t \tilde{u}_\varepsilon \in L^1_{\text{loc}}([0, T] \times \mathbb{R}^d)$. For the $\eta_R \tilde{u}_\varepsilon \partial_t \tilde{v}_\varepsilon$ -term, note that $\tilde{u}_\varepsilon \in L^\infty(Z)$. The boundary terms converge as desired by the continuity of the trace operator $W^{1,1}(Z) \rightarrow L^1(\partial Z)$ and since $\tilde{u}_\varepsilon(0)$ and $\tilde{u}_\varepsilon(T)$ are smooth and therefore in $L^\infty(B_{2R})$.

Substituting (2.45) into (2.44) and integrating by parts in x yields

$$\begin{aligned} - \int_0^T \int_{\mathbb{R}^d} \eta_R \tilde{u}_\varepsilon \partial_t \tilde{v}_\varepsilon \, dx \, dt + \int_{\mathbb{R}^d} \eta_R \tilde{v}_\varepsilon(T) \tilde{u}_\varepsilon(T) \, dx - \int_{\mathbb{R}^d} \eta_R \tilde{v}_\varepsilon(0) \tilde{u}_\varepsilon(0) \, dx \\ + \int_0^T \int_{\mathbb{R}^d} \tilde{u}_\varepsilon (\eta_R \tilde{v}_\varepsilon \text{div } b + \tilde{v}_\varepsilon b \cdot \nabla \eta_R + \eta_R b \cdot \nabla \tilde{v}_\varepsilon) + \eta_R \tilde{v}_\varepsilon cu\beta'(u) \, dx \, dt \\ = \int_0^T \int_{\mathbb{R}^d} \eta_R \tilde{v}_\varepsilon (f\beta'(u) + r_\varepsilon) \, dx \, dt. \end{aligned}$$

By using equation (2.43) for $\tilde{v}_\varepsilon$,

$$\begin{aligned}
& - \int_0^T \int_{\mathbb{R}^d} \eta_R \tilde{u}_\varepsilon (b \cdot \nabla \tilde{v}_\varepsilon + (c + \operatorname{div} b) v \beta'(v) + g \beta'(v) + s_\varepsilon) dx dt \\
& \quad + \int_{\mathbb{R}^d} \eta_R \tilde{v}_\varepsilon(T) \tilde{u}_\varepsilon(T) dx - \int_{\mathbb{R}^d} \eta_R \tilde{v}_\varepsilon(0) \tilde{u}_\varepsilon(0) dx \\
& \quad + \int_0^T \int_{\mathbb{R}^d} \tilde{u}_\varepsilon (\eta_R \tilde{v}_\varepsilon \operatorname{div} b + \tilde{v}_\varepsilon b \cdot \nabla \eta_R + \eta_R b \cdot \nabla \tilde{v}_\varepsilon) + \eta_R \tilde{v}_\varepsilon c u \beta'(u) dx dt \\
& = \int_0^T \int_{\mathbb{R}^d} \eta_R \tilde{v}_\varepsilon (f \beta'(u) + r_\varepsilon) dx dt.
\end{aligned}$$

The terms $\eta_R \tilde{u}_\varepsilon b \cdot \nabla \tilde{v}_\varepsilon$ in the first line and third line cancel out. Letting $\varepsilon \rightarrow 0$, we obtain

$$\begin{aligned}
& - \int_0^T \int_{\mathbb{R}^d} \eta_R \tilde{u} ((c + \operatorname{div} b) v \beta'(v) + g \beta'(v)) dx dt + \int_{\mathbb{R}^d} \eta_R \tilde{v}(T) \tilde{u}(T) dx - \int_{\mathbb{R}^d} \eta_R \tilde{v}(0) \tilde{u}(0) dx \\
& \quad + \int_0^T \int_{\mathbb{R}^d} \tilde{u} (\eta_R \tilde{v} \operatorname{div} b + \tilde{v} b \cdot \nabla \eta_R) + \eta_R \tilde{v} c u \beta'(u) dx dt = \int_0^T \int_{\mathbb{R}^d} \eta_R \tilde{v} f \beta'(u) dx dt. \quad (2.48)
\end{aligned}$$

To justify this, note that $\tilde{u}_\varepsilon(t) \rightharpoonup^* \tilde{u}(t)$ and $\tilde{v}_\varepsilon(t) \rightharpoonup^* \tilde{v}(t)$ in $L^\infty(\mathbb{R}^d)$ for every $t \in [0, T]$. Since $\tilde{u}_\varepsilon(t) \rightarrow \tilde{u}(t)$ and $\tilde{v}_\varepsilon(t) \rightarrow \tilde{v}(t)$ in $L_{\operatorname{loc}}^r(\mathbb{R}^d)$ for all $r < \infty$, we also have $\tilde{u}_\varepsilon(t) \tilde{v}_\varepsilon(t) \rightharpoonup^* \tilde{u}(t) \tilde{v}(t)$ in $L^\infty(\mathbb{R}^d)$. Thus, for almost every $t \in [0, T]$, the integrals over x converge as desired. To conclude the convergence of the space-time integrals, we apply dominated convergence to the time integral.

From now on, assume that $\beta(0) = 0$. By letting $R \rightarrow \infty$ in (2.48), we obtain

$$\begin{aligned}
& - \int_0^T \int_{\mathbb{R}^d} \tilde{u} ((c + \operatorname{div} b) v \beta'(v) + g \beta'(v)) dx dt + \int_{\mathbb{R}^d} \tilde{v}(T) \tilde{u}(T) dx - \int_{\mathbb{R}^d} \tilde{v}(0) \tilde{u}(0) dx \\
& \quad + \int_0^T \int_{\mathbb{R}^d} \tilde{u} \tilde{v} \operatorname{div} b + \tilde{v} c u \beta'(u) dx dt = \int_0^T \int_{\mathbb{R}^d} \tilde{v} f \beta'(u) dx dt. \quad (2.49)
\end{aligned}$$

For all terms except the one with $\nabla \eta_R$, this is justified by dominated convergence, since

$$\tilde{u}_\varepsilon \in L_t^\infty(L_x^\infty \cap L_x^p), \quad \tilde{v} \in L_t^\infty(L_x^\infty \cap L_x^q), \quad \tilde{u}(0), \tilde{u}(T) \in L_x^p, \quad \tilde{v}(0), \tilde{v}(T) \in L_x^q.$$

Since $|\nabla \eta_R| \leq \|\nabla \eta\|_{L^\infty(\mathbb{R}^d)}/R$ and $\operatorname{supp} \nabla \eta_R \subset B_{2R} \setminus B_R$, the $\nabla \eta_R$ -term can be estimated by

$$\left| \int_0^T \int_{\mathbb{R}^d} \tilde{u} \tilde{v} b \cdot \nabla \eta_R dx dt \right| \leq C \int_0^T \int_{B_{2R} \setminus B_R} |\tilde{u} \tilde{v}| \frac{|b|}{1 + |x|} dx dt.$$

Since u and v have dual Hölder exponents and $\beta(0) = 0$, $\tilde{u} \tilde{v} \in L_t^\infty L_x^1 \cap L_t^1 L_x^\infty$. Thus, by the growth condition $(\star)$, the right-hand side converges to 0 as $R \rightarrow \infty$.

Finally, we choose a sequence of admissible functions β_n such that $\beta_n \rightarrow \operatorname{id}_{\mathbb{R}}$, $\beta'_n \rightarrow 1$ pointwise, $|\beta_n(\sigma)| \leq |\sigma|$ and $|\beta'_n(\sigma)| \leq 1$ for every $\sigma \in \mathbb{R}$. Taking $\beta = \beta_n$ in (2.49) and using dominated convergence as $n \rightarrow \infty$, we get

$$\begin{aligned}
& - \int_0^T \int_{\mathbb{R}^d} u ((c + \operatorname{div} b) v + g) dx dt + \int_{\mathbb{R}^d} v(T) u(T) dx - \int_{\mathbb{R}^d} v(0) u(0) dx \\
& \quad + \int_0^T \int_{\mathbb{R}^d} u v \operatorname{div} b + v c u dx dt = \int_0^T \int_{\mathbb{R}^d} v f dx dt.
\end{aligned}$$

The $\operatorname{div} b$ and c terms cancel out, so (2.42) follows. $\square$

Corollary 2 (Uniqueness for $p = \infty$). *Assume $(\star)$ and*

$$b \in L^1(0, T; W_{\operatorname{loc}}^{1,1}(\mathbb{R}^d; \mathbb{R}^d)), \quad c, \operatorname{div} b \in L^1(0, T; L^\infty(\mathbb{R}^d)) \quad (\dagger)$$

If $u \in L^\infty([0, T] \times \mathbb{R}^d)$ is a solution of (2.1) for $u^0 = 0$, then $u = 0$.

Proof. It suffices to show that for every $\phi \in C_c^\infty((0, T) \times \mathbb{R}^d)$

$$\int_0^T \int_{\mathbb{R}^d} u \phi \, dx \, dt = 0. \quad (2.50)$$

To do this, we consider a solution v of

$$\partial_t v - b \cdot \nabla v - (c + \operatorname{div} b)v = \phi, \quad v|_{t=T} = 0. \quad (2.51)$$

By Proposition 1 and Remark 3, there exists a solution to this equation in $L^\infty([0, T] \times \mathbb{R}^d)$. An initial condition at time T can be imposed by reversing time.

To show that also $v \in L^\infty(0, T; L^1(\mathbb{R}^d))$, we look again at the proof of Proposition 1. We constructed a sequence of solutions $v_n \in L^\infty(0, T; C_b^1(\mathbb{R}^d))$ to

$$\partial_t v_n - b_n \cdot \nabla v_n - (c_n + \operatorname{div} b_n)v_n = \phi_n, \quad v_n|_{t=0} = v_n^0$$

where

$$b_n \rightarrow b, \quad c_n \rightarrow c, \quad \operatorname{div} b_n \rightarrow \operatorname{div} b, \quad \phi_n \rightarrow \phi \quad \text{in } L_{\text{loc}}^1([0, T] \times \mathbb{R}^d)$$

and $c_n, \operatorname{div} b_n$ are bounded in $L^1(0, T; L^\infty(\mathbb{R}^d))$. As indicated in Remark 3, the proof of Proposition 1 can be modified to incorporate a right-hand side by setting $\phi_n := (\phi * \rho_{1/n})\eta_n$, where $\rho_{1/n}$ and η_n are defined in (2.5) and (2.6). Then, we used the estimate (2.13) to show that v_n is bounded in $L^\infty([0, T] \times \mathbb{R}^d)$ and obtained v as the weak-* limit of a subsequence of v_n in $L^\infty([0, T] \times \mathbb{R}^d)$. By Step 1 of Proposition 1, v_n is also compactly supported. Thus, by (2.33) and Grönwall's inequality

$$\|v_n(t)\|_{L^1(\mathbb{R}^d)} \leq \exp\left(\|c_n\|_{L^1(0, T; L^\infty(\mathbb{R}^d))}\right) \|\phi_n\|_{L^1(0, T; L^1(\mathbb{R}^d))}.$$

This shows that v_n is bounded in $L^\infty(0, T; L^1(\mathbb{R}^d))$.

Let $\psi \in L^1([0, T])$, $R > 0$ and $\zeta(t, x) := \psi(t)\chi_{B_R}(x)\operatorname{sgn}(v)$. Since $v_n \rightharpoonup^* v$ in $L^\infty([0, T] \times \mathbb{R}^d)$,

$$\int_0^T \int_{\mathbb{R}^d} v_n \zeta \, dx \, dt \xrightarrow{n \rightarrow \infty} \int_0^T \int_{\mathbb{R}^d} v \zeta \, dx \, dt = \int_0^T \psi(t) \int_{B_R} |v| \, dx \, dt \quad (2.52)$$

By Hölder's inequality,

$$\int_0^T \int_{\mathbb{R}^d} v_n \zeta \, dx \, dt \leq \|v_n\|_{L^\infty(0, T; L^1(\mathbb{R}^d))} \|\psi\|_{L^1([0, T])}.$$

Thus, by letting $R \rightarrow \infty$ in (2.52) and using that $\psi \in L^1([0, T])$ is arbitrary, we conclude $v \in L^\infty(0, T; L^1(\mathbb{R}^d))$.

By Proposition 2 (and Remark 6), both u and v are renormalized solutions. Thus (2.50) follows from Theorem 2. $\square$

2.5 Applications to the 2d Euler equations

We pause the general discussion and briefly discuss applications of the theory developed so far to the two-dimensional Euler equations. In vorticity formulation, the Euler equations read

$$\begin{aligned} \partial_t \omega + b \cdot \nabla \omega &= 0 & \text{on } (0, T) \times \mathbb{R}^2 \\ b &= K * \omega \end{aligned} \quad (2.53)$$

The convolution kernel $K : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2$ (also called the Biot-Savart kernel) is defined by

$$K(x) := \frac{1}{2\pi|x|^2} \begin{pmatrix} -x_2 \\ x_1 \end{pmatrix}.$$

We assume that $\omega \in L^\infty(0, T; L^p(\mathbb{R}^2))$ for some $1 \leq p \leq \infty$ and $b \in L^1(0, T; L^q_{\text{loc}}(\mathbb{R}^2))$, where $1/p + 1/q = 1$. Then, the transport equation in (2.53) can be understood in the weak sense.²

Since ∇K is homogeneous of degree -2 , smooth on $\mathbb{R}^2 \setminus \{0\}$, and $\int_{S^1} \nabla K d\mathcal{H}^1 = 0$, by the Calderón-Zygmund theory for singular integrals, the map $f \mapsto \nabla K * f$ is a bounded linear operator $L^p(\mathbb{R}^2) \rightarrow L^p(\mathbb{R}^2; \mathbb{R}^{2 \times 2})$ if $1 < p < \infty$ (see [Ste70], p. 39, Theorem 3). Thus $\nabla b \in L^\infty(0, T; L^p(\mathbb{R}^2))$ for such p .

If $p \geq 2$, this implies $b \in L^1(0, T; W^{1,q}_{\text{loc}})$. Combining this fact with part 2) of Proposition 2 yields the following theorem.

Theorem 3. *Let $2 \leq p < \infty$, and let $\omega \in L^\infty(0, T; L^p(\mathbb{R}^2))$ and $b \in L^1(0, T; L^q_{\text{loc}}(\mathbb{R}^2))$ satisfy the Euler equations (2.53) in the weak sense. Then ω is a renormalized solution.*

The result in Theorem 3 has been observed previously as a consequence of Theorem II.3 of [DL89], see Proposition 1 of [FML05].

For $1 < p < 2$, it was shown in [CS15] that solutions of (2.53) obtained as a limit in the vanishing viscosity approximation are renormalized. The argument in [CS15] uses the integration by parts formula (2.42). They show that every such solution satisfies, for all $\chi \in C_c^\infty((0, T) \times \mathbb{R}^2)$,

$$\int_0^T \int_{\mathbb{R}^2} \chi \omega dx dt = \int_{\mathbb{R}^2} \phi(0, x) \omega(0, x) dx, \quad (2.54)$$

where $\phi \in L^\infty(0, T; L^1 \cap L^\infty)$ is the renormalized solution of

$$\partial_t \phi + b \cdot \nabla \phi = -\chi, \quad \phi|_{t=T} = 0.$$

Equation (2.54) may be used to deduce that ω is renormalized. Note that $\text{div } b = 0$, since $\text{div } K = 0$. Thus by Theorem 2, (2.54) also holds for the renormalized solution of (2.53). Since $\chi \in C_c^\infty((0, T) \times \mathbb{R}^2)$ is arbitrary, ω equals the renormalized solution.

2.6 Stability

2.6.1 The case $c = 0$. In this section, we prove stability results for renormalized solutions of the transport equation. We consider the case $c = 0$ here and more general c in §2.6.2. A consequence of the stability results will be the existence of renormalized solutions for $u^0 \in L(\mathbb{R}^d)$, see Corollary 4.

Theorem 4 (Stability). *Assume b satisfies $(\dagger)$ and $(\star)$ with $c = 0$. For $n \in \mathbb{N}$, let $b_n \in L^1_{\text{loc}}([0, T] \times \mathbb{R}^d; \mathbb{R}^d)$ and $c_n \in L^1_{\text{loc}}([0, T] \times \mathbb{R}^d)$ such that*

$$b_n \rightarrow b \quad \text{in } L^1_{\text{loc}}([0, T] \times \mathbb{R}^d; \mathbb{R}^d), \quad \text{div } b_n \rightarrow \text{div } b, \quad c_n \rightarrow 0 \quad \text{in } L^1_{\text{loc}}([0, T] \times \mathbb{R}^d) \quad (2.55)$$

as $n \rightarrow \infty$. Let $u_n^0, u^0 \in L(\mathbb{R}^d)$ and let $u_n \in L([0, T] \times \mathbb{R}^d)$ be renormalized solutions of (2.1) with (b_n, c_n, u_n^0) .

- 1) *If $u_n^0 \rightarrow u^0$ in $L(\mathbb{R}^d)$, then there exists a renormalized solution $u \in C([0, T]; L(\mathbb{R}^d))$ for $(b, 0, u^0)$, and $u_n \rightarrow u$ in $C([0, T]; L(\mathbb{R}^d))$.*
- 2) *Let $1 \leq p < \infty$ and $u_n \in L^\infty(0, T; L^p(\mathbb{R}^d))$. Assume that $c_n, \text{div } b_n \in L^1(0, T; L^\infty(\mathbb{R}^d))$ and that $(\star)$ holds for all b_n . If $u_n^0 \rightarrow u^0$ in $L^p(\mathbb{R}^d)$ and there exists a function $g \in L^1((0, T))$ such that*

$$\sup_{n \in \mathbb{N}} \|\text{div } b_n(t) + p c_n(t)\|_{L^\infty(\mathbb{R}^d)} \leq g(t) \quad \text{for a.e. } t \in [0, T], \quad (2.56)$$

then $u_n \rightarrow u$ in $C([0, T]; L^p(\mathbb{R}^d))$.

²If $4/3 \leq p < 2$, then $\omega \in L^\infty(0, T; L^p(\mathbb{R}^2))$ already implies $b \in L^1(0, T; L^q_{\text{loc}}(\mathbb{R}^2))$. Since K is homogeneous of degree -1 , by the Hardy-Littlewood-Sobolev theorem $b \in L^\infty(0, T; L^r(\mathbb{R}^2))$, where $1/r = 1/p - 1/2$ ([Ste70], p. 119, Theorem 1). However, for $p > 2$, such a priori estimates are not sufficient to conclude even $K * \omega \in L^1_{\text{loc}}([0, T] \times \mathbb{R}^2)$. Consider for example $\omega(t, x) = \chi_{x_1 > 0}/(1 + |x|)$.

Proof. This proof corresponds to the proofs of Theorems II.3 and II.4 in [DL89]. For the first four steps of the proof, we assume the hypotheses of part 1) of Theorem 4.

Step 1: Weak- convergence of $\beta(u_n)$.* Let β be an admissible function and set

$$v_n = \beta(u_n), \quad v_n^0 = \beta(u_n^0), \quad v^0 = \beta(u^0).$$

We first show that v_n converges weak-* in $L^\infty([0, T] \times \mathbb{R}^d)$ to the unique solution of (2.1) with $c = 0$ and initial data v^0 . Since v_n is bounded in $L^\infty([0, T] \times \mathbb{R}^d)$, there exists a weak-* convergent subsequence v_{k_n} , whose limit we denote by v .

Since u_{k_n} is a renormalized solution, v_{k_n} satisfies

$$\partial_t v_{k_n} - b_{k_n} \cdot \nabla v_{k_n} + c_{k_n} u_{k_n} \beta'(u_{k_n}) = 0 \quad \text{in } (0, T) \times \mathbb{R}^d, \quad v_{k_n}|_{t=0} = v_{k_n}^0 \quad \text{in } \mathbb{R}^d.$$

This means that for every $\phi \in C_c^\infty([0, T] \times \mathbb{R}^d)$

$$\begin{aligned} & - \int_0^T \int_{\mathbb{R}^d} v_{k_n} \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} v_{k_n}^0 \phi(0, x) \, dx \\ & + \int_0^T \int_{\mathbb{R}^d} v_{k_n} (\phi \operatorname{div} b_{k_n} + b_{k_n} \cdot \nabla \phi) + c_{k_n} u_{k_n} \beta'(u_{k_n}) \phi \, dx \, dt = 0. \end{aligned}$$

By letting $n \rightarrow \infty$ in this equation, we now show that v satisfies (2.2) with $c = 0$ and initial data v^0 . By (2.55) and the boundedness of $u_{k_n} \beta'(u_{k_n})$ in $L^\infty([0, T] \times \mathbb{R}^d)$, the first and third integral converge as desired. For the second integral, note that $v_{k_n}^0, v^0 \leq \|\beta\|_{L^\infty(\mathbb{R})}$. Therefore, for every $\varepsilon \geq 0$

$$\int_{\mathbb{R}^d} |v_{k_n}^0 - v^0| |\phi(0, x)| \, dx \leq \|\phi(0, \cdot)\|_{L^1(\mathbb{R}^d)} \left(\varepsilon + 2\|\beta\|_{L^\infty(\mathbb{R})} m^d(\{|v_{k_n}^0 - v^0| \geq \varepsilon\} \cap \operatorname{supp} \phi) \right)$$

Since $v_{k_n}^0 \rightarrow v^0$ in $L(\mathbb{R}^d)$ by the Lipschitz-continuity of β , the right-hand side can be made arbitrarily small by choosing ε appropriately. Hence v is a solution of the transport equation with $(b, 0, v^0)$.

So far, we have only shown the weak-* convergence of the subsequence v_{k_n} to v . However, by uniqueness (Corollary 2) and the above argument, every subsequence of v_n has a subsequence converging to v weak-* in $L^\infty([0, T] \times \mathbb{R}^d)$, which implies the convergence of the whole sequence.

Step 2: $L^2_{\text{loc}}([0, T] \times \mathbb{R}^d)$ convergence of $\beta(u_n)$. We now show that $v_n \rightarrow v$ in $L^2_{\text{loc}}([0, T] \times \mathbb{R}^d)$. Let $K \subset [0, T] \times \mathbb{R}^d$ be compact. By Step 1, $v_n \rightharpoonup v$ in $L^2(K)$, so it suffices to show $\|v_n\|_{L^2(K)} \rightarrow \|v\|_{L^2(K)}$, because $L^2(K)$ is uniformly convex.

Let w be the unique solution of (2.1) with $(b, 0, \beta(u^0)^2)$. By Step 1 with β replaced by β^2 , which is also admissible,

$$v_n^2 \rightharpoonup^* w \quad \text{in } L^\infty([0, T] \times \mathbb{R}^d).$$

By Proposition 2, v is renormalized, hence v^2 is also a solution of (2.1) with $c = 0$ and initial data $\beta(u^0)^2$. Note that v is bounded, so we can choose a suitably cut-off version of x^2 as an admissible function. Thus $v^2 = w$ by Corollary 2. We conclude $\|v_n\|_{L^2(K)} \rightarrow \|v\|_{L^2(K)}$, because $m^{d+1}(K) < \infty$.

Step 3: Uniformity in time. The next goal is to show $v_n \rightarrow v$ in $C([0, T]; L^2_{\text{loc}}(\mathbb{R}^d))$. We first show that for every $\xi \in C_c^\infty(\mathbb{R}^d)$

$$\int_{\mathbb{R}^d} v_n(t, x) \xi(x) \, dx \rightarrow \int_{\mathbb{R}^d} v(t, x) \xi(x) \, dx \quad \text{in } W^{1,1}((0, T)). \quad (2.57)$$

Since u_n is a renormalized solution, for every $\psi \in C_c^\infty((0, T))$

$$\int_0^T \int_{\mathbb{R}^d} v_n \xi \partial_t \psi(t) dx = \int_0^T \int_{\mathbb{R}^d} (v_n (\xi \operatorname{div} b_n + b_n \cdot \nabla \xi) + c_n u_n \beta'(u_n) \xi) \psi(t) dx dt.$$

Because ψ is arbitrary, we deduce that

$$\frac{d}{dt} \int_{\mathbb{R}^d} v_n \xi dx = - \int_{\mathbb{R}^d} v_n (\xi \operatorname{div} b_n + b_n \cdot \nabla \xi) + c_n u_n \beta'(u_n) \xi dx \quad (2.58)$$

in the sense of distributions on $(0, T)$. However, the right-hand side is in $L^1((0, T))$, so (2.58) shows $\int_{\mathbb{R}^d} v_n \xi dx \in W^{1,1}((0, T))$. Similarly,

$$\frac{d}{dt} \int_{\mathbb{R}^d} v \xi dx = - \int_{\mathbb{R}^d} v (\xi \operatorname{div} b + b \cdot \nabla \xi) dx. \quad (2.59)$$

Thus, to prove (2.57), by (2.58) and (2.59) it suffices to show that

$$\int_0^T \int_{\mathbb{R}^d} \xi (v_n \operatorname{div} b_n - v \operatorname{div} b) + (v_n b_n - v b) \cdot \nabla \xi + c_n u_n \beta'(u_n) \xi dx dt \rightarrow 0.$$

The integral of $c_n u_n \beta'(u_n) \xi$ converges to zero since $c_n \rightarrow 0$ in $L_{\text{loc}}^1([0, T] \times \mathbb{R}^d)$ and $u_n \beta'(u_n)$ is bounded in $L^\infty([0, T] \times \mathbb{R}^d)$. The first two terms converge to 0 since $b_n \rightarrow b$ and $\operatorname{div} b_n \rightarrow \operatorname{div} b$ in $L_{\text{loc}}^1([0, T] \times \mathbb{R}^d)$ and $v_n \rightharpoonup^* v$ in $L^\infty([0, T] \times \mathbb{R}^d)$.

By Proposition 3, $v \in C([0, T]; L_{\text{loc}}^2(\mathbb{R}^d))$. Hence, to conclude $v_n \rightarrow v$ in $C([0, T]; L_{\text{loc}}^2(\mathbb{R}^d))$, it suffices to show that for every convergent sequence $t_n \rightarrow t$ in $[0, T]$

$$v_n(t_n, \cdot) \rightarrow v(t, \cdot) \quad \text{in } L_{\text{loc}}^2(\mathbb{R}^d). \quad (2.60)$$

Fix such a sequence t_n . Because $W^{1,1}((0, T))$ embeds continuously into $C([0, T])$, (2.57) implies for every $\xi \in C_c^\infty(\mathbb{R}^d)$

$$\int_{\mathbb{R}^d} v_n(t_n, x) \xi(x) dx \rightarrow \int_{\mathbb{R}^d} v(t, x) \xi(x) dx \quad \text{as } n \rightarrow \infty. \quad (2.61)$$

Let $K \subset \mathbb{R}^d$ be compact. Since $v_n(t_n)$ is bounded in $L^2(K)$, (2.61) implies $v_n(t_n) \rightharpoonup v(t)$ in $L^2(K)$. Since β^2 is also an admissible function, we also have $v_n(t_n)^2 \rightharpoonup v(t)^2$ in $L^2(K)$. Since $\chi_K \in L^2(K)$, we conclude $\|v_n(t_n, \cdot)\|_{L^2(K)} \rightarrow \|v(t, \cdot)\|_{L^2(K)}$, and hence $v_n(t_n, \cdot) \rightarrow v(t, \cdot)$ in $L^2(K)$.

Step 4: Part 1). Choose $\beta = \tanh$ and let v be the limit of $\beta(u_n)$ in $C([0, T]; L_{\text{loc}}^2(\mathbb{R}^d))$. Define $u = \beta^{-1}(v)$. To make sense of this definition, we now show that

$$m^d(\{x \in \mathbb{R}^d : |v(t, x)| = 1\}) = 0 \quad \text{for every } t \in [0, T]. \quad (2.62)$$

For every $t \in [0, T]$, we already know that $|v(t, x)| \leq 1$ for a.e. $x \in \mathbb{R}^d$, since v is the weak-* limit of v_n in $L^\infty([0, T] \times \mathbb{R}^d)$ and $v \in C([0, T]; L_{\text{loc}}^2(\mathbb{R}^d))$. To prove (2.62), for $k \in \mathbb{N}$, let $\theta_k \in C_c^\infty(\mathbb{R})$ be such that

$$0 \leq \theta_k \leq 1, \quad \theta_k|_{[1-1/k, 1]} = 1, \quad \operatorname{supp} \theta_k \subset (1 - 2/k, \infty).$$

Define $\alpha_k(\sigma) = \theta_k(|\sigma|)$ for $\sigma \in \mathbb{R}$ and note that α_k is admissible. Since $|v^0(x)| < 1$ for all $x \in \mathbb{R}^d$, $\alpha_k(v^0) \rightarrow 0$ in $L(\mathbb{R}^d)$ as $k \rightarrow \infty$. Furthermore, since v is renormalized, $\alpha_k(v)$ is the unique renormalized solution of (2.1) with $c = 0$ and initial data $\alpha_k(v^0)$. By applying Steps 1-3 to the sequence $\alpha_k(v)$, $b_k = b$ and $c_k = 0$ for $k \in \mathbb{N}$, we deduce that for every admissible function γ ,

$$\gamma(\alpha_k(v)) \rightarrow \gamma(0) \quad \text{in } C([0, T]; L_{\text{loc}}^2(\mathbb{R}^d)) \text{ as } k \rightarrow \infty.$$

But α_k only takes values in $[0, 1]$, hence we may choose γ such that $\gamma|_{[0,1]} = \text{id}$. Thus

$$\alpha_k(v) \rightarrow 0 \quad \text{in } C([0, T]; L^2_{\text{loc}}(\mathbb{R}^d)) \text{ as } k \rightarrow \infty. \quad (2.63)$$

By the definition of α_k , this implies (2.62).

We now show $u_n \rightarrow u$ in $C([0, T]; L(\mathbb{R}^d))$. Let $K \subset \mathbb{R}^d$ be compact and $\delta > 0$. By (2.63), there exists an $M \in \mathbb{N}$ such that

$$\sup_{t \in [0, T]} m^d(\{x \in K : |v(t, x)| \geq 1 - 1/M\}) < \delta.$$

Let $\varepsilon > 0$. Then

$$\begin{aligned} & \sup_{t \in [0, T]} m^d(\{x \in K : |u_n(t, x) - u(t, x)| \geq \varepsilon\}) \\ & \leq \sup_{t \in [0, T]} m^d(\{x \in K : |u_n(t, x) - u(t, x)| \geq \varepsilon \text{ and } |v(t, x)| < 1 - 1/M\}) + \delta. \end{aligned} \quad (2.64)$$

The supremum on the right-hand side converges to 0 as $n \rightarrow \infty$, since β^{-1} is Lipschitz-continuous on any compact interval $I \subset (-1, 1)$. Explicitly, let $\lambda = \beta(\beta^{-1}(1 - 1/M) + \varepsilon)$ and L be the Lipschitz constant of $\beta^{-1}|_{[-\lambda, \lambda]}$. If $x \in K$ and $|v(t, x)| < 1 - 1/M$, then

$$\{|u_n(t, x) - u(t, x)| \geq \varepsilon\} \quad \text{implies} \quad \{|v_n(t, x) - v(t, x)| \geq \varepsilon/L\}.$$

Since $v_n \rightarrow v$ in $C([0, T]; L(\mathbb{R}^d))$ and $\delta > 0$ is arbitrary, we conclude $u_n \rightarrow u$ in $C([0, T]; L(\mathbb{R}^d))$.

To prove that u is a renormalized solution of (2.1) with $c = 0$ and initial data u^0 , let β be an admissible function. Because β is Lipschitz-continuous, $\beta(u_n)$ converges to $\beta(u)$ in $C([0, T]; L(\mathbb{R}^d))$. By Step 3, $\beta(u_n)$ also converges in $C([0, T]; L^2_{\text{loc}}(\mathbb{R}^d))$ to the unique renormalized solution of (2.1) with $c = 0$ and initial data $\beta(u^0)$.

Step 5: Part 2). By (2.56) and (2.33), u_n is bounded in $L^\infty(0, T; L^p(\mathbb{R}^d))$. By Step 4, u is the limit of u_n in $C([0, T]; L(\mathbb{R}^d))$, hence u belongs to $L^\infty(0, T; L^p(\mathbb{R}^d))$. Thus b and u satisfy the hypothesis of part 2) of Proposition 3, which implies that $u \in C([0, T]; L^p(\mathbb{R}^d))$. So, we only need to show that $u_n(t_n) \rightarrow u(t)$ in $L^p(\mathbb{R}^d)$ for every convergent sequence $t_n \rightarrow t$ in $[0, T]$. Since $u_n \rightarrow u$ in $C([0, T]; L(\mathbb{R}^d))$, we may assume that $u_n(t_n)$ converges to $u(t)$ almost everywhere on $\mathbb{R}^d$. Therefore, by Lemma 5, it suffices to show that $\|u_n(t_n)\|_{L^p(\mathbb{R}^d)} \rightarrow \|u(t)\|_{L^p(\mathbb{R}^d)}$ as $n \rightarrow \infty$.

By Fatou's Lemma,

$$\liminf_{n \rightarrow \infty} \int_{\mathbb{R}^d} |u_n(t_n)|^p dx \geq \int_{\mathbb{R}^d} |u(t)|^p dx. \quad (2.65)$$

For the other inequality, let $a \in W^{1,1}((0, T))$. By (2.33), for a.e. $t \in (0, T)$

$$\frac{d}{dt} \left(e^{-a(t)} \int_{\mathbb{R}^d} |u(t)|^p dx \right) = -e^{-a(t)} \int_{\mathbb{R}^d} (\text{div } b(t) + a'(t)) |u(t)|^p dx. \quad (2.66)$$

Integrating (2.66) from 0 to t yields

$$e^{-a(t)} \int_{\mathbb{R}^d} |u(t)|^p dx = \int_{\mathbb{R}^d} |u^0|^p dx - \int_0^t \int_{\mathbb{R}^d} (\text{div } b + a') e^{-a(s)} |u|^p dx ds. \quad (2.67)$$

Since the assumptions of Proposition 3 are also satisfied for u_n , (2.66) also holds with u_n, b_n, c_n instead of $u, b, 0$. We integrate from 0 to t_n :

$$e^{-a(t_n)} \int_{\mathbb{R}^d} |u_n(t_n)|^p dx = \int_{\mathbb{R}^d} |u_n^0|^p dx - \int_0^{t_n} \int_{\mathbb{R}^d} (\text{div } b_n + p c_n + a') e^{-a(s)} |u_n|^p dx ds \quad (2.68)$$

Since $b_n \rightarrow b$ and $c_n \rightarrow 0$ in $L^1_{\text{loc}}([0, T] \times \mathbb{R}^d)$, after extracting a subsequence, we may assume they converge pointwise almost everywhere on $[0, T] \times \mathbb{R}^d$. We now choose a such that a is continuous, $a(0) = 0$ and $a' = g$, where $g \in L^1((0, T))$ is from the assumptions of Theorem 4, part 2). Due to our choice of a' , the integrand on the right-hand side in (2.68) is nonnegative. Thus, by the assumption that $u_n^0 \rightarrow u^0$ in $L^p(\mathbb{R}^d)$ and another application of Fatou's Lemma,

$$\begin{aligned} & \limsup_{n \rightarrow \infty} \left(e^{-a(t_n)} \int_{\mathbb{R}^d} |u_n(t_n)|^p dx \right) \\ &= \int_{\mathbb{R}^d} |u^0|^p dx - \liminf_{n \rightarrow \infty} \int_0^{t_n} \int_{\mathbb{R}^d} (\operatorname{div} b_n + pc_n + a') e^{-a(s)} |u_n|^p dx ds \\ &\leq e^{-a(t)} \int_{\mathbb{R}^d} |u(t)|^p dx. \end{aligned} \quad (2.69)$$

Since a is continuous, the left-hand side is equal to $e^{-a(t)} \limsup_{n \rightarrow \infty} \int_{\mathbb{R}^d} |u_n(t_n)|^p dx$. Thus we conclude $\|u_n(t_n)\|_{L^p(\mathbb{R}^d)} \rightarrow \|u(t)\|_{L^p(\mathbb{R}^d)}$. $\square$

2.6.2 More general c . We now prove stability results for $c \in L^1(0, T; L^\infty(\mathbb{R}^d))$ by reducing to the case $c = 0$. Let $a \in L^1(0, T; L^\infty(\mathbb{R}^d))$ and consider the equation

$$\partial_t \Phi - b \cdot \nabla \Phi = a, \quad \Phi|_{t=0} = 0.$$

By Proposition 1 and Remark 3, if $b \in L^1_{\text{loc}}([0, T] \times \mathbb{R}^d; \mathbb{R}^d)$ and $\operatorname{div} b \in L^1(0, T; L^\infty(\mathbb{R}^d))$, there exists a solution $\Phi \in L^\infty([0, T] \times \mathbb{R}^d)$ to this equation. If $b \in L^1(0, T; W^{1,1}_{\text{loc}}(\mathbb{R}^d))$, then Φ is renormalized by Proposition 2.

Lemma 6. *Assume $b \in L^1(0, T; W^{1,1}_{\text{loc}}(\mathbb{R}^d))$ and $c \in L^1(0, T; L^1_{\text{loc}}(\mathbb{R}^d))$. Let $u^0 \in L(\mathbb{R}^d)$. Then, $u \in L([0, T] \times \mathbb{R}^d)$ is a renormalized solution of (2.1) with (b, c, u^0) if and only if $e^{-\Phi} u$ is a renormalized solution of (2.1) with $(b, c + a, u^0)$.*

Proof. This proof is from [DL89], p. 526, Lemma II.2.

It suffices to show just one direction due to the symmetry of the statement. Assume that u is a renormalized solution of the transport equation for the initial condition u^0 .

Formally, the lemma would follow from the chain rule. To make this argument rigorous, we invoke the regularization result, Lemma 2. Let β be an admissible function and set $v = \beta(u)$. Then by Lemma 2

$$\begin{aligned} \partial_t v_\varepsilon - b \cdot \nabla v_\varepsilon + cu\beta'(u) &= r_\varepsilon & v_\varepsilon|_{t=0} &= v_\varepsilon^0 \\ \partial_t \Phi_\varepsilon - b \cdot \nabla \Phi_\varepsilon &= a + s_\varepsilon & \Phi_\varepsilon|_{t=0} &= 0 \end{aligned} \quad (2.70)$$

Here, $v_\varepsilon = v * \rho_\varepsilon$, $v_\varepsilon^0 = \beta(u^0) * \rho_\varepsilon$, $\Phi_\varepsilon = \Phi * \rho_\varepsilon$ and $r_\varepsilon, s_\varepsilon \rightarrow 0$ in $L^1_{\text{loc}}([0, T] \times \mathbb{R}^d)$. Let γ be an admissible function and define $\omega^\varepsilon := \gamma(e^{-\Phi_\varepsilon} v_\varepsilon)$. By the product and chain rules for Sobolev functions,

$$\begin{aligned} \partial_t \omega^\varepsilon &= \gamma'(e^{-\Phi_\varepsilon} v_\varepsilon) e^{-\Phi_\varepsilon} (-\partial_t \Phi_\varepsilon v_\varepsilon + \partial_t v_\varepsilon) \\ &= \gamma'(e^{-\Phi_\varepsilon} v_\varepsilon) e^{-\Phi_\varepsilon} (-(b \cdot \nabla \Phi_\varepsilon + a + s_\varepsilon) v_\varepsilon + b \cdot \nabla v_\varepsilon - cu\beta'(u) + r_\varepsilon). \end{aligned}$$

Even though the derivative of the exponential function is not bounded on $\mathbb{R}$, the chain rule for Sobolev functions applies here since $\Phi_\varepsilon \in L^\infty([0, T] \times \mathbb{R}^d)$. Since

$$\nabla \omega^\varepsilon = \gamma'(e^{-\Phi_\varepsilon} v_\varepsilon) e^{-\Phi_\varepsilon} (-\nabla \Phi_\varepsilon + \nabla v_\varepsilon),$$

we have

$$\partial_t \omega^\varepsilon - b \cdot \nabla \omega^\varepsilon = \gamma'(e^{-\Phi_\varepsilon} v_\varepsilon) e^{-\Phi_\varepsilon} (-(a + s_\varepsilon) v_\varepsilon - cu\beta'(u) + r_\varepsilon). \quad (2.71)$$

Let $\phi \in C_c^\infty([0, T] \times \mathbb{R}^d)$. By integrating (2.71) over $[0, T] \times \mathbb{R}^d$ against ϕ and using integration by parts, we get

$$\begin{aligned} & - \int_0^T \int_{\mathbb{R}^d} \gamma(e^{-\Phi_\varepsilon} v_\varepsilon) \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} \gamma(v_\varepsilon^0) \phi(0, x) \, dx + \int_0^T \int_{\mathbb{R}^d} \gamma(e^{-\Phi_\varepsilon} v_\varepsilon) \operatorname{div}(b\phi) \, dx \, dt \\ & = \int_0^T \int_{\mathbb{R}^d} \gamma'(e^{-\Phi_\varepsilon} v_\varepsilon) e^{-\Phi_\varepsilon} (-(a + s_\varepsilon) v_\varepsilon - cu\beta'(u) + r_\varepsilon) \phi \, dx \, dt. \end{aligned} \quad (2.72)$$

We let $\varepsilon \rightarrow 0$ in (2.72). Since $r_\varepsilon, s_\varepsilon \rightarrow 0$ in $L_{\text{loc}}^1([0, T] \times \mathbb{R}^d)$ and $e^{-\Phi_\varepsilon}, v_\varepsilon$ are bounded uniformly in ε ,

$$\int_0^T \int_{\mathbb{R}^d} \gamma'(e^{-\Phi_\varepsilon} v_\varepsilon) e^{-\Phi_\varepsilon} s_\varepsilon v_\varepsilon \phi \, dx \, dt \rightarrow 0, \quad \int_0^T \int_{\mathbb{R}^d} \gamma'(e^{-\Phi_\varepsilon} v_\varepsilon) e^{-\Phi_\varepsilon} r_\varepsilon \phi \, dx \, dt \rightarrow 0$$

as $\varepsilon \rightarrow 0$. For all other terms, we notice that $v_\varepsilon \rightarrow v$ and $\Phi_\varepsilon \rightarrow \Phi$ almost everywhere on $[0, T] \times \mathbb{R}^d$. Thus, by the dominated convergence theorem

$$\begin{aligned} & - \int_0^T \int_{\mathbb{R}^d} \gamma(e^{-\Phi} v) \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} \gamma(v^0) \phi(0, x) \, dx + \int_0^T \int_{\mathbb{R}^d} \gamma(e^{-\Phi} v) \operatorname{div}(b\phi) \, dx \, dt \\ & + \int_0^T \int_{\mathbb{R}^d} \gamma'(e^{-\Phi} v) e^{-\Phi} (av + cu\beta'(u)) \phi \, dx \, dt = 0. \end{aligned} \quad (2.73)$$

Next, we choose a sequence of admissible functions β_n such that $\beta_n \rightarrow \text{id}_{\mathbb{R}}$, $\beta'_n \rightarrow 1$ pointwise and

$$|\beta_n(\sigma)| \leq |\sigma|, \quad |\sigma \beta'_n(\sigma)| \leq |\beta_n(\sigma)| \quad \text{for every } \sigma \in \mathbb{R}.$$

For instance, $\beta_n(\sigma) = n \tanh(\sigma/n)$ has these properties. Since $v = \beta(u)$, by choosing $\beta = \beta_n$ in (2.73) and letting $n \rightarrow \infty$, we get

$$\begin{aligned} & - \int_0^T \int_{\mathbb{R}^d} \gamma(e^{-\Phi} u) \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} \gamma(u^0) \phi(0, x) \, dx + \int_0^T \int_{\mathbb{R}^d} \gamma(e^{-\Phi} u) \operatorname{div}(b\phi) \, dx \, dt \\ & + \int_0^T \int_{\mathbb{R}^d} \gamma'(e^{-\Phi} u) e^{-\Phi} u(a + c) \phi \, dx \, dt = 0. \end{aligned} \quad (2.74)$$

This is justified by the dominated convergence theorem, using the fact that $\gamma(\sigma)$ and $\gamma'(\sigma)(1 + |\sigma|)$ are bounded on $\mathbb{R}$. To get an integrable majorant for the term with $cu\beta'(u)$, we use that $|\sigma \beta'_n(\sigma)| \leq |\beta(\sigma)|$:

$$|\gamma'(e^{-\Phi} \beta_n(u)) e^{-\Phi} cu\beta'_n(u)| \leq |\gamma'(e^{-\Phi} \beta_n(u)) e^{-\Phi} \beta_n(u) c| \leq C_\gamma |c|.$$

Since γ is an arbitrary admissible function, (2.74) implies that $e^{-\Phi} u$ is a renormalized solution of the transport equation with $(b, c + a, u^0)$. $\square$

The following is a stability result for $c \in L^1(0, T; L^\infty(\mathbb{R}^d))$. The hypotheses are the same as in Theorem 4, except that we require $b_n \in L^1(0, T; W_{\text{loc}}^{1,1}(\mathbb{R}^d))$ in order to use Lemma 6.

Corollary 3. *Assume b and c satisfy $(\dagger)$ and $(\star)$. We keep the notations and assumptions of Theorem 4, except that we replace $c_n \rightarrow 0$ by $c_n \rightarrow c$ in $L_{\text{loc}}^1([0, T] \times \mathbb{R}^d)$, and assume in addition that $b_n \in L^1(0, T; W_{\text{loc}}^{1,1}(\mathbb{R}^d))$ for every $n \in \mathbb{N}$.*

- 1) *If $u_n^0 \rightarrow u^0$ in $L(\mathbb{R}^d)$, then there exists a renormalized solution $u \in C([0, T]; L(\mathbb{R}^d))$ for (b, c, u^0) , and $u_n \rightarrow u$ in $C([0, T]; L(\mathbb{R}^d))$.*
- 2) *Part 2) of Theorem 4 remains true under its additional assumptions.*

Proof. Let $\Phi \in L^\infty([0, T] \times \mathbb{R}^d)$ be a solution to

$$\partial_t \Phi - b \cdot \nabla \Phi = -c, \quad \Phi|_{t=0} = 0.$$

By Proposition 1, such a Φ exists. By Lemma 6, $\tilde{u}_n := e^{-\Phi} u_n$ is a renormalized solution of

$$\partial_t \tilde{u}_n - b_n \cdot \nabla \tilde{u}_n + (c_n - c) \tilde{u}_n = 0, \quad \tilde{u}_n|_{t=0} = u_n^0.$$

By part 1) of Theorem 4, $\tilde{u}_n$ converges in $C([0, T]; L(\mathbb{R}^d))$ to a renormalized solution $\tilde{u}$ of (2.1) with $c = 0$ and initial data u^0 . Using again Lemma 6, we deduce that $u := e^\Phi \tilde{u}$ is a renormalized solution of

$$\partial_t u - b \cdot \nabla u + cu = 0, \quad u|_{t=0} = u^0.$$

To conclude the proof of part 1) of Corollary 3, we need to show that $e^\Phi \tilde{u}_n \rightarrow e^\Phi \tilde{u}$ in $C([0, T]; L(\mathbb{R}^d))$. Since $\Phi \in L^\infty([0, T] \times \mathbb{R}^d)$, we also have $e^\Phi \in L^\infty([0, T] \times \mathbb{R}^d)$. Thus for every compact $K \subset \mathbb{R}^d$ and $\varepsilon > 0$

$$\begin{aligned} & \sup_{t \in [0, T]} m^d \left(\left\{ x \in K : |e^{\Phi(t, x)} \tilde{u}_n(t, x) - e^{\Phi(t, x)} \tilde{u}(t, x)| \geq \varepsilon \right\} \right) \\ & \leq \sup_{t \in [0, T]} m^d \left(\left\{ x \in K : |\tilde{u}_n(t, x) - \tilde{u}(t, x)| \geq \varepsilon / \|e^\Phi\|_{L^\infty([0, T] \times \mathbb{R}^d)} \right\} \right). \end{aligned}$$

Since $\tilde{u}_n \rightarrow \tilde{u}$ in $C([0, T]; L(\mathbb{R}^d))$, the right-hand side converges to 0 as $n \rightarrow \infty$.

For part 2), by assumption, we know that (2.56) holds for some $g \in L^1((0, T))$. Then, since $c(t) \in L^\infty(\mathbb{R}^d)$ for almost every $t \in [0, T]$,

$$\sup_{n \in \mathbb{N}} \|\operatorname{div} b_n(t) + p(c_n - c)(t)\|_{L^\infty(\mathbb{R}^d)} \leq g(t) + p\|c(t)\|_{L^\infty(\mathbb{R}^d)} \quad \text{for a.e. } t \in [0, T].$$

Therefore, the sequence $\tilde{u}_n$ satisfies the assumptions of Theorem 4 part 2), so $\tilde{u}_n \rightarrow \tilde{u}$ in $C([0, T]; L^p(\mathbb{R}^d))$. Since $e^\Phi \in L^\infty([0, T] \times \mathbb{R}^d)$, multiplication by e^Φ preserves convergence in $C([0, T]; L^p(\mathbb{R}^d))$, hence also $u_n \rightarrow u$ in $C([0, T]; L^p(\mathbb{R}^d))$. $\square$

Corollary 4 (Existence & Uniqueness). *Assume $(\dagger)$ and $(\star)$. If $u^0 \in L(\mathbb{R}^d)$, then there exists a unique renormalized solution u of (2.1). In addition,*

- 1) $u \in C([0, T]; L(\mathbb{R}^d))$.
- 2) If $u^0 \in L^\infty(\mathbb{R}^d)$, then $u \in L^\infty([0, T] \times \mathbb{R}^d)$ and $u \in C([0, T]; L_{\text{loc}}^r(\mathbb{R}^d))$ for every $r < \infty$.
- 3) If $u^0 \in L^p(\mathbb{R}^d)$ for some $1 \leq p < \infty$, then $u \in C([0, T]; L^p(\mathbb{R}^d))$.

Proof. For the uniqueness, by Lemma 6, we may assume that $c = 0$. Then, uniqueness follows from Corollary 2 by choosing an injective admissible function.

For existence, let $\beta_n(\sigma) = n \tanh(\sigma/n)$ and $u_n^0 = \beta_n(u^0)$. By Proposition 1, there exists a weak solution $u_n \in L^\infty([0, T] \times \mathbb{R}^d)$ of the transport equation with (b, c, u_n^0) for every $n \in \mathbb{N}$. Since $b \in L^1(0, T; W_{\text{loc}}^{1,1}(\mathbb{R}^d))$, u_n is renormalized by Proposition 2. Furthermore, $u_n^0 \rightarrow u^0$ in $L(\mathbb{R}^d)$, so the existence of u and part 1) follow from Corollary 3.

Part 2) follows from Propositions 1 and 3.

For part 3), we choose a sequence $b_n \in L^1(0, T; C_b^1(\mathbb{R}^d))$ as in Lemma 1 and set $c_n := (c * \rho_{1/n})\eta_n$, $u_n^0 := (u^0 * \rho_{1/n})\eta_n$ for $n \in \mathbb{N}$. Then, by Step 1 of Proposition 1, there exists a sequence $u_n \in L^\infty(0, T; C_c^1(\mathbb{R}^d))$ satisfying

$$\partial_t u_n - b_n \cdot \nabla u_n + c_n u_n = 0, \quad u_n|_{t=0} = u_n^0.$$

By Proposition 2, the u_n are renormalized solutions. Since $u_n^0 \rightarrow u^0$ in $L^p(\mathbb{R}^d)$ and (2.56) is satisfied by (2.4), we conclude by part 2) of Corollary 3. $\square$

2.7 More properties of renormalized solutions

Lemma 7. *Let $b \in L^1(0, T; W_{\text{loc}}^{1,1}(\mathbb{R}^d))$, $c = 0$ and $u_1, \dots, u_m \in L([0, T] \times \mathbb{R}^d)$ be renormalized solutions of (2.1) with initial conditions $u_1^0, \dots, u_m^0 \in L(\mathbb{R}^d)$. If $F : \mathbb{R}^m \rightarrow \mathbb{R}$ is continuous, then $F(u_1, \dots, u_m)$ is a renormalized solution of (2.1) with initial condition $F(u_1^0, \dots, u_m^0)$.*

Proof. Let β be an admissible function and $v_i := \beta(u_i)$, $v_i^0 := \beta(u_i^0)$. Then, by Lemma 2, $v_{i,\varepsilon} := v_i * \rho_\varepsilon$ satisfies

$$\partial_t v_{i,\varepsilon} - b \cdot \nabla v_{i,\varepsilon} = r_{i,\varepsilon}, \quad v_{i,\varepsilon}|_{t=0} = v_{i,\varepsilon}^0$$

where $r_{i,\varepsilon} \rightarrow 0$ in $L^1(0, T; L_{\text{loc}}^1(\mathbb{R}^d))$. In the following, we write $u = (u_i)_{i=1}^m$, $v_\varepsilon = (v_{i,\varepsilon})_{i=1}^m$ etc. for the vector-valued functions.

For now, assume that $F \in C_b^1(\mathbb{R}^m)$ and let γ be admissible. Then, by the chain rule for Sobolev functions

$$\begin{aligned} \partial_t (\gamma(F(v_\varepsilon))) &= \gamma'(F(v_\varepsilon)) \nabla F(v_\varepsilon) \cdot (b \cdot \nabla v_\varepsilon + r_\varepsilon) \\ &= \nabla (\gamma(F(v_\varepsilon))) \cdot b + \gamma'(F(v_\varepsilon)) \nabla F(v_\varepsilon) \cdot r_\varepsilon. \end{aligned}$$

We integrate against a test function $\phi \in C_c^\infty([0, T] \times \mathbb{R}^d)$:

$$\begin{aligned} & - \int_0^T \int_{\mathbb{R}^d} \gamma(F(v_\varepsilon)) \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} \gamma(F(v_\varepsilon^0)) \phi(0, x) \, dx \\ & + \int_0^T \int_{\mathbb{R}^d} \gamma(F(v_\varepsilon)) (\phi \operatorname{div} b + b \cdot \nabla \phi) = \int_0^T \int_{\mathbb{R}^d} \gamma'(F(v_\varepsilon)) \nabla F(v_\varepsilon) \cdot r_\varepsilon \phi \, dx \, dt \end{aligned}$$

Next, we let $\varepsilon \rightarrow 0$ to obtain

$$- \int_0^T \int_{\mathbb{R}^d} \gamma(F(v)) \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} \gamma(F(v^0)) \phi(0, x) \, dx + \int_0^T \int_{\mathbb{R}^d} \gamma(F(v)) (\phi \operatorname{div} b + b \cdot \nabla \phi) = 0.$$

On the left-hand side of the equation, this is justified by dominated convergence, since $v_\varepsilon \rightarrow v$ almost everywhere. The right-hand side converges to 0, because $r_\varepsilon \rightarrow 0$ in $L_{\text{loc}}^1([0, T] \times \mathbb{R}^d; \mathbb{R}^d)$, and γ' and ∇F are bounded.

By letting β converge pointwise to the identity on $\mathbb{R}$, using dominated convergence and the boundedness of γ , we get

$$- \int_0^T \int_{\mathbb{R}^d} \gamma(F(u)) \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} \gamma(F(u^0)) \phi(0, x) \, dx + \int_0^T \int_{\mathbb{R}^d} \gamma(F(u)) (\phi \operatorname{div} b + b \cdot \nabla \phi) = 0.$$

This concludes the proof for $F \in C_b^1(\mathbb{R}^m)$. For $F \in C(\mathbb{R}^m)$, we choose a sequence $F_n \in C_b^1(\mathbb{R}^m)$ converging pointwise to F everywhere on $\mathbb{R}^m$. Then we set $F = F_n$ in the last equation, let $n \rightarrow \infty$, and use dominated convergence. $\square$

Lemma 7 implies that renormalized solutions of the transport equation are “additive”: If

$$b \in L^1(0, T; W_{\text{loc}}^{1,1}(\mathbb{R}^d)), \quad c \in L^1(0, T; L^\infty(\mathbb{R}^d)),$$

and u, v are renormalized solutions of (2.1) for initial conditions u^0, v^0 , then $u + v$ is a renormalized solution of (2.1) for the initial condition $u^0 + v^0$. By Lemma 7, this holds for $c = 0$, and via Lemma 6, also for $c \in L^1(0, T; L^\infty(\mathbb{R}^d))$.

3 Regular Lagrangian flows

3.1 Existence

We now prove the existence of a regular Lagrangian flow for a vector field b satisfying

$$b \in W_{\text{loc}}^{1,1}(\mathbb{R}^d; \mathbb{R}^d), \quad \operatorname{div} b \in L^\infty(\mathbb{R}^d), \quad \frac{|b|}{1+|x|} \in L^1(\mathbb{R}^d) + L^\infty(\mathbb{R}^d). \quad (3.1)$$

For simplicity, we restrict ourselves to the autonomous case. This restriction is not necessary, but it allows us to work with a flow depending on a single time variable instead of two time variables, which simplifies the proof.

Theorem 5. *Assume (3.1). Then there exists a map $X \in C(\mathbb{R}; L(\mathbb{R}^d; \mathbb{R}^d))$ with the following properties.*

1) *For almost all $x \in \mathbb{R}^d$, $X(\cdot, x) \in W_{\text{loc}}^{1,1}(\mathbb{R}; \mathbb{R}^d)$ and*

$$\partial_t X(t, x) = b(X(t, x)) \quad \text{on } \mathbb{R}, \quad X(t, x)|_{t=0} = x. \quad (3.2)$$

2) *For all $t \in \mathbb{R}$*

$$e^{-|t|D} m^d \leq X(t, \cdot)_{\#} m^d \leq e^{|t|D} m^d, \quad \text{where } D := \|\operatorname{div} b\|_{L^\infty(\mathbb{R}^d)}. \quad (3.3)$$

3) *If $u^0 \in L(\mathbb{R}^d)$, then $u^0(X(t, x))$ is the unique renormalized solution of (2.1) with $c = 0$ for the initial data u^0 .*

4) *The following group property holds: For all $s, t \in \mathbb{R}$*

$$X(s+t, x) = X(s, X(t, x)) \quad \text{a.e. on } \mathbb{R}^d. \quad (3.4)$$

In particular, Theorem 5 implies that for every $T > 0$, the restriction $X|_{[0,T] \times \mathbb{R}^d}$ is a regular Lagrangian flow for b .

In the proof of Theorem 5, we will use the following estimate of the “superlevels” of the flow, due to Crippa and DeLellis [CL08]. If X is a regular Lagrangian flow, we define

$$G_\lambda^X := \{x \in \mathbb{R}^d : |X(t, x)| \leq \lambda \text{ for all } t \in [0, T]\}. \quad (3.5)$$

In Lemma 8, we assume that the growth condition $(\star)$ holds. We write $|b|/(1+|x|) = b^1 + b^2$, where $b^1 \in L^1(0, T; L^1(\mathbb{R}^d))$ and $b^2 \in L^1(0, T; L^\infty(\mathbb{R}^d))$.

Lemma 8 ([CL08], Proposition 3.2). *Let $b \in L_{\text{loc}}^1([0, T] \times \mathbb{R}^d)$, assume $(\star)$, and let X be a regular Lagrangian flow for b . Then for every $R > 0$*

$$m^d(B_R(0) \setminus G_\lambda^X) \leq g(R, \lambda).$$

The function g only depends on $\|b^1\|_{L^1(0,T;L^1(\mathbb{R}^d))}$, $\|b^2\|_{L^1(0,T;L^\infty(\mathbb{R}^d))}$ and the compressibility constant of X . For fixed R , $\lim_{\lambda \rightarrow \infty} g(R, \lambda) = 0$.

Proof. By property F1) of a regular Lagrangian flow,

$$\begin{aligned} \int_{B_R} \sup_{t \in [0, T]} \log \left(\frac{1 + |X(t, x)|}{1 + R} \right) dx &\leq \int_{B_R} \int_0^T \frac{|\partial_t X(t, x)|}{1 + |X(t, x)|} dt dx \\ &= \int_{B_R} \int_0^T \frac{|b(t, X(t, x))|}{1 + |X(t, x)|} dt dx. \end{aligned}$$

Using the decomposition $|b|/(1+|x|) = b^1 + b^2$, we obtain

$$\begin{aligned} \int_{B_R} \sup_{t \in [0, T]} \log \left(\frac{1 + |X(t, x)|}{1 + R} \right) dx &\leq \int_0^T \int_{B_R} |b^1(t, X(t, x))| + |b^2(t, X(t, x))| dt dx \\ &\leq K \|b^1\|_{L_t^1 L_x^1} + m^d(B_R) \|b^2\|_{L_t^1 L_x^\infty}. \end{aligned}$$

Here K is the compressibility constant of X from Definition 1. By Chebyshev's inequality,

$$m^d(B_R \setminus G_\lambda^X) \leq \left(\log \left(\frac{1 + \lambda}{1 + R} \right) \right)^{-1} \left(K \|b^1\|_{L_t^1 L_x^1} + m^d(B_R) \|b^2\|_{L_t^1 L_x^\infty} \right).$$

Now, the function on the right-hand side has the required properties of g . $\square$

Proof of Theorem 5. By Corollary 4, there exists a $X \in C(\mathbb{R}; L(\mathbb{R}^d; \mathbb{R}^d))$ such that for each $1 \leq j \leq d$, X_j is the unique renormalized solution of

$$\partial_t X_j - b \cdot \nabla X_j = 0 \quad \text{on } \mathbb{R} \times \mathbb{R}^d, \quad X_j|_{t=0} = x_j. \quad (3.6)$$

Even though Corollary 4 is formulated for a finite time interval $[0, T]$, we get a solution $X \in C(\mathbb{R}; L(\mathbb{R}^d; \mathbb{R}^d))$ as follows: For every $T > 0$, there exists a unique solution $X_j^T \in C([0, T]; L(\mathbb{R}^d; \mathbb{R}^d))$ of (3.6) on $[0, T] \times \mathbb{R}^d$. Therefore, if $S < T$, then $X_j^T|_{[0, S] \times \mathbb{R}^d} = X_j^S$ almost everywhere. Hence, by inductively choosing X_j^n for $n \in \mathbb{N}$ such that $X_j^n|_{[0, m] \times \mathbb{R}^d} = X_j^m$ for $m < n$ everywhere on $[0, m] \times \mathbb{R}^d$, we can define $X_j(t, x) := X_j^n(t, x)$ for some $n > t$. The construction for negative times is the same.

Part 2). By Lemma 7, for every $u^0 \in C(\mathbb{R}^d)$, $u^0(X)$ is the unique renormalized solution of (2.1) with $c = 0$ for the initial data u^0 . Combining this with (2.33), we see that for every $u^0 \in C(\mathbb{R}^d) \cap L^1(\mathbb{R}^d)$

$$\frac{d}{dt} \int_{\mathbb{R}^d} |u^0(X(t, x))| dx = - \int_{\mathbb{R}^d} (\operatorname{div} b) |u^0(X(t, x))| dx \quad \text{for a.e. } t \in \mathbb{R}. \quad (3.7)$$

Let $D := \|\operatorname{div} b\|_{L^\infty}$. By Grönwall's inequality and (3.7), for every $t \in \mathbb{R}$

$$e^{-|t|D} \int_{\mathbb{R}^d} |u^0| dx \leq \int_{\mathbb{R}^d} |u^0(X(t, x))| dx \leq e^{|t|D} \int_{\mathbb{R}^d} |u^0| dx. \quad (3.8)$$

Let $A \subset \mathbb{R}^d$ be a measurable set. Letting $u^0 \uparrow \chi_A$ in (3.8) and the monotone convergence theorem imply (3.3).

Part 3). We now show that also for $u^0 \in L(\mathbb{R}^d)$, $u^0(X)$ is the unique renormalized solution of (2.1) with $c = 0$ and initial data u^0 . By Lusin's theorem, $C(\mathbb{R}^d) \subset L(\mathbb{R}^d)$ is dense with respect to local convergence in measure. Let $u_n^0 \in C(\mathbb{R}^d)$ be a sequence converging to u^0 in $L(\mathbb{R}^d)$. By passing to a subsequence, we may in addition assume that $u_n^0 \rightarrow u^0$ almost everywhere on $\mathbb{R}^d$. By part 1) of Theorem 4, $u_n^0(X) \rightarrow \tilde{u}$ in $L(\mathbb{R} \times \mathbb{R}^d)$, where $\tilde{u}$ is the unique renormalized solution of (2.1) with $(b, 0, u^0)$. To show that $\tilde{u} = u^0(X)$, it suffices to show that $u_n^0(X) \rightarrow u^0(X)$ a.e. on $\mathbb{R}^d$ for a.e. every $t \in \mathbb{R}$. This is the case because for a.e. $t \in \mathbb{R}$

$$m^d \left(X(t, \cdot)^{-1} \left(\left\{ x \in \mathbb{R}^d : u_n^0(x) \not\rightarrow u^0(x) \right\} \right) \right) = 0$$

by (3.3).

Part 4). Let $s \in \mathbb{R}$. By the previous paragraph, $X(s, X(t, x))$ is the unique renormalized solution of

$$\partial_t u - b \cdot \nabla u = 0, \quad u|_{t=0} = X(s, \cdot).$$

However, $X(s+t, x)$ is also a renormalized solution of this equation. So by uniqueness (Corollary 4), $X(s+t, x) = X(s, X(t, x))$ for almost every $x \in \mathbb{R}^d$.

Part 1). Let $b_n := b * \rho_{1/n}$, where $\rho_{1/n}$ is a regularizing kernel defined in (2.5). By the growth condition in (3.1), we can write $|b| = (1 + |x|)(b^1 + b^2)$, where $b^1 \in L^1(\mathbb{R}^d)$, $b^2 \in L^\infty(\mathbb{R}^d)$ and $b^1, b^2 \geq 0$. Using that $\rho_{1/n}$ is supported in $B_{C_\rho/n}(0)$ for some $C_\rho > 0$, we compute

$$\begin{aligned} \frac{|b_n|}{1 + |x|} &= \frac{1}{1 + |x|} \left| \int_{\mathbb{R}^d} b(y) \rho_{1/n}(y - x) dy \right| \\ &\leq \frac{1}{1 + |x|} \int_{\mathbb{R}^d} (b^1(y) + b^2(y))(1 + |y|) \rho_{1/n}(y - x) dy \\ &\leq \tilde{C}_\rho \int_{\mathbb{R}^d} (b^1(y) + b^2(y)) \rho_{1/n}(y - x) dy \\ &= \tilde{C}_\rho (b^1 * \rho_{1/n} + b^2 * \rho_{1/n}). \end{aligned} \tag{3.9}$$

Here $\tilde{C}_\rho$ is some constant only depending on ρ .

We claim that there exists a flow $X_n \in C^1(\mathbb{R} \times \mathbb{R}^d; \mathbb{R}^d)$ for b_n . By the local version of the Cauchy-Lipschitz theorem, for every $x^0 \in \mathbb{R}^d$, there exists a $\tilde{T} > 0$ and a solution $\gamma \in C^1([0, \tilde{T}]; \mathbb{R}^d)$ to the ODE (1.1) with b replaced by b_n . By (3.9), $b_n/(1 + |x|) \in L^\infty(\mathbb{R}^d)$, since $\|b^1 * \rho_{1/n}\|_{L^\infty} \leq \|b^1\|_{L^1} \|\rho_{1/n}\|_{L^\infty}$. Therefore the solution γ can be extended to a maximal solution on $[0, \infty)$. The same argument applies for negative times. Thus, there exists a global flow map $X_n : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$. Since $b_n \in C^1(\mathbb{R}^d)$, by Proposition A.1, $X_n(t, \cdot) \in C^1(\mathbb{R}^d; \mathbb{R}^d)$ for every $t \in \mathbb{R}$. Since b is autonomous, we also have $X(\cdot, x) \in C^1(\mathbb{R}; \mathbb{R}^d)$ for every $x \in \mathbb{R}^d$, and hence $X \in C^1(\mathbb{R} \times \mathbb{R}^d; \mathbb{R}^d)$.

Since b_n is locally Lipschitz continuous, X_n has the group property (3.4). We compute

$$0 = \frac{d}{dt} (X_n(t, X_n(-t, x))) = \partial_t X_n(t, X_n(-t, x)) - DX_n(t, X_n(-t, x)) \partial_t X_n(-t, x).$$

Since $\partial_t X_n(-t, x) = b_n(X_n(-t, x))$, we see that X_n satisfies the transport equation with $c = 0$ in the classical sense:

$$\partial_t X_n - b_n \cdot \nabla X_n = 0, \quad X_n|_{t=0} = \text{id}_{\mathbb{R}^d}$$

By the chain rule, for every $\beta \in C^1(\mathbb{R}^d)$,

$$\partial_t \beta(X_n) - b_n \cdot \nabla (\beta(X_n)) = 0, \quad \beta(X_n)|_{t=0} = \beta. \tag{3.10}$$

Thus, for every $1 \leq j \leq d$, the j -th component $X_{n,j}$ is a renormalized solution of (2.1) with $(b_n, 0, x_j)$. By part 1) of Theorem 4, for every $T > 0$, $X_n \rightarrow X$ in $C([-T, T]; L(\mathbb{R}^d; \mathbb{R}^d))$. In particular, $X_n \rightarrow X$ in $L(\mathbb{R} \times \mathbb{R}^d; \mathbb{R}^d)$.

We now show that for every $\beta \in C^1(\mathbb{R}^d)$ such that β and $|\nabla \beta|(1 + |x|)$ are bounded, the following equation holds in the weak sense:

$$\partial_t \beta(X) = b(X) \cdot \nabla \beta(X) \quad \text{on } \mathbb{R} \times \mathbb{R}^d, \quad \beta(X)|_{t=0} = \beta \quad \text{on } \mathbb{R}^d \tag{3.11}$$

For simplicity, we only prove this on $[0, \infty) \times \mathbb{R}^d$. The argument for negative times is the same. Since $\partial_t X_n = b_n(X_n)$ and $X_n \in C^1(\mathbb{R} \times \mathbb{R}^d; \mathbb{R}^d)$, (3.11) holds with X and b replaced by X_n and b_n . Thus, for every $\phi \in C_c^\infty([0, \infty) \times \mathbb{R}^d)$

$$- \int_0^\infty \int_{\mathbb{R}^d} \beta(X_n(t, x)) \partial_t \phi dx dt - \int_{\mathbb{R}^d} \phi(0) \beta dx = \int_0^\infty \int_{\mathbb{R}^d} b_n(X_n(t, x)) \cdot \nabla \beta(X_n(t, x)) \phi dx dt.$$

Our goal is to prove

$$-\int_0^\infty \int_{\mathbb{R}^d} \beta(X(t, x)) \partial_t \phi \, dx \, dt - \int_{\mathbb{R}^d} \phi(0) \beta \, dx = \int_0^\infty \int_{\mathbb{R}^d} b(X(t, x)) \cdot \nabla \beta(X(t, x)) \phi \, dx \, dt. \quad (3.12)$$

Since $X_n \rightarrow X$ in $L(\mathbb{R} \times \mathbb{R}^d)$ and β is bounded and Lipschitz continuous,

$$\int_0^\infty \int_{\mathbb{R}^d} \beta(X_n(t, x)) \partial_t \phi \, dx \, dt \xrightarrow{n \rightarrow \infty} \int_0^\infty \int_{\mathbb{R}^d} \beta(X(t, x)) \partial_t \phi \, dx \, dt.$$

Thus, it remains to show that

$$\int_0^\infty \int_{\mathbb{R}^d} b_n(X_n(t, x)) \cdot \nabla \beta(X_n(t, x)) \phi \, dx \, dt \xrightarrow{n \rightarrow \infty} \int_0^\infty \int_{\mathbb{R}^d} b(X(t, x)) \cdot \nabla \beta(X(t, x)) \phi \, dx \, dt. \quad (3.13)$$

By (3.10), X_n is a renormalized solution of the transport equation with $(b_n, 0, \text{id}_{\mathbb{R}^d})$. Since $\text{div } b_n \in L^\infty(\mathbb{R}^d)$, by the computation in the proof of part 2), the measure $X_n(t, \cdot)_{\#} m^d$ is absolutely continuous with respect to m^d and

$$e^{-|t|D_n} m^d \leq X_n(t, \cdot)_{\#} m^d \leq e^{|t|D_n} m^d, \quad \text{where } D_n := \|\text{div } b_n\|_{L^\infty(\mathbb{R}^d)}. \quad (3.14)$$

Let $h_n(t)$ be the density of $X_n(t, \cdot)_{\#} m^d$, and let $T > 0$ such that $\text{supp } \phi \subset [0, T) \times \mathbb{R}^d$. Since

$$\|\text{div } b_n\|_{L^\infty} = \|(\text{div } b) * \rho_{1/n}\|_{L^\infty} \leq \|\text{div } b\|_{L^\infty},$$

$\{h_n\}_{n \in \mathbb{N}}$ is bounded in $L^\infty([0, T] \times \mathbb{R}^d)$ by (3.14). By (3.3), also $X(t, \cdot)_{\#}$ is absolutely continuous with respect to m^d , and we denote the density of $X(t, \cdot)_{\#}$ by $h(t, \cdot)$.

By changing variables and using the group property, (3.13) is equivalent to

$$\int_0^\infty \int_{\mathbb{R}^d} b_n \cdot \nabla \beta \phi(t, X_n(-t, x)) h_n \, dx \, dt \xrightarrow{n \rightarrow \infty} \int_0^\infty \int_{\mathbb{R}^d} b \cdot \nabla \beta \phi(t, X(-t, x)) h \, dx \, dt. \quad (3.15)$$

Since ϕ is bounded and h_n is bounded in $L^\infty([0, T] \times \mathbb{R}^d)$, by the Banach-Alaoglu theorem, there exists a subsequence $\phi(t, X_{k_n}(-t, x)) h_{k_n}$ of $\phi(t, X_n(-t, x)) h_n$ converging weak-* in $L^\infty([0, T] \times \mathbb{R}^d)$. To identify the weak-* limit, we test against $\psi \in C_c^\infty([0, T] \times \mathbb{R}^d)$.

$$\begin{aligned} \int_0^T \int_{\mathbb{R}^d} \psi \phi(t, X_{k_n}(-t, x)) h_{k_n} \, dx \, dt &= \int_0^T \int_{\mathbb{R}^d} \psi(t, X_{k_n}(t, x)) \phi \, dx \, dt \\ &\xrightarrow{n \rightarrow \infty} \int_0^T \int_{\mathbb{R}^d} \psi(t, X(t, x)) \phi \, dx \, dt = \int_0^T \int_{\mathbb{R}^d} \psi \phi(t, X(-t, x)) h \, dx \, dt. \end{aligned}$$

The convergence follows from $X_{k_n} \rightarrow X$ in $L(\mathbb{R} \times \mathbb{R}^d; \mathbb{R}^d)$, the compact support of ϕ , and because ψ is bounded and Lipschitz. Since this argument also applies to any subsequence of $\phi(t, X_n(-t, x)) h_n$, we conclude

$$\phi(t, X_n(-t, x)) h_n \rightharpoonup^* \phi(t, X(-t, x)) h \quad \text{in } L^\infty([0, T] \times \mathbb{R}^d). \quad (3.16)$$

Let $\lambda > 0$. We write

$$\begin{aligned} &\left| \int_0^\infty \int_{\mathbb{R}^d} b_n \cdot \nabla \beta \phi(t, X_n(-t, x)) h_n \, dx \, dt - \int_0^\infty \int_{\mathbb{R}^d} b \cdot \nabla \beta \phi(t, X(-t, x)) h \, dx \, dt \right| \\ &\leq \left| \int_0^\infty \int_{B_\lambda} b_n \cdot \nabla \beta \phi(t, X_n(-t, x)) h_n - b \cdot \nabla \beta \phi(t, X(-t, x)) h \, dx \, dt \right| + r_n(\lambda) + r(\lambda), \end{aligned} \quad (3.17)$$

where

$$r_n(\lambda) := \left| \int_0^\infty \int_{\mathbb{R}^d \setminus B_\lambda} b_n \cdot \nabla \beta \phi(t, X_n(-t, x)) h_n dx dt \right| \quad \text{for } n \in \mathbb{N},$$

$$r(\lambda) := \left| \int_0^\infty \int_{\mathbb{R}^d \setminus B_\lambda} b \cdot \nabla \beta \phi(t, X(-t, x)) h dx dt \right|.$$

To conclude (3.15), we show that the right-hand side of (3.17) can be made arbitrarily small by choosing λ appropriately and n large enough. By (3.16) and since $b_n \rightarrow b$ in $L^1_{\text{loc}}(\mathbb{R}^d)$, for every fixed $\lambda > 0$, the first term in the second line of (3.17) converges to 0 as $n \rightarrow \infty$.

Since $|\nabla \beta|(1 + |x|)$ is bounded and $(\star)$ holds, $b \cdot \nabla \beta \in L^1(\mathbb{R}^d) + L^\infty(\mathbb{R}^d)$. By (3.3) and since ϕ and h are bounded on $[0, T] \times \mathbb{R}^d$,

$$\phi(t, X(-t, x))h \in L^1(0, T; L^1) \cap L^1(0, T; L^\infty).$$

Thus $r(\lambda) \rightarrow 0$ as $\lambda \rightarrow \infty$. We finish the proof of (3.15) by showing that $r_n(\lambda) \rightarrow 0$ uniformly in n as $\lambda \rightarrow \infty$. By (3.9),

$$\begin{aligned} r_n(\lambda) &\leq C_{\rho, \beta} \int_0^\infty \int_{\mathbb{R}^d \setminus B_\lambda} (b^1 * \rho_{1/n} + b^2 * \rho_{1/n}) |\phi(t, X_n(-t, x))| h_n dx dt \\ &\leq C_{\rho, \beta} \|\phi\|_{L^\infty_{t,x}} \|h_n\|_{L^\infty([0, T] \times \mathbb{R}^d)} T \int_{\mathbb{R}^d \setminus B_\lambda} b^1 * \rho_{1/n} dx \\ &\quad + C_{\rho, \beta} \|b^2\|_{L^\infty(\mathbb{R}^d)} \int_0^T \int_{X_n(-t, \mathbb{R}^d \setminus B_\lambda)} |\phi| dx dt. \end{aligned}$$

In the last line, we changed variables $x' = X(-t, x)$. Since $\text{supp } \rho_{1/n} \subset B_{C_\rho/n}$ for some $C_\rho > 0$, and since h_n is bounded in $L^\infty([0, T] \times \mathbb{R}^d)$, the first term on the right-hand side converges to 0 as $\lambda \rightarrow \infty$, uniformly in n . Let $R > 0$ such that $\text{supp } \phi \subset [0, T] \times B_R(0)$. By the group property, $X_n(-t, x) = X_n(t, \cdot)^{-1}(x)$. Therefore

$$\begin{aligned} \int_0^T \int_{X_n(-t, \mathbb{R}^d \setminus B_\lambda)} |\phi| dx dt &\leq \|\phi\|_{L^\infty_{t,x}} \int_0^T m^d(\{x \in B_R(0) : |X_n(t, x)| \geq \lambda\}) dt \\ &\leq \|\phi\|_{L^\infty_{t,x}} T m^d(B_R(0) \setminus G_\lambda^{X_n}). \end{aligned}$$

Here $G_\lambda^{X_n}$ is the set defined in (3.5). We now want to apply Lemma 8. By (3.9),

$$\frac{|b_n|}{1 + |x|} \leq \tilde{C}_\rho (b^1 * \rho_{1/n} + b^2 * \rho_{1/n}),$$

and $b^1 * \rho_{1/n} \in L^1(\mathbb{R}^d)$, $b^2 * \rho_{1/n} \in L^\infty(\mathbb{R}^d)$. Thus, by Lemma 8,

$$m^d(B_R(0) \setminus G_\lambda^{X_n}) \leq g_n(R, \lambda)$$

where the function g_n depends only on the compressibility constant K_n of X_n as well as the norms $\|b^1 * \rho_{1/n}\|_{L^1(0, T; L^1)}$ and $\|b^2 * \rho_{1/n}\|_{L^1(0, T; L^\infty)}$. Since $\{K_n\}_{n \in \mathbb{N}}$ is bounded by (3.14) and

$$\|b^1 * \rho_{1/n}\|_{L^1(0, T; L^1)} \leq \|b^1\|_{L^1(0, T; L^1)}, \quad \|b^2 * \rho_{1/n}\|_{L^1(0, T; L^\infty)} \leq \|b^2\|_{L^1(0, T; L^\infty)},$$

we conclude that $g_n(R, \lambda) \rightarrow 0$ as $\lambda \rightarrow \infty$, uniformly in n . Therefore $r_n(\lambda) \rightarrow 0$ as $\lambda \rightarrow \infty$, uniformly in n .

We now show that for almost every $x \in \mathbb{R}^d$, $X(\cdot, x) \in W^{1,1}_{\text{loc}}(\mathbb{R})$ and $\partial_t X(t, x) = b(X(t, x))$. Since $b \cdot \nabla \beta \in L^1(\mathbb{R}^d) + L^\infty(\mathbb{R}^d)$, by (3.3)

$$b(X) \cdot \nabla \beta(X) \in L^1(-T, T; L^1(\mathbb{R}^d)) + L^1(-T, T; L^\infty(\mathbb{R}^d))$$

for every $T > 0$. In particular, $b(X) \cdot \nabla \beta(X) \in L^1_{\text{loc}}(\mathbb{R} \times \mathbb{R}^d)$.

Thus

$$b(X(t, x)) \cdot \nabla \beta(X(t, x)) \in L^1_{\text{loc}}(\mathbb{R}) \quad \text{for a.e. } x \in \mathbb{R}^d. \quad (3.18)$$

This is true for the “loc”-spaces, because both $\mathbb{R}$ and $\mathbb{R}^d$ can be covered by a countable number of compact sets.

Let $\psi \in C_c^\infty([0, \infty))$ and $\xi \in C_c^\infty(\mathbb{R}^d)$. We choose $\phi(t, x) = \xi(x)\psi(t)$ in (3.12).

$$-\int_0^\infty \int_{\mathbb{R}^d} \beta(X) \xi \partial_t \psi \, dx \, dt - \psi(0) \int_{\mathbb{R}^d} \beta \xi \, dx = \int_0^\infty \int_{\mathbb{R}^d} b(X) \cdot \nabla \beta(X) \xi \psi \, dx \, dt.$$

Since ξ is arbitrary, for a.e. $x \in \mathbb{R}^d$,

$$-\int_0^\infty \beta(X(t, x)) \partial_t \psi(t) \, dt - \psi(0) \beta(x) = \int_0^\infty b(X(t, x)) \cdot \nabla \beta(X(t, x)) \psi(t) \, dt. \quad (3.19)$$

Fix $x \in \mathbb{R}^d$ such that both (3.18) and (3.19) hold. Then $\beta(X(t, x)) \in W^{1,1}_{\text{loc}}(\mathbb{R})$. We choose $\beta = \tanh(x_i)$, where x_i is the i -th coordinate function. Thus, $\tanh(X_i(t, x)) \in C(\mathbb{R})$ after modifying $X_i(\cdot, x)$ on a m^1 -negligible set. It is possible to achieve this via a modification of $X_i(\cdot, x)$, since $\tanh' > 0$ on $\mathbb{R}$. Since $\tanh$ is strictly increasing, this implies $X_i(t, x) \in C(\mathbb{R})$.

Let $T > 0$ and $1 \leq i \leq d$. By the time-continuity of X just shown, $X([-T, T], x) \subset \mathbb{R}^d$ is compact, so we may choose β such that $\beta(X(t, x)) = X_i(t, x)$ for all $t \in [-T, T]$. Then, (3.19) shows that $X(\cdot, x) \in W^{1,1}_{\text{loc}}(\mathbb{R}; \mathbb{R}^d)$ and $\partial_t X(t, x) = b(X(t, x))$. $\square$

3.2 Uniqueness

Instead of working with regular Lagrangian flows, DiPerna and Lions [DL89] use a different characterization of the flow map. For time-dependent vector fields, they introduce a map $X : [0, T]^2 \times \mathbb{R}^d \rightarrow \mathbb{R}^d$, where (informally speaking) $X(s, t, x)$ is the position at time s of a particle that was at x at time t . This flow belongs to $C([0, T]^2; L(\mathbb{R}^d; \mathbb{R}^d))$ and is uniquely determined by the following three properties:

- 1) Let $\beta \in C^1(\mathbb{R}^d)$ such that β and $|\nabla \beta|(1 + |x|)$ are bounded. Then for every $t \in [0, T]$, the following equation holds in the weak sense:

$$\partial_s \beta(X) = b(s, X) \cdot \nabla \beta(X) \quad \text{on } (0, T) \times \mathbb{R}^d, \quad \beta(X)|_{s=t} = \beta \quad \text{on } \mathbb{R}^d$$

- 2) There exists a $K > 0$ such that $K^{-1}m^d \leq X(t, \cdot)_\# m^d \leq Km^d$ for all $t \in [0, T]$.
- 3) For all $s, t, r \in [0, T]$,

$$X(s, t, x) = X(s, r, X(r, t, x)) \quad \text{for a.e. } x \in \mathbb{R}^d. \quad (3.20)$$

The third property is the analogue of the group property (3.4) if b is time-dependent. However, subsequently it has been shown that (3.20) and the lower bound on the pushforward measure $X(t, \cdot)_\# m^d$ are not required for uniqueness of the flow. This is a consequence of more general principles connecting uniqueness for the continuity equation with uniqueness of regular Lagrangian flows.

The continuity equation for b is the following PDE. It is a special case of the transport equation (2.1) with $c = \text{div } b$ and b replaced by $-b$.

$$\partial_t u + \text{div}(bu) = 0 \quad \text{on } (0, T) \times \mathbb{R}^d, \quad u|_{t=0} = u^0 \quad \text{on } \mathbb{R}^d \quad (3.21)$$

These principles were introduced by Ambrosio in [Amb04]. A comprehensive treatment of them can be found in [AC08], p. 11–19. Below, we use the connection with the continuity equation to give a direct proof of the uniqueness of regular Lagrangian flows.

Lemma 9. Assume that $(\star)$ holds and let X be a regular Lagrangian flow for b . Let $u^0 \in C_c^\infty(\mathbb{R}^d)$ and $g(t, \cdot)$ be the density of $X(t, \cdot)_\#(u^0 dm^d)$ with respect to m^d . Then g is a weak solution of the continuity equation (3.21) with initial data u^0 .

By the second property of a regular Lagrangian flow, the measure $X(t, \cdot)_\#(u^0 dm^d)$ is absolutely continuous with respect to m^d and its density belongs to $L^\infty([0, T] \times \mathbb{R}^d)$. Hence the weak formulation of the continuity equation is well defined with $u = g$.

Proof. Let $\phi \in C_c^\infty([0, T] \times \mathbb{R}^d)$. Since $X(\cdot, x) \in W^{1,1}(0, T; \mathbb{R}^d)$ for almost every $x \in \mathbb{R}^d$, by the chain rule for Sobolev functions

$$\frac{d}{dt}(\phi(t, X(t, x))) = \partial_t \phi(t, X(t, x)) + b(t, X(t, x)) \cdot \nabla \phi(t, X(t, x)) \quad \text{for a.e. } x \in \mathbb{R}^d.$$

Integrating over $[0, T]$ yields

$$-\phi(0, x) = \int_0^T \partial_t \phi(t, X(t, x)) dt + \int_0^T b(t, X(t, x)) \cdot \nabla \phi(t, X(t, x)) dt \quad \text{for a.e. } x \in \mathbb{R}^d. \quad (3.22)$$

Since $b \cdot \nabla \phi \in L^1(0, T; L^1 + L^\infty)$, by property F2) of a regular Lagrangian flow, also

$$b(t, X(t, x)) \cdot \nabla \phi(t, X(t, x)) \in L^1(0, T; L^1 + L^\infty).$$

Thus, we may integrate (3.22) over $\mathbb{R}^d$ after multiplying with u^0 ,

$$\begin{aligned} - \int_{\mathbb{R}^d} u^0 \phi(0, x) dx &= \int_0^T \int_{\mathbb{R}^d} u^0 \partial_t \phi(t, X(t, x)) dx dt \\ &\quad + \int_0^T \int_{\mathbb{R}^d} u^0 b(t, X(t, x)) \cdot \nabla \phi(t, X(t, x)) dx dt \end{aligned}$$

By definition of g , this equation is the weak form of (3.21) with $u = g$. $\square$

Theorem 6. If $b \in L^1(0, T; W_{\text{loc}}^{1,1}(\mathbb{R}^d; \mathbb{R}^d))$, $\text{div } b \in L^1(0, T; L^\infty(\mathbb{R}^d))$ and $(\star)$ holds, then the regular Lagrangian flow for b is unique.

Proof. Let $X, Y : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ be two regular Lagrangian flows for b . Let $u^0 \in C_c^\infty(\mathbb{R}^d)$ and $g_X, g_Y \in L^\infty([0, T] \times \mathbb{R}^d)$ be the densities of $X(t, \cdot)_\#(u^0 dm^d)$ and $Y(t, \cdot)_\#(u^0 dm^d)$. By Lemma 9, both g_X and g_Y are weak solutions of the continuity equation (3.21) for the initial data u^0 . Since weak solutions of (3.21) are unique by Corollary 2, we have $g_X = g_Y$ a.e. on $[0, T] \times \mathbb{R}^d$.

By Proposition 3, $g_X, g_Y \in C([0, T]; L_{\text{loc}}^r(\mathbb{R}^d))$ for every $r < \infty$. Thus $g_X(t, \cdot) = g_Y(t, \cdot)$ a.e. on $\mathbb{R}^d$ for every $t \in [0, T]$. Hence, for every $t \in [0, T]$

$$X(t, \cdot)_\#(u^0 dm^d) = Y(t, \cdot)_\#(u^0 dm^d)$$

as measures on $\mathbb{R}^d$. Testing against $\xi \in C_c^\infty(\mathbb{R}^d)$ yields

$$\int_{\mathbb{R}^d} u^0 \xi(X(t, x)) dx = \int_{\mathbb{R}^d} u^0 \xi(Y(t, x)) dx.$$

Since u^0 is arbitrary, $\xi(X(t, x)) = \xi(Y(t, x))$ a.e. on $\mathbb{R}^d$. Since ξ is arbitrary, we conclude $X(t, x) = Y(t, x)$ a.e. on $\mathbb{R}^d$ for every $t \in [0, T]$. Because $X(\cdot, x)$ and $Y(\cdot, x)$ are continuous on $[0, T]$ for a.e. $x \in \mathbb{R}^d$, this implies $X(\cdot, x) = Y(\cdot, x)$ on $[0, T]$ for a.e. $x \in \mathbb{R}^d$. $\square$

4 Counterexamples

4.1 The growth condition

To illustrate the necessity of the growth condition $(\star)$ for uniqueness of solutions to the transport equation, consider the following example in two dimensions. Let

$$b(x, y) = \begin{cases} \begin{pmatrix} x^2 \\ -2xy \end{pmatrix} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Since b is smooth, $b \in W_{\text{loc}}^{1,1}(\mathbb{R}^2)$. Furthermore, $\text{div } b = 0$. Note that $u \in C^1([0, \infty) \times \mathbb{R}^2)$ is a solution of the transport equation (2.1) with $c = 0$ if and only if it is constant along all characteristics of $-b$.

The flow map of $-b$ is

$$X(t, x, y) = \begin{cases} \begin{pmatrix} 1/(x^{-1} + t) \\ y(1 + tx)^2 \end{pmatrix} & x > 0 \\ \begin{pmatrix} x \\ y \end{pmatrix} & x \leq 0 \end{cases}$$

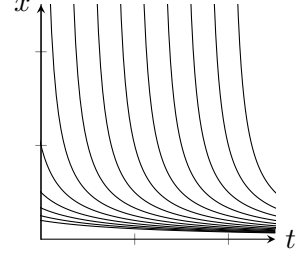


Figure 1: The characteristic curves of $-b$, projected to the x - t -plane.

Every characteristic curve intersecting the plane $t = 0$ stays in the region $\{(t, x, y) \in [0, \infty) \times \mathbb{R}^2 : x < 1/t\}$ for all $t \geq 0$. Hence, the initial condition at $t = 0$ does not affect the region $\{x \geq 1/t\}$, so there are infinitely many solutions to the transport equation. For instance, let $h \in C^1(\mathbb{R})$ such that $\text{supp } h \subset (0, \infty)$. Then

$$u(t, x, y) = \begin{cases} h(t - x^{-1}) & x \geq 1/t \\ 0 & x < 1/t \end{cases}$$

is a solution in $C^1([0, \infty) \times \mathbb{R}^2)$.

4.2 Integrability of the derivatives of the vector field

We will construct an autonomous vector field b on $\mathbb{R}^2$ such that

$$\begin{aligned} b &\in W_{\text{loc}}^{s,1}(\mathbb{R}^2) \quad \text{for every } s \in (0, 1), \\ b &\in L^p(\mathbb{R}^2) + L^\infty(\mathbb{R}^2) \quad \text{for every } 1 \leq p < 2, \quad \text{div } b = 0 \text{ on } \mathbb{R}^2 \text{ as a distribution,} \end{aligned} \tag{4.1}$$

and there are two distinct regular Lagrangian flows for b . Here, $W^{s,1}$ is a fractional Sobolev space which will be explained later. This example is from section IV.2 in [DL89], except that DiPerna and Lions define b slightly differently for $|x| \geq |y|$.

We define b as follows:

$$b(x, y) := \begin{pmatrix} -\frac{x}{y^2} \text{sgn}(y) \\ -\frac{1}{|y|} \end{pmatrix} \quad \text{if } |x| < |y|, \quad b(x, y) = 0 \quad \text{if } |x| \geq |y|. \tag{4.2}$$

Note that b comes from the Hamiltonian

$$H(x, y) = \begin{cases} -\frac{x}{|y|} & |x| < |y| \\ -1 & x \geq |y| \\ 1 & x \leq -|y| \end{cases} \quad \text{i.e.} \quad b_x = -\frac{\partial H}{\partial y}, \quad b_y = \frac{\partial H}{\partial x}.$$

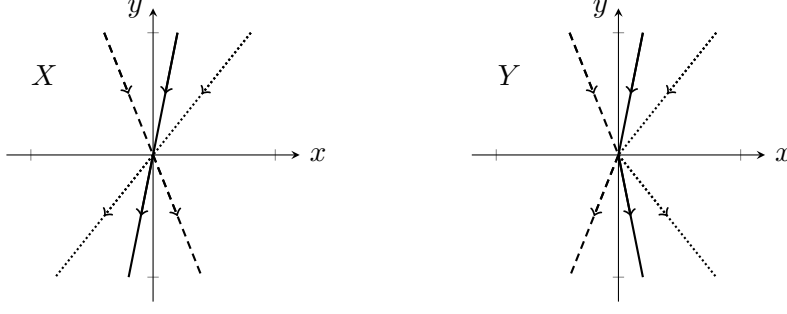


Figure 2: Flow lines of X and Y .

Therefore, $\operatorname{div} b = 0$ as a distribution on $\mathbb{R}^2$.

Before verifying the other properties listed in (4.1), we construct the regular Lagrangian flows. For $(x, y) \in \mathbb{R} \times [0, \infty)$, define

$$Z(t, x, y) := \begin{pmatrix} \frac{x}{y} \sqrt{y^2 - 2t} \\ \sqrt{y^2 - 2t} \end{pmatrix} \quad \text{if } |x| < y \text{ and } t \leq y^2/2. \quad (4.3)$$

If $|x| \geq y$, we define $Z(t, x, y) = (x, y)$ for all $t \in \mathbb{R}$. We do not define Z for $|x| < y$ and $t > y^2/2$. Note that

$$\partial_t Z(t, x, y) = b(Z(t, x, y)) \quad \text{if } t < y^2/2.$$

Hence, for each $(x, y) \in \mathbb{R} \times [0, \infty)$, the curve $t \mapsto Z(t, x, y)$ is the trajectory along b until the origin is reached. Moreover,

$$\det DZ(t, x, y) = 1 \quad \text{if } y > 0, |x| < y, t < y^2/2. \quad (4.4)$$

Beyond the origin, we extend the curves in the following two ways: We define the x -components of two flows $\tilde{X}, \tilde{Y}$ for $(x, y) \in \mathbb{R} \times [0, \infty)$ by

$$\begin{aligned} \tilde{X}_x(t, x, y) &= \begin{cases} Z_x(t, x, y) & t \leq y^2/2 \\ -Z_x(y^2 - t, x, y) & t > y^2/2 \end{cases} \\ \tilde{Y}_x(t, x, y) &= \begin{cases} Z_x(t, x, y) & t \leq y^2/2 \\ Z_x(y^2 - t, x, y) & t > y^2/2 \end{cases} \end{aligned}$$

The y -components of $\tilde{X}$ and $\tilde{Y}$ are defined identically:

$$\tilde{X}_y(t, x, y) = \tilde{Y}_y(t, x, y) = \begin{cases} Z_y(t, x, y) & t \leq y^2/2 \\ -Z_y(y^2 - t, x, y) & t > y^2/2 \end{cases}$$

Next, we extend $\tilde{X}$ and $\tilde{Y}$ to $y < 0$ using the symmetry of b . Because $b(x, y) = b(-x, -y)$, we set

$$X(t, x, y) := \begin{cases} \tilde{X}(t, x, y) & y \geq 0 \\ -\tilde{X}(-t, -x, -y) & y < 0 \end{cases}.$$

We also extend $\tilde{Y}$ to $y < 0$ by the same formula and call the resulting flow Y .

Let $(x, y) \in \mathbb{R}^2$ such that $|x| < |y|$. Then, for all but one $t \in \mathbb{R}$, $X(\cdot, x, y)$ and $Y(\cdot, x, y)$ are differentiable and solve the ODE (1.1) with initial data (x, y) . Due to (4.4), X and Y are measure-preserving.

We now check the properties in (4.1), beginning with $b \in L^p(\mathbb{R}^2) + L^\infty(\mathbb{R}^2)$ for every $1 \leq p < 2$. Since b is bounded on $\mathbb{R}^2 \setminus B_1$, it suffices to show that $b \in L^p(B_1)$ for every $1 \leq p < 2$. First, consider $b_x = -x \operatorname{sgn}(y)/|y|^2$. Since $|b_x|$ is invariant under negation of y , it suffices to show that $b_x \in L^p(B_1 \cap \{y > 0\})$. Because $b = 0$ if $|x| > |y|$,

$$\begin{aligned} \|b_x\|_{L^p(B_1 \cap \{y > 0\})}^p &\leq \int_0^1 \int_{-y}^y |b_x|^p dx dy = \int_0^1 \int_{-y}^y \frac{|x|^p}{y^{2p}} dx dy \\ &\leq \int_0^1 \int_{-y}^y \frac{1}{y^p} dx dy = 2 \int_0^1 \frac{1}{y^{p-1}} dy < \infty. \end{aligned}$$

The computation for $b_y = -1/|y|$ is similar.

Next, we show that $b \in W_{\text{loc}}^{s,1}(\mathbb{R}^2)$ for every $0 < s < 1$. For an open set $\Omega \subset \mathbb{R}^d$, the fractional Sobolev space $W^{s,1}(\Omega)$ is defined as (see [DPV12], p. 524)

$$W^{s,1}(\Omega) = \left\{ u \in L^1(\Omega) : \frac{|u(z) - u(w)|}{|z - w|^{d+s}} \in L^1(\Omega \times \Omega) \right\}.$$

We also write

$$[u]_{W^{s,1}(\Omega)} := \left\| \frac{|u(z) - u(w)|}{|z - w|^{d+s}} \right\|_{L^1(\Omega \times \Omega)} \quad \text{and} \quad \|u\|_{W^{s,1}(\Omega)} := \|u\|_{L^1(\Omega)} + [u]_{W^{s,1}(\Omega)}.$$

The notation $b \in W_{\text{loc}}^{s,1}(\mathbb{R}^2)$ means that $b \in W^{s,1}(\Omega)$ for every bounded, open set $\Omega \subset \mathbb{R}^2$. Because $b \in L^1(\mathbb{R}^2) + L^\infty(\mathbb{R}^2)$, we have $b \in L_{\text{loc}}^1(\mathbb{R}^2)$, so we only need to show that $[u]_{W^{s,1}(B_R)} < \infty$ for every $R > 0$.

First, we show that b is of bounded variation on $B_4 \setminus B_1$. For simplicity, we will only consider $(B_4 \setminus B_1) \cap \{y > 0\}$. In fact, since b is symmetric around $y = 0$ up to signs,

$$\|b\|_{BV(B_4 \setminus B_1)} = 2\|b\|_{BV((B_4 \setminus B_1) \cap \{y > 0\})}.$$

The total variation of b splits into the following two parts:

$$\|b\|_{BV((B_4 \setminus B_1) \cap \{y > 0\})} = \|b\|_{BV((B_4 \setminus B_1) \cap \{|x| < |y|, y > 0\})} + 2\sqrt{2} \int_{1/\sqrt{2}}^{4/\sqrt{2}} |b(x, x)| dx \quad (4.5)$$

The second term comes from the discontinuities at the lines $x = y$ and $x = -y$. Since

$$b \in C_b^1((B_4 \setminus B_1) \cap \{|x| < |y|, y > 0\}; \mathbb{R}^2),$$

the first term in (4.5) is finite. The integral in (4.5) is

$$\int_{1/\sqrt{2}}^{4/\sqrt{2}} |b(x, x)| dx = \sqrt{2} \int_{1/\sqrt{2}}^{4/\sqrt{2}} \frac{1}{x} dx < \infty.$$

Therefore $b \in BV(B_4 \setminus B_1)$.

For every bounded open set $\Omega \subset \mathbb{R}^d$ with Lipschitz boundary, there is a continuous linear embedding $BV(\Omega) \hookrightarrow W^{s,1}(\Omega)$ for all $0 < s < 1$, see Proposition A.2. Therefore also $b \in W^{s,1}(B_4 \setminus B_1)$.

For $k \in \mathbb{Z}$, let $A_k := \{z \in \mathbb{R}^2 : 2^k < |z| < 2^{k+1}\}$ and define for $k, \ell \in \mathbb{Z}$

$$I_{k,\ell} := \int_{A_k} \int_{A_\ell} \frac{|b(z) - b(w)|}{|z - w|^{2+s}} dw dz. \quad (4.6)$$

Then, for all $m \in \mathbb{Z}$

$$[b]_{W^{s,1}(B_{2^m})} = \sum_{k < m} \sum_{\ell < m} I_{k,\ell}. \quad (4.7)$$

We will now estimate $I_{k,\ell}$. If $k = \ell$, we have

$$I_{k,k} = [b]_{W^{s,1}(A_k)} = \int_{A_k} \int_{A_k} \frac{|b(z) - b(w)|}{|z - w|^{2+s}} dw dz.$$

Using that b is homogeneous of degree -1 , we can express this in terms of $[b]_{W^{s,1}(A_0)}$. By changing variables $\zeta = 2^{-k}z$ and $\eta = 2^{-k}w$,

$$I_{k,k} = \int_{A_0} \int_{A_0} \frac{2^{-k}|b(\zeta) - b(\eta)|}{2^{k(2+s)}|\zeta - \eta|^{2+s}} 2^{4k} d\zeta d\eta = 2^{k(1-s)}[b]_{W^{s,1}(A_0)}.$$

By the above discussion, we know that $b \in W^{s,1}(A_0)$. Moreover, the sum $\sum_{k < m} 2^{k(1-s)}$ is finite since $s < 1$. Therefore, the contribution of the diagonal terms $I_{k,k}$ to (4.7) is finite.

Next, we look at the case $\ell = k + 1$. We note that $I_{k,k+1} \leq [b]_{W^{s,1}(A_k \cup A_{k+1})}$. Again by homogeneity of b ,

$$[b]_{W^{s,1}(A_k \cup A_{k+1})} = 2^{k(1-s)}[b]_{W^{s,1}(A_0 \cup A_1)}.$$

Because $A_0 \cup A_1 = B_4 \setminus B_1$ and $b \in W^{s,1}(B_4 \setminus B_1)$, the contribution of the terms $I_{k,k+1}$ to (4.7) is finite. By symmetry, also the contribution of the terms $I_{k,k-1}$ is finite.

The remaining case is when $|k - \ell| \geq 2$. Assume that $k < \ell$. If $z \in A_k$ and $w \in A_\ell$, we have

$$|z - w| \geq 2^{\ell-1}, \quad |b(z)| \leq 2^{-k+1}, \quad |b(w)| \leq 2^{-\ell+1} \leq 2^{-k+1}.$$

Therefore

$$\begin{aligned} I_{k,\ell} &= \int_{A_k} \int_{A_\ell} \frac{|b(z) - b(w)|}{|z - w|^{2+s}} dw dz \leq \int_{A_k} \int_{A_\ell} \frac{2^{-k+2}}{2^{(\ell-1)(2+s)}} dw dz \\ &\leq C_s 2^{-\ell(2+s)-k} \cdot 2^{2k} \cdot 2^{2\ell} = C_s 2^{-\ell s+k}. \end{aligned}$$

Since $s < 1$,

$$\sum_{\ell < m} \sum_{k \leq \ell-2} 2^{-\ell s+k} = \sum_{\ell < m} 2^{-\ell s+\ell-1} < \infty,$$

so the contribution of the terms $I_{k,\ell}$ ($k \leq \ell - 2$) to (4.7) is finite. By symmetry, also the contribution from $k \geq \ell + 2$ is finite. This concludes the proof that $b \in W_{\text{loc}}^{s,1}(\mathbb{R}^2)$.

4.3 Integrability of the solution

Consider the transport equation with $c = 0$.

$$\partial_t u - b \cdot \nabla u = 0, \quad u|_{t=0} = u^0 \tag{4.8}$$

For simplicity, we set $T = 1$ in this section. Let $1 \leq p, q, r \leq \infty$ such that $1/p + 1/q = 1$ and

$$b \in L^1(0, 1; W_{\text{loc}}^{1,r}(\mathbb{R}^d; \mathbb{R}^d) \cap L_{\text{loc}}^q), \quad \text{div } b \in L^1(0, 1; L^\infty(\mathbb{R}^d))$$

such that $(\star)$ holds. Then, by Corollaries 1 and 2, weak solutions of (4.8) in $L^\infty(0, 1; L^p(\mathbb{R}^d))$ are unique provided that

$$\frac{1}{p} + \frac{1}{r} \leq 1. \tag{4.9}$$

Outside the range (4.9), uniqueness of (4.8) may fail. In this section, we demonstrate this by a counterexample of Bruè, Colombo and Kumar [BCK26]. They showed that if $d \geq 2$ and

$$\frac{1}{p} + \frac{d-1}{dr} > 1, \quad (4.10)$$

then there exists a compactly supported vector field

$$b \in C([0, 1]; W^{1,r}(\mathbb{R}^d; \mathbb{R}^d)) \cap C([0, 1]; L^q(\mathbb{R}^d; \mathbb{R}^d)),$$

for which there exist two nonnegative solutions of (4.8) in $C([0, 1]; L^p(\mathbb{R}^d))$.

The full construction for the range (4.10) is not presented here, but only the simpler version from section 2 of [BCK26]. It covers the case $p = 1$ and $r < d$.

Theorem 7 ([BCK26], Theorem 2.1). *Let $d \geq 2$ and $r < d$. Then there exists $0 < \alpha < 1$ and a compactly supported vector field*

$$b \in C([0, 1]; W^{1,r}(\mathbb{R}^d; \mathbb{R}^d)) \cap C^\alpha([0, 1] \times \mathbb{R}^d; \mathbb{R}^d), \quad \operatorname{div} b = 0,$$

for which there are two nonnegative solutions of (4.8) in $L^\infty(0, 1; L^1(\mathbb{R}^d))$ with the same initial condition.

Here, C^α is the space of α -Hölder continuous functions with the seminorm

$$\|f\|_{C^\alpha} := \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\alpha}.$$

4.3.1 The self-similarity of the vector field. First, let us introduce some notation. For $\ell > 0$ and $x \in \mathbb{R}^d$, we denote by $Q_\ell(x)$ the closed cube of side length ℓ centered at x , that is

$$Q_\ell(x) := \left\{ y \in \mathbb{R}^d : \max \{|x_j - y_j| : 1 \leq j \leq d\} \leq \ell/2 \right\}. \quad (4.11)$$

Let $g \in \mathbb{N}_{\geq 1}$ and $k \in \{-1, 1\}^g$. We define the centers of dyadic cubes of the g -th generation in one dimension by

$$c_g^1(k) := \frac{1}{2} \sum_{j=1}^g k_j 2^{-j}.$$

The numbers $c_g^1(k)$ are evenly spaced in $[-1/2, 1/2]$ with distance 2^{-g} . For example, if $g = 2$ they are $-\frac{3}{8}, -\frac{1}{8}, \frac{1}{8}, \frac{3}{8}$. For $k \in (\{-1, 1\}^g)^d$, we define the centers of dyadic cubes in d dimensions by

$$c_g(k) := (c_g^1(k_j))_{j=1}^d. \quad (4.12)$$

For notational convenience, instead of directly constructing b , we will construct a vector field $a(t, \cdot) = -b(1 - t, \cdot)$ and two solutions to

$$\partial_t u - a \cdot \nabla u = 0, \quad u|_{t=1} = u^1. \quad (4.13)$$

Their common value at $t = 1$ is $u^1 := \chi_{Q_1(0)}$. Since a will be supported in $Q_1(0)$, one solution to (4.13) is the constant solution $\chi_{Q_1(0)}(x)$. The second solution is denoted by u and will satisfy

$$u|_{t=0} = u^0 := 2^{\nu d} \chi_{P(2^{-(\nu+1)}, 1)}, \quad \text{where } P(s, \ell) := \bigcup_{k \in \{-1, 1\}^d} Q_s(\ell c_1(k)). \quad (4.14)$$

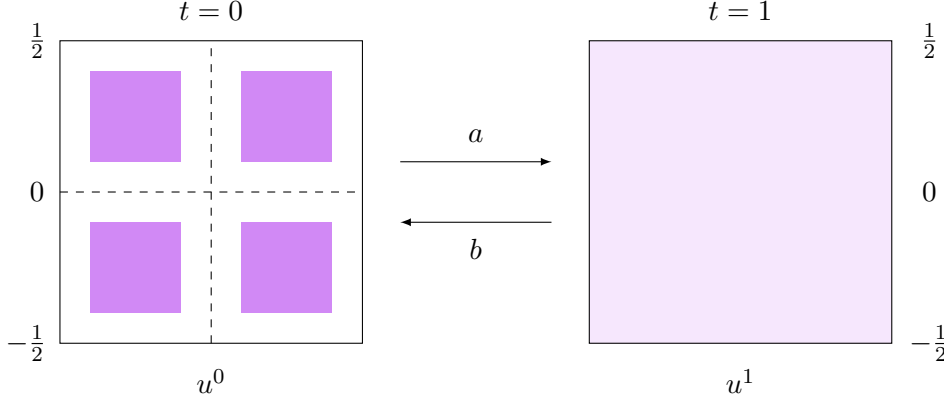


Figure 3: The solution u at $t = 0$ and $t = 1$. The intensity of the color indicates the magnitude of u . (From [BCK26], Figure 2)

for some $\nu > 0$. Note that $\|u^0\|_{L^1} = \|u^1\|_{L^1} = 1$. Figure 3 shows the initial and final state of u for $d = 2$.

The vector field a “diffuses” the 2^d cubes of u^0 by repeatedly breaking them into smaller cubes while preserving the L^1 -norm. To formulate this precisely, we define

$$t_i := 1 - 2^{-(i-1)\beta}, \quad t_i^m := \frac{t_i + t_{i+1}}{2} \quad \text{for } i \in \mathbb{N}_{\geq 1}.$$

At time t_i , the solution u is concentrated on 2^{di} cubes of side length $2^{-(\nu+1)i}$, centered at $c_i(k)$ for $k \in (\{-1, 1\}^i)^d$. The value of u on these cubes is $2^{\nu di}$ and 0 everywhere else, so that $\|u(t_i)\|_{L^1} = 1$. From time t_i to t_i^m , each cube is broken into 2^d smaller cubes of side length $2^{-(\nu+1)(i+1)}$, and the magnitude of u increases so that the L^1 -norm is preserved. At time t_i^m , the centers of the $2^{d(i+1)}$ cubes do not align with the centers of the dyadic cubes of generation $i + 1$. This is fixed by moving the cubes to the centers from time t_i^m to t_{i+1} . The process is shown in Figure 4.

Translating cubes which are separated by a positive distance is relatively straightforward. In §4.3.4, we explicitly construct a vector field $\hat{a}$ for this task, and we define a in terms of $\hat{a}$ on $[t_i^m, t_{i+1}]$.

The crucial part is subdividing each cube into smaller cubes. The key observation of [BCK26] is that breaking a cube into 2^d smaller cubes is exactly the original problem, reversed in time. Thus, a should be defined on $[t_i, t_i^m]$ as a collection of 2^{di} time-reversed and scaled copies of itself.

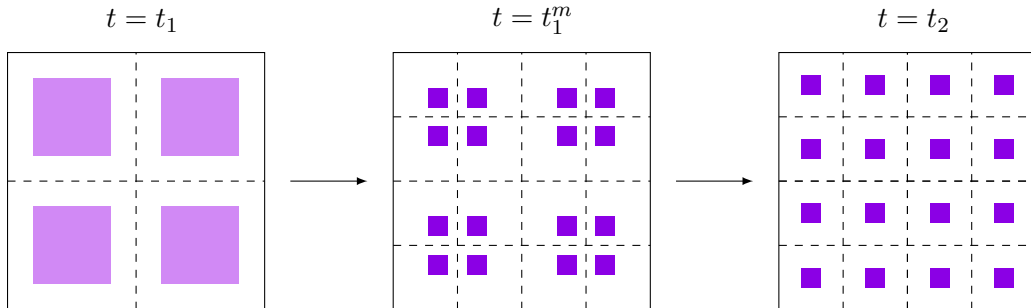


Figure 4: The evolution from $t = t_1$ to $t = t_1^m$ consists of four scaled and time-reversed copies of the original problem. From time t_1^m to t_2 , the vector field $\hat{a}$ is used to translate the cubes, so their centers align with $c_2(k)$ for $k \in (\{-1, 1\}^2)^2$. (From [BCK26], Figure 5)

4.3.2 *The contraction map.* The self-similar nature of a suggests constructing a as the fixed point of a contraction on a suitable complete metric space. Let A be the space of all

$$a \in C([0, 1]; W^{1,r}(\mathbb{R}^d; \mathbb{R}^d)) \cap C^\alpha([0, 1] \times \mathbb{R}^d; \mathbb{R}^d) \quad (4.15)$$

such that

$$a|_{t=0} = a|_{t=1} = 0, \quad \operatorname{div} a = 0, \quad \operatorname{supp} a \subset [0, 1] \times Q_1(0). \quad (4.16)$$

We equip A with the norm

$$\|a\|_A := \|\nabla a\|_{C([0,1]; L^r(\mathbb{R}^d))} + \|a\|_{C^\alpha([0,1] \times \mathbb{R}^d)}. \quad (4.17)$$

Lemma 10. *If $r < d$ and $0 < \alpha < 1$, then A is a Banach space.*

Proof. Since $r < d$, by Sobolev inequalities and $\operatorname{supp} a \subset [0, 1] \times Q_1(0)$,

$$\|a\|_{L^r(\mathbb{R}^d)} \leq C_{d,r} \|\nabla a\|_{L^r(\mathbb{R}^d)}. \quad (4.18)$$

Thus, if $\|a\|_A = 0$, then $a = 0$. Since the triangle inequality and homogeneity of $\|\cdot\|_A$ are clear from the definition, $\|\cdot\|_A$ indeed defines a norm.

To show that A is complete, let $a_n \in A$ be a Cauchy sequence with respect to (4.17). By (4.18) and the completeness of $C([0, 1]; W^{1,r}(\mathbb{R}^d))$, a_n converges in $C([0, 1]; W^{1,r}(\mathbb{R}^d))$ to some a_W .

Next, we observe that

$$A \hookrightarrow C^{0,\alpha}([0, 1] \times Q_1(0); \mathbb{R}^d), \quad (4.19)$$

where $C^{0,\alpha}$ is the space of functions f for which the norm

$$\|f\|_{C^{0,\alpha}} := \|f\|_{C_b} + \|f\|_{C^\alpha}$$

is finite. The embedding (4.19) is continuous, since for every $a \in A$ and $(t, x) \in (0, 1] \times Q_1(0)$

$$|a(t, x)| \leq \frac{|a(t, x) - a(0, x)|}{t^\alpha} \leq \|a\|_{C^\alpha([0,1] \times \mathbb{R}^d)}.$$

The first inequality holds because $a|_{t=0} = 0$ and $0 < t \leq 1$. Since $[0, 1] \times Q_1(0)$ is closed and bounded, $C^{0,\alpha}([0, 1] \times Q_1(0); \mathbb{R}^d)$ is a Banach space (see [Eva10], p. 255). Hence, a_n also converges to some a_α in $C^{0,\alpha}([0, 1] \times Q_1(0); \mathbb{R}^d)$.

The two limits a_W and a_α must agree almost everywhere, since a_n converges to a_α uniformly on $[0, 1] \times \mathbb{R}^d$, hence also in $L^1_{\text{loc}}([0, 1] \times \mathbb{R}^d; \mathbb{R}^d)$.

Finally, $a_W = a_\alpha$ also satisfies (4.16) because the maps

$$\begin{aligned} C^{0,\alpha}([0, 1] \times Q_1(0); \mathbb{R}^d) &\rightarrow C^{0,\alpha}(Q_1(0); \mathbb{R}^d) & a &\mapsto a|_{t=0} \text{ or } a|_{t=1}, \\ W^{1,r}(\mathbb{R}^d; \mathbb{R}^d) &\rightarrow L^r(\mathbb{R}^d) & a &\mapsto \operatorname{div} a \end{aligned}$$

are continuous. □

We now define a contraction F on A . To this end, we introduce the following notations.

$$\Delta_i := t_i^m - t_i = 2^{-i\beta} \left(\frac{2^\beta - 1}{2} \right), \quad \mathcal{K}_i^d := (\{-1, 1\}^i)^d \quad (4.20)$$

If $t \in [t_i, t_i^m)$, we define

$$F(a)(t, x) := \frac{1}{2^{(\nu+1)i} \Delta_i} \sum_{k \in \mathcal{K}_i^d} a \left(\frac{t_i^m - t}{\Delta_i}, 2^{(\nu+1)i} (x - c_i(k)) \right) \quad (4.21)$$

The scaling factor in space is $2^{-(\nu+1)i}$, since each block at time t_i has side length $2^{-(\nu+1)i}$. If $t \in [t_i^m, t_{i+1})$, we define

$$F(a)(t, x) := \frac{1}{2^i \Delta_i} \sum_{k \in \mathcal{K}_i^d} \hat{a} \left(\frac{t - t_i^m}{\Delta_i}, 2^i(x - c_i(k)); 2^{-\nu(i+1)-1}, 2^{-\nu i} \right) \quad (4.22)$$

Here $\hat{a}(t, x; s, \ell)$ is the vector field from Proposition 4 below, which translates 2^d cubes of side length s , evenly arranged in $Q_\ell(0)$, to the centers of the first-generation dyadic cubes, namely $c_1(k)$ for $k \in \mathcal{K}_1^d$. The cubes should be moved to the centers of the $(i+1)$ -th generation dyadic cubes, hence the space variable is scaled by 2^i . Since our cubes have side length $2^{-(\nu+1)(i+1)}$, we set the parameter s of $\hat{a}$ to $2^{-(\nu+1)(i+1)}/2^{-i} = 2^{-\nu(i+1)-1}$. Finally, we define $F(a)(1, x) = 0$.

To prove that F maps A to A and is a contraction, we need estimates for $\hat{a}$. These are collected in Proposition 4, together with properties of the corresponding solution $\hat{u}$. The proof is deferred to §4.3.4. As described above, the values of $\hat{u}$ at $t = 0$ and $t = 1$ are

$$\hat{u}^0 = \chi_{P(s, \ell)}, \quad \hat{u}^1 = \chi_{P(s, 1)}, \quad \text{where } P(s, \ell) := \bigcup_{k \in \{-1, 1\}^d} Q_s(\ell c_1(k)). \quad (4.23)$$

Proposition 4. *Let $0 < 4s < \ell < 1$. Then there exists*

$$\hat{a}(t, x; s, \ell) \in C_c^\infty([0, 1] \times \mathbb{R}^d; \mathbb{R}^d), \quad \text{such that } \operatorname{div} \hat{a} = 0, \quad \operatorname{supp} \hat{a} \subset [1/3, 2/3] \times Q_1(0)$$

and $\hat{u}(t, x; s, \ell) \in L^\infty([0, 1] \times \mathbb{R}^d)$, $\hat{u} \geq 0$ such that

$$\partial_t \hat{u} - \hat{a} \cdot \nabla \hat{u} = 0 \quad \text{on } [0, 1] \times \mathbb{R}^d, \quad \hat{u}|_{t \in [0, 1/3]} = \hat{u}^0, \quad \hat{u}|_{t \in [2/3, 1]} = \hat{u}^1. \quad (4.24)$$

Furthermore

$$\begin{aligned} m^d(\operatorname{supp} \hat{a}(t, \cdot)) &\leq 4^d s^d \quad \text{for all } t \in [0, 1], \\ \|\hat{a}\|_{L_{t,x}^\infty} &\leq C_d, \quad \|\nabla_{(t,x)} \hat{a}\|_{L_{t,x}^\infty} \leq C_d/s. \end{aligned} \quad (4.25)$$

Proposition 5. *Let $d \geq 2$ and $0 < \alpha, \beta < 1$. If*

$$\nu > 1 - \log_2(2^\beta - 1), \quad r < \frac{\nu d}{\nu + \beta}, \quad \alpha < \frac{1 - \beta}{1 + \nu}, \quad (4.26)$$

then F maps A to A and is a contraction.

Proof. Let $a \in A$. We first estimate $\|\nabla F(a)(t)\|_{L_x^r}$. If $t \in [t_i, t_{i+1}^m)$, by the chain rule

$$\nabla F(a)(t) = \frac{1}{\Delta_i} \sum_{k \in \mathcal{K}_i^d} \nabla a \left(\frac{t_i^m - t}{\Delta_i}, 2^{(\nu+1)i}(x - c_i(k)) \right).$$

Since the summands have disjoint support,

$$\|\nabla F(a)(t)\|_{L^r} = \frac{1}{\Delta_i} \left(\sum_{k \in \mathcal{K}_i^d} \int_{\mathbb{R}^d} \left| \nabla a \left(\frac{t_i^m - t}{\Delta_i}, 2^{(\nu+1)i}(x - c_i(k)) \right) \right|^r dx \right)^{1/r}.$$

By changing variables $x' = 2^{(\nu+1)i}(x - c_i(k))$ and using that $|\mathcal{K}_i^d| = 2^{id}$, we obtain

$$\|\nabla F(a)(t)\|_{L^r} = \frac{1}{\Delta_i} \left(\sum_{k \in \mathcal{K}_i^d} \frac{1}{2^{(\nu+1)id}} \|\nabla a((t_i^m - t)/\Delta_i)\|_{L_x^r}^r \right)^{1/r} = \frac{1}{\Delta_i 2^{\nu di/r}} \|\nabla a\|_{C_t L_x^r}. \quad (4.27)$$

Substituting the definition of Δ_i (4.20), the constant on the right-hand side is

$$\frac{1}{\Delta_i 2^{\nu di/r}} = 2^{i(\beta - \nu d/r)} \frac{2}{2^\beta - 1} < 2^{-i\nu} \frac{2}{2^\beta - 1} < 1. \quad (4.28)$$

Here, we used that $\nu d/r > \nu + \beta$ and $\nu > 1 - \log_2(2^\beta - 1)$.

Let $a, a' \in A$. By definition, $F(a)(t) = F(a')(t)$ for every t that is not in $[t_i, t_i^m]$ for some $i \in \mathbb{N}_{\geq 1}$. Therefore, by (4.27) and (4.28)

$$\|\nabla(F(a) - F(a'))\|_{C_t L_x^r} = \|\nabla F(a - a')\|_{C_t L_x^r} < \|\nabla(a - a')\|_{C_t L_x^r}. \quad (4.29)$$

If $t \in [t_i^m, t_{i+1})$, then by a similar computation as above and estimates for the building block vector field (4.25)

$$\begin{aligned} \|\nabla F(a)(t)\|_{L^r} &\leq \frac{1}{\Delta_i} \left\| \nabla \hat{a}\left(t, x; 2^{-\nu(i+1)-1}, 2^{-\nu i}\right) \right\|_{C_t L_x^r} \leq C_{\nu, d, r} \frac{2^{\nu i(1-d/r)}}{\Delta_i} \\ &= C_{\nu, d, r} 2^{i(\nu - \nu d/r + \beta)} \frac{2}{2^\beta - 1} < C_{\nu, d, r} \frac{2}{2^\beta - 1}. \end{aligned} \quad (4.30)$$

In the last line, we used that $r < \frac{\nu d}{\nu + \beta}$. From (4.27), (4.28) and (4.30), we conclude

$$\|\nabla F(a)\|_{C_t L_x^r} \leq C_{\nu, d, r, \beta} + \|\nabla a\|_{C_t L_x^r}. \quad (4.31)$$

We now turn to the Hölder part of the norm (4.17). For this, we note that for any two functions f, g ,

$$\text{supp } f \cap \text{supp } g = \emptyset \implies \|f + g\|_{C^\alpha} \leq \max\{\|f\|_{C^\alpha}, \|g\|_{C^\alpha}\}. \quad (4.32)$$

Since $\text{supp}_t \hat{a} \subset [1/3, 2/3]$,

$$\|F(a)\|_{C^\alpha([0,1] \times \mathbb{R}^d)} \leq \sup_{i \in \mathbb{N}} \left(\max \left\{ \|F(a)\|_{C^\alpha([t_i, t_i^m] \times \mathbb{R}^d)}, \|F(a)\|_{C^\alpha([t_i^m, t_{i+1}] \times \mathbb{R}^d)} \right\} \right). \quad (4.33)$$

We first look at the intervals $[t_i, t_i^m]$. Since $\nu > 0$, the summands in (4.21) have disjoint support in space. Hence it suffices to estimate them separately by (4.32). For every $k \in \mathcal{K}_i^d$,

$$\begin{aligned} &\left\| a\left(\frac{t_i^m - t}{\Delta_i}, 2^{(\nu+1)i}(x - c_i(k))\right) \right\|_{C^\alpha([t_i, t_i^m] \times \mathbb{R}^d)} \\ &= \sup_{\substack{s, t \in [t_i, t_i^m], s \neq t \\ x \neq y}} \frac{\left| a\left(\frac{t_i^m - t}{\Delta_i}, 2^{(\nu+1)i}(x - c_i(k))\right) - a\left(\frac{t_i^m - s}{\Delta_i}, 2^{(\nu+1)i}(y - c_i(k))\right) \right|}{(|t - s|^2 + |x - y|^2)^{\alpha/2}} \\ &= \sup_{\substack{s, t \in [0,1], s \neq t \\ x \neq y}} \frac{|a(t, x) - a(s, y)|}{(\Delta_i^2 |t - s|^2 + 2^{-2(\nu+1)i} |x - y|^2)^{\alpha/2}} \\ &\leq \max \left\{ \frac{1}{\Delta_i}, 2^{(\nu+1)i} \right\}^\alpha \|a\|_{C^\alpha([0,1] \times \mathbb{R}^d)}. \end{aligned}$$

Therefore, by the definition of F on $[t_i, t_i^m]$,

$$\|F(a)\|_{C^\alpha([t_i, t_i^m] \times \mathbb{R}^d)} \leq \frac{1}{2^{(\nu+1)i} \Delta_i} \max \left\{ \frac{1}{\Delta_i}, 2^{(\nu+1)i} \right\}^\alpha \|a\|_{C^\alpha([0,1] \times \mathbb{R}^d)}. \quad (4.34)$$

We estimate the two arguments of the maximum separately. For $1/\Delta_i$, by the definition of Δ_i in (4.20) and since $2^\nu > 2/(2^\beta - 1)$,

$$\frac{1}{2^{(\nu+1)i} \Delta_i^{1+\alpha}} < 2^{-(1+\nu)i} 2^{\beta(1+\alpha)i} 2^{\nu(1+\alpha)i}.$$

Note that $\nu > 1$ since $\beta < 1$. Since $\beta(1 + \alpha) < 2$, the above expression is largest for $i = 1$. Using that $\beta\alpha < \alpha$, we obtain

$$\frac{1}{2^{(\nu+1)i}\Delta_i^{1+\alpha}} < 2^{-(1+\nu)+\beta+\beta\alpha+\nu+\nu\alpha} < 2^{-1+\beta+\alpha+\nu\alpha} < 1. \quad (4.35)$$

The last inequality follows from $\alpha < \frac{1-\beta}{1+\nu}$. For the argument $2^{(\nu+1)i}$ of the maximum in (4.34), by $\alpha < \frac{1-\beta}{1+\nu}$ we have

$$\frac{1}{2^{(\nu+1)(1-\alpha)i}\Delta_i} < 2^{(1-\beta)i} 2^{-(\nu+1)i} 2^{\beta i} \frac{2}{2^\beta - 1} = 2^{-\nu i} \frac{2}{2^\beta - 1} < 1. \quad (4.36)$$

From (4.34), (4.35) and (4.36) we conclude

$$\|F(a)\|_{C^\alpha([t_i, t_i^m] \times \mathbb{R}^d)} \leq C_{\nu, \beta} \|a\|_{C^\alpha([0, 1] \times \mathbb{R}^d)}, \quad (4.37)$$

where $C_{\nu, \beta} < 1$ is independent of i .

Let $a, a' \in A$. Because $F(a)(t) = F(a')(t)$ for every t that is not in $[t_i, t_i^m]$ for some i ,

$$\begin{aligned} \|F(a) - F(a')\|_{C_{t,x}^\alpha} &= \sup_{i \in \mathbb{N}} \|F(a) - F(a')\|_{C^\alpha([t_i, t_i^m] \times \mathbb{R}^d)} \\ &= \sup_{i \in \mathbb{N}} \|F(a - a')\|_{C^\alpha([t_i, t_i^m] \times \mathbb{R}^d)} < \|a - a'\|_{C^\alpha([0, 1] \times \mathbb{R}^d)} \end{aligned} \quad (4.38)$$

It remains to estimate the C^α -norm of $F(a)$ on the intervals $[t_i^m, t_{i+1}]$. By definition of C^α and (4.22),

$$\begin{aligned} \|F(a)\|_{C^\alpha([t_i^m, t_{i+1}] \times \mathbb{R}^d)} &\leq 2\|F(a)\|_{L^\infty([t_i^m, t_{i+1}] \times \mathbb{R}^d)}^{1-\alpha} \|\nabla_{(t,x)} F(a)\|_{L^\infty([t_i^m, t_{i+1}] \times \mathbb{R}^d)}^\alpha \\ &\leq \frac{2}{2^i \Delta_i} \|\hat{a}\|_{L_{t,x}^\infty}^\alpha \max\left\{\frac{1}{\Delta_i}, 2^i\right\}^\alpha \left\|\nabla_{(t,x)} \hat{a}(t, x; 2^{-\nu(i+1)-1}, 2^{-\nu i})\right\|_{L_{t,x}^\infty}^\alpha \end{aligned}$$

We estimate $\max\{1/\Delta_i, 2^i\} \leq C_\beta 2^i$ and use (4.25),

$$\begin{aligned} \|F(a)\|_{C^\alpha([t_i^m, t_{i+1}] \times \mathbb{R}^d)} &\leq C_{d,\alpha,\beta} 2^{-i} 2^{i\beta} 2^{i\alpha} 2^{(\nu(i+1)+1)\alpha} \\ &\leq C_{d,\alpha,\nu,\beta} 2^{-i(1-\alpha-\beta-\alpha\nu)}. \end{aligned} \quad (4.39)$$

Since $\alpha < \frac{1-\beta}{1+\nu}$, this is largest for $i = 1$.

We can now complete the proof of the Proposition. By (4.31), (4.33), (4.37) and (4.39), F maps A to A . By (4.29) and (4.38), F is a contraction. $\square$

4.3.3 Construction of the solution u . Analogously to the construction of the vector field, we obtain u as the fixed point of a contraction. Let $1 < \gamma, \lambda < \infty$ and let U be the space of all

$$u \in L^\gamma(0, 1; L^1(\mathbb{R}^d)) \cap \text{Lip}([0, 1]; W^{-1,\lambda}(\mathbb{R}^d))$$

such that

$$u|_{t=0} = u^0, \quad u|_{t=1} = u^1, \quad u \geq 0 \text{ a.e.}, \quad \text{supp } u \subset [0, 1] \times Q_1(0).$$

Here $\text{Lip}([0, 1]; W^{-1,\lambda}(\mathbb{R}^d))$ is the set of Lipschitz-continuous functions $[0, 1] \rightarrow W^{-1,\lambda}(\mathbb{R}^d)$. We define the following seminorm on $\text{Lip}([0, 1]; W^{-1,\lambda}(\mathbb{R}^d))$.

$$\|u\|_{\text{Lip}([0, 1]; W^{-1,\lambda}(\mathbb{R}^d))} := \sup_{s, t \in [0, 1], t \neq s} \frac{\|u(s, \cdot) - u(t, \cdot)\|_{W^{-1,\lambda}(\mathbb{R}^d)}}{|s - t|}$$

The norm of U is given by

$$\|u\|_U := \|u\|_{L^\gamma(0,1;L^1(\mathbb{R}^d))} + \|u\|_{\text{Lip}([0,1];W^{-1,\lambda}(\mathbb{R}^d))}.$$

We define a map G on U similarly to F . Let $u \in U$. If $t \in [t_i, t_i^m]$, we set

$$G(u)(t, x) = 2^{\nu d i} \sum_{k \in \mathcal{K}_i^d} u\left(\frac{t_i^m - t}{\Delta_i}, 2^{(\nu+1)i}(x - c_i(k))\right). \quad (4.40)$$

If $t \in [t_i^m, t_{i+1}]$, we set

$$G(u)(t, x) = 2^{\nu d(i+1)} \sum_{k \in \mathcal{K}_i^d} \hat{u}\left(\frac{t - t_i^m}{\Delta_i}, 2^i(x - c_i(k)); 2^{-\nu(i+1)-1}, 2^{-\nu i}\right), \quad (4.41)$$

where $\hat{u}$ is from Proposition 4. If $t = 1$, we define $G(u)(t, x) = u^1(x) = \chi_{Q_1(0)}$.

Proposition 6. *Let $d \geq 2$ and $1 < \gamma, \lambda < \infty$. If $\nu \geq 2$ and*

$$\frac{\nu d}{1 - \beta} < \frac{\lambda}{\lambda - 1}, \quad (4.42)$$

then G maps U to U and is a contraction.

We omit the proof of Proposition 6. It can be found in [BCK26], Proposition 2.4.

Proof of Theorem 7. For every $0 < \beta < 1$, (4.26) is satisfied for large enough ν since $r < d$. By choosing $\lambda > 1$ close enough to 1, also (4.42) holds. By the Banach fixed-point theorem and Propositions 5 and 6, there exist unique fixed points $a \in A$ and $u \in U$ of F and G respectively.

Let $\phi \in C_c^\infty(\mathbb{R}^d)$. To prove that a and u satisfy (4.13), we need to show that $\int_{\mathbb{R}^d} u \phi dx$ is almost everywhere differentiable in $[0, 1]$ and for a.e. $t \in [0, 1]$

$$\frac{d}{dt} \int_{\mathbb{R}^d} u \phi dx = - \int_{\mathbb{R}^d} u a \cdot \nabla \phi dx. \quad (4.43)$$

Since

$$\left| \int_{\mathbb{R}^d} u(t, x) \phi dx - \int_{\mathbb{R}^d} u(s, x) \phi dx \right| \leq \|u(t) - u(s)\|_{W^{-1,\lambda}} \|\phi\|_{W^{1,\lambda}},$$

and $u \in \text{Lip}([0, 1]; W^{-1,\lambda})$, the derivative on the left-hand side of (4.43) exists for almost every $t \in [0, 1]$. Let $M \subset [0, 1]$ be the set where $\int_{\mathbb{R}^d} u \phi dx$ is differentiable and (4.43) holds. Since $a \in C_c([0, 1] \times \mathbb{R}^d)$ and $u \in L^\gamma(0, 1; L^1(\mathbb{R}^d))$, the right-hand side of (4.43) is a measurable function on $[0, 1]$, hence M is measurable.

By definitions (4.22) and (4.41), on $[t_i^m, t_{i+1}]$ for all $i \in \mathbb{N}_{\geq 1}$, $F(a)$ and $G(u)$ consist of scaled and translated copies of $\hat{a}$ and $\hat{u}$. Thus $\bigcup_{i \in \mathbb{N}_{\geq 1}} (t_i^m, t_{i+1}) \subset M$ by Proposition 4 and since a and u are fixed points. On the intervals $[t_i, t_i^m]$, $F(a)$ and $G(u)$ are defined as scaled and translated copies of a and u . Because a and u are fixed points and the measure of $\bigcup_{i \in \mathbb{N}_{\geq 1}} [t_i, t_i^m]$ is $1/2$, we conclude

$$m^1(M) \geq \frac{1}{2} + \frac{m^1(M)}{2}.$$

Thus $m^1(M) = 1$.

It remains to show that $u \in L^\infty(0, 1; L^1(\mathbb{R}^d))$. By choosing ϕ in (4.43) such that $\phi|_{Q_1(0)} = 1$, we see that

$$\int_{\mathbb{R}^d} u(t, x) dx = \int_{\mathbb{R}^d} u^1 dx = 1$$

for every $t \in [0, 1]$. Since $u \geq 0$, also $\|u(t)\|_{L_x^1} = 1$ for every $t \in [0, 1]$. $\square$

4.3.4 *Translating cubes.* We now show how to construct $\hat{a}$ and $\hat{u}$ with the properties listed in Proposition 4. The construction here differs from the one in section 7 of [BCK26]. The extension of the two-dimensional vector field $\mathbf{b}_2$ used there is not a vector field on $\mathbb{R}^d$ when $d \geq 3$.

Let $\xi \in \mathbb{R}^d$ such that $|\xi| = 1$. Let $\psi = \eta(2|x|) \in C_c^\infty((-1, 1))$, where η is a smooth cutoff function defined in (2.6). With $\Psi(x) := \prod_{i=1}^d \psi(x_i)$, we define the tensor

$$\omega_{ij}^\xi(x) := (\xi_i x_j - \xi_j x_i) \Psi(x),$$

and the vector field $\Gamma^\xi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ by

$$\Gamma_i^\xi := \frac{1}{d-1} \sum_{j=1}^d \partial_j \omega_{ij}^\xi. \quad (4.44)$$

Since $\text{supp } \Psi \subset Q_2(0)$, also Γ^ξ is supported in $Q_2(0)$. Moreover,

$$\Gamma_i^\xi|_{Q_1(0)} = \frac{1}{d-1} \sum_{j=1}^d \partial_j (\xi_i x_j - \xi_j x_i) = \frac{1}{d-1} \sum_{j=1}^d (\xi_i - \xi_j \delta_{ij}) = \xi_i.$$

Since ω is antisymmetric and smooth,

$$\text{div } \Gamma^\xi = \frac{1}{d-1} \sum_{i=1}^d \partial_i \sum_{j=1}^d \partial_j \omega_{ij} = 0.$$

Next, we define a vector field $\tau(t, x; s, z^0, z^1)$ that moves a cube of side length s from $z^0 \in \mathbb{R}^d$ to $z^1 \in \mathbb{R}^d$. Let $\xi := (z^1 - z^0)/|z^1 - z^0|$ and $\zeta \in C^\infty([0, 1]; [0, 1])$ such that $\zeta|_{[0, 1/3]} = 0$ and $\zeta|_{[2/3, 1]} = 1$. We set

$$z(t) := z^0 + \zeta(t)(z^1 - z^0)$$

and define

$$\tau(t, x; s, z^0, z^1) := |z'(t)| \Gamma^\xi \left(\frac{x - z(t)}{s} \right) \quad \text{for } t \in [0, 1], x \in \mathbb{R}^d. \quad (4.45)$$

Finally, we arrange 2^d copies of τ :

$$\hat{a}(t, x; s, \ell) := \sum_{k \in \mathcal{K}_1^d} \tau(t, x; s, \ell c_1(k), c_1(k)). \quad (4.46)$$

The solution $\hat{u}$ is defined by

$$\hat{u}(t, x) := \chi_{P(s, \ell(1-\zeta(t))+\zeta(t))}(x), \quad \text{where } P(s, \ell) := \bigcup_{k \in \{-1, 1\}^d} Q_s(\ell c_1(k)).$$

By (4.45),

$$|\partial_t \tau| \leq C_{\zeta, \Gamma^\xi} \left(|z_1 - z_0| + \frac{|z_1 - z_0|^2}{s} \right), \quad |\nabla_x \tau| \leq C_{\zeta, \Gamma^\xi} \frac{|z_1 - z_0|}{s}.$$

In the definition of $\hat{a}$, we set $z^0 = \ell c_1(k)$ and $z_1 = c_1(k)$. Since $|c_1(k)| \leq \sqrt{d/2}$ for all $k \in \mathcal{K}_1^d$, and since $0 < \ell < 1$,

$$|\nabla_{(t,x)} \hat{a}| \leq 2^d C_{\zeta, \Gamma^\xi} \left(|1 - \ell| \sqrt{d/2} + \frac{|1 - \ell| \sqrt{d/2} + |1 - \ell|^2 d/2}{s} \right) \leq C_{\zeta, \Gamma^\xi, d} \frac{1}{s}.$$

The last inequality follows because $0 < s < 1$. Since $m^d(\text{supp } \Gamma^\xi) \leq 2^d$ and $|\mathcal{K}_1^d| = 2^d$, we have $m^d(\text{supp } \hat{a}(t, \cdot)) \leq 4^d s^d$ for all $t \in [0, 1]$.

Appendix

In the following Proposition, we denote by $L^1(0, T; C_{\text{loc}}^1(\mathbb{R}^d; \mathbb{R}^d))$ the space of functions $f : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that $f \in L^1(0, T; C_b^1(K; \mathbb{R}^d))$ for every compact $K \subset \mathbb{R}^d$.

Proposition A.1. *Let $b \in L^1(0, T; C_{\text{loc}}^1(\mathbb{R}^d; \mathbb{R}^d))$. Let $\Omega \subset \mathbb{R}^d$ be open and $X : [0, T] \times \Omega \rightarrow \mathbb{R}^d$ be a flow for b , i.e. $X(\cdot, x)$ is an absolutely continuous solution to (1.1) with $x^0 = x$ for every $x \in \Omega$. Then $X(t) \in C^1(\Omega; \mathbb{R}^d)$ for every $t \in [0, T]$. Furthermore, $DX(\cdot, x)$ is absolutely continuous on $[0, T]$ for every $x \in \Omega$ and*

$$\partial_t DX(t, x) = Db(t, X(t, x))DX(t, x). \quad (\text{A.1})$$

Proof. For every $x \in \Omega$, let $J(\cdot, x) \in W^{1,1}((0, T); \mathbb{R}^{d \times d})$ be a solution of the linear differential equation

$$\partial_t J(t, x) = Db(t, X(t, x))J(t, x), \quad J(0, x) = \text{id}_{\mathbb{R}^d}. \quad (\text{A.2})$$

For the existence of such a solution, note that for fixed t and x ,

$$Db(t, X(t, x)) : \mathbb{R}^{d \times d} \rightarrow \mathbb{R}^{d \times d}, \quad J \mapsto Db(t, X(t, x))J$$

is Lipschitz continuous with Lipschitz constant $\|Db(t, X(t, x))\|_{\mathbb{R}^d \rightarrow \mathbb{R}^d}$. Here $\|\cdot\|_{\mathbb{R}^d \rightarrow \mathbb{R}^d}$ is the matrix norm induced by the euclidean norm on $\mathbb{R}^d$. Since $X([0, T], x) \subset \mathbb{R}^d$ is compact, $\|Db(t, X(t, x))\|_{\mathbb{R}^d \rightarrow \mathbb{R}^d} \in L^1([0, T])$. Thus the existence and uniqueness of a solution to (A.2) follow from the Cauchy-Lipschitz theorem. Since $Db(t, X(t, x))$ depends continuously on x , also $J(t, x)$ is continuous in x .

Let $x \in \Omega$ and let $\varepsilon > 0$ such that $\overline{B_\varepsilon(x)} \subset \Omega$. For $h \in B_\varepsilon(0)$, we define

$$r_h(t, x) := X(t, x + h) - X(t, x) - J(t, x)h.$$

Our goal is to show that $\lim_{h \rightarrow 0} r_h(t, x)/|h| = 0$ for all $t \in [0, T]$. By (A.2) and since $\partial_t X(t, x) = b(t, X(t, x))$,

$$\frac{d}{dt}|r_h(t, x)| \leq |b(t, X(t, x + h)) - b(t, X(t, x)) - Db(t, X(t, x))J(t, x)h|. \quad (\text{A.3})$$

Writing $k = X(t, x + h) - X(t, x)$, we have

$$b(t, X(t, x) + k) - b(t, X(t, x)) = \int_0^1 Db(t, X(t, x) + sk)k \, ds.$$

Thus

$$\begin{aligned} & |b(t, X(t, x) + k) - b(t, X(t, x)) - Db(t, X(t, x))k| \\ & \leq |k| \int_0^1 \|Db(t, X(t, x) + sk) - Db(t, X(t, x))\|_{\mathbb{R}^d \rightarrow \mathbb{R}^d} \, ds \\ & \leq |k| A_t(X(t, x), |k|), \end{aligned} \quad (\text{A.4})$$

where

$$A_t(z, \lambda) := \sup_{y \in B_\lambda(z)} \|Db(t, y) - Db(t, z)\|_{\mathbb{R}^d \rightarrow \mathbb{R}^d}.$$

Define $K := X([0, T] \times \overline{B_\varepsilon(x)}) \subset \mathbb{R}^d$. Since X is continuous, K is compact. By combining (A.4) with (A.3),

$$\begin{aligned} \frac{d}{dt}|r_h(t, x)| & \leq |Db(t, X(t, x))(k - J(t, x)h)| + |k| A_t(X(t, x), |k|) \\ & \leq \|Db(t, X(t, x))\|_{\mathbb{R}^d \rightarrow \mathbb{R}^d} |r_h(t, x)| + C|h| A_t(X(t, x), C|h|) \end{aligned}$$

Here $C = \sup_{t \in [0, T]} \text{Lip}(X(t)|_{B_\varepsilon(x)})$, which is finite by Remark 2. Thus, by Grönwall's inequality and $r_h(0, x) = 0$

$$|r_h(t, x)| \leq \exp\left(\|Db\|_{L^1(0, T; C_b(K))}\right) C|h| \int_0^t A_s(X(s, x), C|h|) ds.$$

Since $A_s(X(s, x), C|h|) \leq 2\|Db(s)\|_{C_b(K')}$, where $K' := K + B_{C|h|}(0)$, by the dominated convergence theorem

$$\lim_{h \rightarrow 0} \int_0^t A_s(X(s, x), C|h|) ds = 0.$$

Thus $\lim_{h \rightarrow 0} r_h/|h| = 0$ and X is differentiable in x with derivative $DX(t, x) = J(t, x)$. $\square$

Proposition A.2. *Let $\Omega \subset \mathbb{R}^d$ be open and bounded, with Lipschitz boundary $\partial\Omega$. If $u \in BV(\Omega)$, then $u \in W^{s,1}(\Omega)$ for every $0 < s < 1$ and $\|u\|_{W^{s,1}(\Omega)} \leq C_{\Omega,s}(\|u\|_{BV(\Omega)} + \|u\|_{L^1(\Omega)})$.*

Proof. We extend u by 0 to $\mathbb{R}^d$ and denote the extension also by u . By [EG15], Theorem 5.6 (p. 204) and Theorem 5.8 (p. 210),

$$\|u\|_{BV(\mathbb{R}^d)} \leq C_\Omega \|u\|_{BV(\Omega)}.$$

Therefore, it suffices to show that

$$\|u\|_{W^{s,1}(\mathbb{R}^d)} \leq C_{d,s} \left(\|u\|_{BV(\mathbb{R}^d)} + \|u\|_{L^1(\mathbb{R}^d)} \right).$$

First, we change variables $w = z + h$ in the definition of $[u]_{W^{s,1}(\mathbb{R}^d)}$.

$$[u]_{W^{s,1}(\mathbb{R}^d)} = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \frac{|u(z) - u(w)|}{|z - w|^{d+s}} dw dz = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \frac{|u(z) - u(z+h)|}{|h|^{d+s}} dh dz$$

Next, we split the integral over h for $|h| < 1$ and $|h| \geq 1$.

$$[u]_{W^{s,1}(\mathbb{R}^d)} \leq \int_{B_1} \frac{1}{|h|^{d+s-1}} \int_{\mathbb{R}^d} \frac{|u(z) - u(z+h)|}{|h|} dz dh + 2\|u\|_{L^1(\mathbb{R}^d)} \int_{\mathbb{R}^d \setminus B_1} \frac{1}{|h|^{d+s}} dh$$

Since $s > 0$, the integral of $1/|h|^{d+s}$ over $\mathbb{R}^d \setminus B_1$ is a constant only depending on d and s . By applying Hölder's inequality to the integral over B_1 , we get

$$\begin{aligned} [u]_{W^{s,1}(\mathbb{R}^d)} &\leq \int_{B_1} \frac{1}{|h|^{d+s-1}} dh \sup_{h \in B_1} \left(\int_{\mathbb{R}^d} \frac{|u(z) - u(z+h)|}{|h|} dz \right) + C_{s,d} \|u\|_{L^1(\mathbb{R}^d)} \\ &\leq \tilde{C}_{s,d} \|u\|_{BV(\mathbb{R}^d)} + C_{s,d} \|u\|_{L^1(\mathbb{R}^d)}. \end{aligned}$$

Here, we used that

$$\|u - u(\cdot + h)\|_{L^1(\mathbb{R}^d)} \leq |h| \|u\|_{BV(\mathbb{R}^d)}. \quad (\text{A.5})$$

The estimate (A.5) follows from an approximation by smooth functions, see [EG15], p. 199. Furthermore, $s < 1$ implies that the integral of $1/|h|^{d+s-1}$ over B_1 is finite. $\square$

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